AN ORDINAL-PRIME TRANSFORM OF THE MÖBIUS FUNCTION

Contents

1. Purpose and scope 2
2. Notation and elementary lemmas 4
3. The transform 5
4. The signed prime block 5
5. The prime scale, with explicit error 6
6. Transfer of the Mertens criterion 7
7. The zeta ingredients 7
8. Exact information content 8
9. The residual and its correct centering 8
10. Exact antisymmetric gap representation 9
11. Two variance models 10
12. Numerical verification and measured laws 11
12.1. Protocol 11
12.2. A. Exact statements verified 11
12.3. B. Global residual scale, $100 \le n \le 10^6$ 11
12.4. C. The $h^{3/2}$ law and its stationarity 12
12.5. D. Which variance model is right 12
12.6. E. Distribution of the standardised residual 13
12.7. F. The points are not independent 14
12.8. G. Relative size of the residual 15
12.9. H. Largest excursions 15
12.10. I. The packet envelope 15
12.11. J. Spectrum on the logarithmic coordinate: a classical check 16
12.12. K. Prime-counting variance comparison 17
12.13. L. Cross-coupling test 17
13. Status of the claims 17
14. Ordinal pairing and the quadratic route 18
14.1. Quadratic energy on a balanced prefix 19
14.2. Mean-square criterion and closure of the quadratic route 20
15. Local ordinal correlations beyond quadratic energy 22
16. Generating function, parity cancellation, and higher-order filters 25
16.1. The mod–4 root-of-unity filter 26
17. Excursion blocks, active pairs, and prime residue channels 27
17.1. Survival kernel of the ordinal pairs 29
18. Universal decoding, square-root recovery, and endpoint verification 30
19. Self-iteration of the ordinal-prime transform 32
20. Collisions and elementary modular rigidity of SS 35
21. Level crossings, returns, and negative-excursion energy 36
22. Removing the primes: exact ordinal and positional controls 37
23. What the construction does and does not deliver 39
24. A variant that is not an endpoint statistic (speculative) 40

Date: Version 12 — 7 September 2026.

25. Novelty status 40
26. Summary 41
27. Research programme through alpha: exact pairing and analytic obligations 44
27.1. Step 0: independent notation and target 44
27.2. Step 1: decomposition and genuine pole statements 44
27.3. Step 2: exact pairing without linearisation 45
27.4. Step 3: the nonpolynomial duration observable 46
27.4.1. What fixed-lag arithmetic input actually controls 47
27.5. Step 4: holomorphy for a fixed duration cutoff 48
27.6. Step 5: long pairs and the sign involution 48
27.7. Step 6: the precise original-order target 49
27.7.1. The unfinished-pair term 49
27.7.2. Unresolved Cauchy estimate 49
27.8. Step 7: the rigorous transfer to RH 49
27.9. Step 8: the separate continuation-and-poles route 50
27.9.1. 8A: contours 50
27.9.2. 8B: positivity and the correctly shifted Li map 51
27.9.3. 8C: spectral realisation 51
27.10. Step 9: corrected rules and excluded shortcuts 51
27.11. Step 10: comparison with established research programmes 52
27.11.1. 10A: Mertens criteria and conditional moments 52
27.11.2. 10B: Nyman–Beurling and Báez-Duarte 52
27.11.3. 10C: Li–Weil positivity 52
27.11.4. 10D: Hilbert–Pólya, Berry–Keating, and Connes 52
27.11.5. 10E: de Bruijn–Newman deformation 53
27.11.6. 10F: local correlations and short intervals 53
27.11.7. 10G: rigorous computation and statistical models 53
27.12. Step 11: precise distinctiveness and limits of novelty 53
27.13. Step 12: complete execution order and present status 54
27.14. Step 13: the energy of SS and its exact critical threshold 54
27.15. Step 14: explicit poles and an attempted isolation of delta's singularities 55
27.15.1. 14A: what the two explicit zeta blocks actually imply 56
27.15.2. 14B: a differentiated ordinal block 56
27.15.3. 14C: why a relative asymptotic does not prove meromorphy 57
27.16. Step 15: audit of five routes suggested by the Connes survey 57
27.16.1. 15A: stability of poles is local and conditional 57
27.16.2. 15B: Voronin universality does not forbid a polynomial lower bound 57
27.16.3. 15C: a finite prime window is not yet Weil positivity 58
27.16.4. 15D: an exact determinant with entries zero or signed primes 58
27.16.5. 15E: what regularising the pair measure actually gives 59
27.16.6. 15F: the survey's current programme and the remaining work here 60
References 60

1. Purpose and scope

This note develops the ordinal-prime transform of the Möbius function and determines exactly how much arithmetic information it carries. It is self-contained and uses no other construction.

The starting point is the Möbius function $\mu(n) \in \{-1, 0, +1\}$. The occurrences of -1 and of $+1$ are numbered separately, in increasing order of n, and the corresponding ordinal prime is attached

to each occurrence; zeros are mapped to 1. The resulting signed sequence is $S(n)$ and its partial sum is $SS(n) = \sum_{k \le n} S(k)$.

The conclusions are:

(i) $SS(n)$ has an exact closed form in terms of the Mertens function $M(n)$, the squarefree counting function $Q(n)$, and the sum of the first m primes (Proposition 3.3).

(ii) After subtracting the deterministic zero contribution $n - Q(n)$, the remainder $T(n) := SS(n) - n + Q(n)$ is *exactly* a signed sum of $|M(n)|$ consecutive primes, occupying a window of prime indices symmetric about $(Q(n) + 1)/2$ (Proposition 4.1).

(iii) Consequently $T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}(1 + \theta(n))$ with $\theta(n) \to 0$ effectively, and the Mertens criterion transfers to T in both directions (Theorem 6.2).

(iv) Every endpoint object built in Sections 3–10 — SS, T, the residual E, and the centered residual E_{mid} — is a deterministic function of the pair $(M(n), Q(n))$ alone. The transform therefore creates no independent arithmetic constraint on $M(n)$. This applies to the residuals as well and is stated as Proposition 8.1.

(v) The nonlinearity of the transform splits exactly into a central part linear in $M(n)$ and an *antisymmetric weighted sum of prime gaps* with triangular weights (Theorem 10.1). The weight norm has a closed form (Lemma 10.2), which forces the exponent $3/2$ in the observed law $|E_{\text{mid}}(n)| \approx 0.155\, |M(n)|^{3/2} \log n$.

(vi) The measured size of that residual is quantitatively accounted for by the local covariance structure of prime gaps. A covariance-corrected model predicts its standard deviation to within 0.3% with no free parameter, whereas an independent-gap model underpredicts by a factor 0.853 (Section 12, part D).

(vii) The packet structure visible in E_{mid} is an amplitude effect: its moving RMS follows $|M(n)|^{3/2} \log n$. On the logarithmic coordinate $u = \log n$, the spectrum of $M(e^u)e^{-u/2}$ displays the ordinates γ_k of zeta zeros, as predicted by the classical explicit formula. This spectral observation is a sanity check, not new information about the zeros; the finite u-window has frequency resolution about 0.91.

(viii) A separate test asks whether the prime-gap fluctuation factor is coupled to the sign or zeta phase of M. No such coupling is detected at the present scale. With effective sample about 1.7×10^4, the tests only exclude linear effects larger than roughly 0.015 and do not constitute a theorem of independence.

(ix) Versions 7–9 introduce the inverse-counting displacement $d_j = \beta_j - \alpha_j$. Its logarithmic two-point squarefree-time correlation has an unconditional theorem, but several apparently stronger uses are shown to collapse: odd parity is an involutive algebraic cancellation rather than a Mertens bound; a universal coercive functional can be evenised; a simple real-axis lower bound for its generating function fails by an exact finite counterexample; and the natural two-cut survival kernel depends only on the two endpoint heights.

(x) Version 9 proves a square-root recovery theorem: the genuine prefix $SS(0), \ldots, SS(\lfloor \sqrt{N} \rfloor)$ together with the endpoint $SS(N)$ determines $Q(N)$ and then $M(N)$ uniquely. The theorem is a recovery identity, not a cryptographic authentication statement: an independent verifier must still check that the decoded short prefix is the actual Möbius function.

(xi) Version 9 also studies the self-map $\varphi(n) = |S(n)|$ on the squarefree integers. Its functional digraph is classified exactly: the first ten primes $2, 3, \ldots, 29$ are the only periodic points and are fixed; every other squarefree input reaches a prime at least 31 after one step and then increases strictly to infinity. On prime indices the map is $f(j) = N_-(p_j) = j + C_-(p_j)$, is strictly increasing, and its image has counting function asymptotic to $(\pi^2/3)J/\log J$; hence a density-one set of prime indices starts escaping rays. Escaping orbits satisfy an unconditional prime- number-theorem growth law.

(xii) Version 11 gives exact formulae for every crossing and every return to a fixed Mertens level. Returns to $h > 0$ force SS upward, returns to $h = 0$ change it by exactly the number of intervening nonsquarefree integers, and returns to $h < 0$ contain a strictly positive prime-separation deficit.

(xiii) For a complete negative squarefree excursion of depth H, the sum $\mathcal{A}$ of these prime-separation deficits satisfies the unconditional coercive inequality $\mathcal{A} \geq (H^3 - H)/3$. This is a genuine one-sided use of the negative levels, although an independent upper bound for $\mathcal{A}$ would still be needed to deduce a new bound for M.

(xiv) Version 11 separates position, ordinal growth, and primality exactly. Replacing p_j by j gives $SS_{\mathrm{ord}} = n - Q + M(Q+1)/2$, while the genuinely prime contribution is a signed block of the positive displacements $p_j - j$. The prime number theorem then gives an amplitude amplification asymptotic to $\log(Q/2)$ and a squared energy amplification asymptotic to $\log^2(Q/2)$.

(xv) Version 12 adds the independent research chapter Section 27. It proves the exact Dirichlet decomposition, two-cut duration identity, and holomorphy of every bounded-duration paired subseries. It states the unresolved original-order Cauchy condition with its RH transfer, and compares its precise obligations with established research programmes. This is a programme with proved intermediate statements, not a proof of RH or an assertion of universal priority.

The numerical pipeline was regenerated in Python for $1 \leq n \leq 10^6$ where stated. The package contains the scripts, intermediate arrays, textual outputs, and figure-generation code. Historical follow-up values from earlier versions were replaced by the regenerated values whenever the original computation was not reproducible from a supplied script.

Remark 1.1 (Version 5 audit scope). Version 12 retains the section-based equation numbering and complete equation labelling introduced in Version 5. It also retains the boundary and status corrections made there: the multiplicative prime scale starts at $n = 2$; the residuals are not claimed to be monotone in M; prime gaps are not asserted pointwise to be asymptotic to $\log p$; the cases $|M| \leq 2$ are separated in the gap formula; the prime-counting integral includes the $a = 0$ endpoint convention; all sums used in that integral and in the κ identity are explicitly finite; the Montgomery–Soundararajan comparison is identified as a reference scale for weighted prime counts rather than a theorem about the unweighted π count; and the shared-counter non-endpoint claim is stated as a functional statement with an explicit pair of histories.

2. Notation and elementary lemmas

Let $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ denote the primes in increasing order, and set $p_0 := 0$ by convention (used only to make the product forms below well defined at $n = 1$).

Write

$$M(n) := \sum_{k \leq n} \mu(k), \qquad Q(n) := \sum_{k \leq n} \mu(k)^2, \qquad \mathcal{P}(m) := \sum_{j=1}^{m} p_j, \quad \mathcal{P}(0) := 0. \tag{2.1}$$

Define the cumulative occurrence counts

$$N_-(n) = \sum_{k \leq n} \frac{\mu(k)^2 - \mu(k)}{2}, \qquad N_0(n) = \sum_{k \leq n} (1 - \mu(k)^2), \qquad N_+(n) = \sum_{k \leq n} \frac{\mu(k)^2 + \mu(k)}{2}. \tag{2.2}$$

Thus $N_-(n)$, $N_0(n)$, $N_+(n)$ count the occurrences of $\mu = -1$, $\mu = 0$, $\mu = +1$ up to n, and the counts are inclusive of n itself.

Lemma 2.1 (Counter identities)**.** *For every $n \geq 1$,*

$$N_+(n) + N_-(n) = Q(n), \qquad N_+(n) - N_-(n) = M(n), \qquad N_0(n) = n - Q(n), \tag{2.3}$$

hence

$$N_+(n) = \frac{Q(n) + M(n)}{2}, \qquad N_-(n) = \frac{Q(n) - M(n)}{2}. \tag{2.4}$$

Lemma 2.2 (Parity)**.** *$Q(n) \equiv M(n) \pmod 2$ for every $n \geq 1$. Consequently the indices $(Q(n) \pm M(n))/2$ occurring throughout are integers.*

Proof. $Q = N_+ + N_-$ and $M = N_+ - N_-$, so $Q - M = 2N_-$. □

Lemma 2.2 is used silently in every subsequent formula and was absent from version 1; without it the central identities are not well formed.

3. The transform

Definition 3.1. The three rows are

$$R_-(n) = -\frac{\mu(n)^2 - \mu(n)}{2}\, p_{N_-(n)}, \qquad R_0(n) = 1 - \mu(n)^2, \qquad R_+(n) = \frac{\mu(n)^2 + \mu(n)}{2}\, p_{N_+(n)}, \tag{3.1}$$

with the convention $p_0 = 0$ (needed only at $n = 1$, where $N_-(1) = 0$ and the coefficient vanishes). The ordinal-prime transform is

$$S(n) := R_-(n) + R_0(n) + R_+(n) = \begin{cases} -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \\ +p_{N_+(n)}, & \mu(n) = +1. \end{cases} \tag{3.2}$$

Along the occurrences of $\mu = -1$ the values are $-2, -3, -5, -7, -11, \ldots$; along the occurrences of $\mu = +1$ they are $2, 3, 5, 7, 11, \ldots$, in a separate ordinal count. Explicitly,

$$S(1), \ldots, S(12) = 2,\ -2,\ -3,\ 1,\ -5,\ 3,\ -7,\ 1,\ 1,\ 5,\ -11,\ 1. \tag{3.3}$$

Definition 3.2. $SS(n) := \sum_{k=1}^{n} S(k)$ and $T(n) := SS(n) - \big(n - Q(n)\big)$.

Proposition 3.3 (Closed form)**.** *For every $n \geq 1$,*

$$\boxed{\begin{aligned} SS(n) &= \mathcal{P}\big(N_+(n)\big) - \mathcal{P}\big(N_-(n)\big) + N_0(n) \\ &= \mathcal{P}\Big(\frac{Q(n) + M(n)}{2}\Big) - \mathcal{P}\Big(\frac{Q(n) - M(n)}{2}\Big) + n - Q(n). \end{aligned}} \tag{3.4}$$

Proof. The $k \leq n$ with $\mu(k) = +1$ receive, in order, the primes $p_1, \ldots, p_{N_+(n)}$, contributing $\mathcal{P}(N_+(n))$; those with $\mu(k) = -1$ contribute $-\mathcal{P}(N_-(n))$; each of the $N_0(n)$ zeros contributes 1. The second form is Lemma 2.1, the indices being integers by Lemma 2.2. □

4. The signed prime block

Proposition 4.1 (Block representation)**.** *Write $W(n)$ for the set of prime indices strictly between $N_-(n)$ and $N_+(n)$, inclusive of the larger endpoint; $W(n)$ has exactly $|M(n)|$ elements and is symmetric about the index $(Q(n) + 1)/2$. Then*

$$T(n) = \operatorname{sgn}\big(M(n)\big) \sum_{j \in W(n)} p_j = \begin{cases} \displaystyle\sum_{j=(Q-M)/2+1}^{(Q+M)/2} p_j, & M(n) > 0, \\ 0, & M(n) = 0, \\ \displaystyle -\sum_{j=(Q-|M|)/2+1}^{(Q+|M|)/2} p_j, & M(n) < 0. \end{cases} \tag{4.1}$$

In particular

$$\operatorname{sgn} T(n) = \operatorname{sgn} M(n), \qquad T(n) = 0 \iff M(n) = 0. \tag{4.2}$$

Proof. Immediate from Proposition 3.3: $T = \mathcal{P}(N_+) - \mathcal{P}(N_-)$ is a telescoping difference of prefix sums of the primes. The window endpoints are $\min(N_\pm) + 1$ and $\max(N_\pm)$, whose midpoint is $(N_+ + N_- + 1)/2 = (Q+1)/2$. □

Corollary 4.2 (Exact bounds)**.**

$$|M(n)|\, p_{\frac{Q(n)-|M(n)|}{2}+1} \;\le\; |T(n)| \;\le\; |M(n)|\, p_{\frac{Q(n)+|M(n)|}{2}}. \tag{4.3}$$

Both inequalities are trivial (each of the $|M(n)|$ primes in the block lies between the smallest and the largest one), but the *lower* bound is what makes the transfer of Mertens bounds an equivalence rather than an implication; see Theorem 6.2.

5. The prime scale, with explicit error

We record the multiplicative estimate in a form that is valid at every n, including all $n \ge 2$ with $M(n) = 0$. The case $n = 1$ is excluded only because $\lfloor Q(1)/2 \rfloor = 0$ and the reference prime $p_0 = 0$ cannot be used as a multiplicative scale. (Version 1 wrote $T(n) \sim M(n) p_{Q(n)/2}$; that assertion is empty on the zero set of M, which contains 5361 values of $n \le 10^6$.)

Proposition 5.1. *For $n \ge 2$, set $\theta(n) = 0$ when $M(n) = 0$. When $M(n) \ne 0$, define $\theta(n)$ by equation* (5.1)*. Then*

$$T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big). \tag{5.1}$$

$$|\theta(n)| \;\le\; \max_{|j - Q(n)/2| \le |M(n)|/2+1} \left| \frac{p_j}{p_{\lfloor Q(n)/2 \rfloor}} - 1 \right|, \tag{5.2}$$

and $\theta(n) \to 0$ as $n \to \infty$.

Proof. The bound is Proposition 4.1 together with the triangle inequality. For the limit: by the prime number theorem in the form $M(n) = o(n)$, and by $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, one has $|M(n)|/Q(n) \to 0$; hence the window endpoints are $(1 + o(1))Q(n)/2$ on the index scale, and $p_{(1+o(1))m}/p_m \to 1$ by the prime number theorem again. □

Both hypotheses invoked here are named: $M(n) = o(n)$ is equivalent to the prime number theorem [11], and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$ is the standard squarefree-counting estimate [1].

Remark 5.2 (The asymptotic constant is a poor numerical guide). From $p_m = m(\log m + \log\log m - 1 + o(1))$ and $Q(n)/2 \sim \frac{3}{\pi^2} n$ one gets

$$\begin{aligned} p_{\lfloor Q(n)/2 \rfloor} &= \frac{3}{\pi^2}\, n \log n \\ &\quad \times \left(1 + \frac{\log\log n - 1 + \log(3/\pi^2)}{\log n} + o\Big(\tfrac{1}{\log n}\Big) \right), \end{aligned} \tag{5.3}$$

so the correction to the leading constant decays only like $1/\log n$. Measured values are

$$\frac{T(n)}{(3/\pi^2) n \log n\, M(n)} = 0.9558,\ 0.9954,\ 1.0151,\ 1.0281 \tag{5.4}$$

at $n = 10^3, 10^4, 10^5, 10^6$; they are not monotone and remain 2.8% off at $n = 10^6$. By contrast,

$$\frac{T(n)}{M(n) p_{\lfloor Q(n)/2 \rfloor}} = 1.00200,\ 1.00071,\ 1.000097,\ 0.999998 \tag{5.5}$$

at the same points. Any quantitative use should keep $p_{\lfloor Q(n)/2 \rfloor}$ and not the closed constant.

6. Transfer of the Mertens criterion

Proposition 6.1. *$|T(n)| = O(n \log n\, |M(n)|)$ and hence $SS(n) = O(n \log n\, |M(n)| + n)$. Unconditionally $SS(n) = o(n^2 \log n)$, by $M(n) = o(n)$ (prime number theorem).*

Theorem 6.2. *The following are equivalent.*

(a) The Riemann hypothesis.
(b) $M(n) = O_\varepsilon(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$.
(c) $SS(n) - n + Q(n) = O_\varepsilon(n^{3/2+\varepsilon})$ for every $\varepsilon > 0$.

Proof. (a) $\iff$ (b) is the classical Littlewood criterion; see [7, 11].

(b)$\Rightarrow$(c): fix $\varepsilon > 0$ and apply (b) with $\eta = \varepsilon/2$. By Corollary 4.2 and the prime number theorem, $|T(n)| \ll |M(n)|\, n \log n \ll_\varepsilon n^{3/2+\varepsilon/2} \log n = O_\varepsilon(n^{3/2+\varepsilon})$.

(c)$\Rightarrow$(b): this is the direction version 1 asserted without proof. By the prime number theorem $|M(n)| = o(n)$ and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, so

$$\frac{Q(n) - |M(n)|}{2} + 1 = (1 + o(1))\frac{3}{\pi^2} n, \tag{6.1}$$

whence the smallest prime of the block satisfies $p_{(Q-|M|)/2+1} \geq c\, n \log n$ for some $c > 0$ and all large n. The lower bound of Corollary 4.2 then gives

$$|M(n)| \leq \frac{|T(n)|}{c\, n \log n} \ll_\varepsilon \frac{n^{3/2+\varepsilon}}{n \log n} \ll_\varepsilon n^{1/2+\varepsilon}. \tag{6.2}$$

□

Remark 6.3. Theorem 6.2 is a reformulation, not a proof mechanism. Note also that (b) must not be strengthened to $M(n) = O(\sqrt{n})$: the Riemann hypothesis does not imply the Mertens conjecture, which is false [5].

7. The zeta ingredients

The two Möbius-derived ingredients have the classical Dirichlet series [1]

$$\sum_{m \geq 1} \frac{\mu(m)}{m^s} = \frac{1}{\zeta(s)}, \qquad \sum_{m \geq 1} \frac{\mu(m)^2}{m^s} = \frac{\zeta(s)}{\zeta(2s)}, \qquad \Re s > 1, \tag{7.1}$$

and the prime weights are governed by $\log \zeta(s)$ through the prime-counting function, via the exact Abel identity

$$\mathcal{P}(m) = m\, p_m - \int_2^{p_m} \pi(t)\, dt, \tag{7.2}$$

(verified here for $m \leq 100$; e.g. $\mathcal{P}(3) = 10 = 3 \cdot 5 - 5$).

Version 1 substituted Perron integrals for M and Q into the arguments of $\mathcal{P}$. That is not a functional identity and it is dropped here, for two reasons. First, $\mathcal{P}$ is defined only on nonnegative integers, so an integral placed inside its argument is notation for "the integer that this integral evaluates to", not an analytic representation. Second, at integer x the untruncated Perron integral returns $M(x) - \frac{1}{2}\mu(x)$, not $M(x)$.

The honest statement is the following. Equations (7.1) and (7.2) show that the transform composes three classical objects,

(7.3)

$$\frac{1}{\zeta(s)} \ \text{(signed Möbius imbalance)}, \qquad \frac{\zeta(s)}{\zeta(2s)} \ \text{(squarefree count)}, \qquad \log \zeta(s) \ \text{(prime weights)},$$

through the nonlinear expression $\mathcal{P}\big(\frac{Q+M}{2}\big) - \mathcal{P}\big(\frac{Q-M}{2}\big) + n - Q$. None of the three components is new, and the composition is not a Dirichlet series of anything.

8. Exact information content

Proposition 8.1. *For each n, $SS(n)$ is a deterministic function of the pair $(M(n), Q(n))$. With $Q(n)$ held fixed and $M(n)$ restricted to the admissible parity class $M(n) \equiv Q(n) \pmod 2$, $T(n)$ is strictly increasing in $M(n)$. Every residual defined below, including $E(n)$ and $E_{\mathrm{mid}}(n)$, is likewise a deterministic function of $(M(n), Q(n))$; no monotonicity claim is made for the residuals.*

Proof. The first claim is Lemma 2.1 plus Proposition 3.3. For the second, increasing M by 2 at fixed Q raises N_+ by one and lowers N_- by one, changing T by $p_{N_++1} + p_{N_-} > 0$. The residuals E and E_{mid} are T minus explicit functions of (M, Q), hence are also functions of (M, Q). This last statement is only a dependence statement; subtracting the central term does not preserve the preceding monotonicity automatically. □

Two consequences deserve to be stated plainly.

First, $SS(n)$ does not retain the internal ordering of the ± 1 values of μ below n: two Möbius sign histories with the same endpoint counts produce the same $SS(n)$. It is a nonlinear endpoint statistic.

Second — and this corrects a claim in version 1 — the residual $E(n)$ is *not* structurally distinct from $SS(n)$ in the informational sense. Its window is determined by $(M(n), Q(n))$, so the argument above applies to it verbatim. What is genuinely different about the residual is not its information content but its *size*, which turns out to be governed by prime-gap statistics; that is the subject of the next two sections.

There are two distinct meanings of "information" that must not be mixed. In the logical sense, Proposition 8.1 is final: $E_{\mathrm{mid}}(n) = G(M(n), Q(n))$ for an explicit deterministic G, so no experiment can make E_{mid} contain arithmetic information absent from the endpoint pair. In the analytic or *leverage* sense, one can still ask whether a separately proved bound for the prime-gap fluctuation factor could combine with the identity to constrain M. A bound for T itself cannot do this: at fixed Q, T is strictly monotone in M and the two-sided block bounds make a Mertens bound and the corresponding T bound equivalent up to the explicit prime scale. What remains testable is only whether some independently controlled feature of the gap fluctuation is coupled to the sign, phase, or trajectory of M. Section 12.13 measures a small family of such couplings and finds none at the present resolution; this is a negative numerical result, not a logical strengthening of Proposition 8.1.

9. The residual and its correct centering

Definition 9.1. Let

$$p^*(n) := \tfrac{1}{2}\Big(p_{\lfloor (Q(n)+1)/2 \rfloor} + p_{\lceil (Q(n)+1)/2 \rceil}\Big) \tag{9.1}$$

be the linear interpolation of the prime sequence at the exact window centre $(Q(n) + 1)/2$ of Proposition 4.1. Define

$$E(n) := T(n) - M(n)\, p_{\lfloor Q(n)/2 \rfloor}, \qquad E_{\mathrm{mid}}(n) := T(n) - M(n)\, p^*(n), \tag{9.2}$$

so that the exact decomposition

$$\boxed{T(n) = M(n)\, p^*(n) + E_{\mathrm{mid}}(n)} \tag{9.3}$$

holds for every n.

When $Q(n)$ is odd, $(Q+1)/2$ is an integer and $p^* = p_{(Q+1)/2}$; when $Q(n)$ is even, p^* is the mean of the two central primes.

Proposition 9.2 (Centering offset)**.** *$E(n) - E_{\mathrm{mid}}(n) = M(n)(p^*(n) - p_{\lfloor Q(n)/2 \rfloor})$, where, for $Q(n) \geq 2$, the bracket equals one half of a prime gap when $Q(n)$ is even and one full prime gap when $Q(n)$ is odd. At $n = 1$ the identity still holds under $p_0 = 0$, with bracket $p_1 - p_0 = 2$. It is positive. Pointwise it is exactly one half or one full prime gap; no uniform two-sided estimate $\asymp \log p^*(n)$*

is asserted. Its empirical typical scale in the computed range is comparable with $\log p^*(n)$*, so* E *carries a deterministic component linear in* $M(n)$ *which* E_{mid} *does not.*

The identity was verified exactly for all $n \le 10^6$; the median of $(p^* - p_{\lfloor Q/2 \rfloor})/\log p_{\lfloor Q/2 \rfloor}$ over $10^4 \le n \le 10^6$ is 0.5318. Proposition 9.2 explains the correlation structure of the two residuals: the offset is a fixed positive multiple of $M(n)$, so E correlates with M and E_{mid} does not. Measured over $100 \le n \le 10^6$,

$$\mathrm{corr}(E, M) = +0.2573, \qquad \mathrm{corr}(E_{\mathrm{mid}}, M) = +0.0045. \tag{9.4}$$

E_{mid} is therefore the object to study; E is E_{mid} plus a bookkeeping artefact. Version 1 studied only E.

10. Exact antisymmetric gap representation

Write $g_i := p_{i+1} - p_i$ and $m(n) := \lfloor (Q(n)+1)/2 \rfloor$.

Theorem 10.1 (Gap representation)**.** *Let* $h := |M(n)|$*. If* $h \le 2$*, then* $E_{\mathrm{mid}}(n) = 0$*. For* $h \ge 3$*,*

$$\boxed{E_{\mathrm{mid}}(n) = \mathrm{sgn}(M(n)) \sum_i w_i \, g_i} \tag{10.1}$$

where the weights w_i *are supported on finitely many gaps symmetrically placed about* $m = m(n)$*, are antisymmetric in sign about the centre, and are given explicitly by*

$$h \text{ even}: \quad w_{m+k} = \tfrac{h}{2} - k, \quad w_{m-k} = -(\tfrac{h}{2} - k), \quad 1 \le k \le \tfrac{h}{2} - 1; \tag{10.2}$$

$$h \text{ odd}: \quad w_{m+k} = \tfrac{h-1}{2} - k \ \ (0 \le k \le \tfrac{h-3}{2}), \qquad w_{m-k} = -(\tfrac{h+1}{2} - k) \ \ (1 \le k \le \tfrac{h-1}{2}). \tag{10.3}$$

For even h*, the single gap* g_m *between the two central primes cancels. For odd* h*, the centre is the prime* p_m *and there is no single central gap; the two adjacent gaps belong to the two antisymmetric sides. In either parity case the nonzero weight magnitudes on each side are* $\{1, 2, \ldots, \lfloor (h-1)/2 \rfloor\}$*. The cases* $h = 0, 1, 2$ *are the zero-residual cases stated above.*

Proof. The cases $h = 0, 1, 2$ follow directly from the definition: the block is empty, a single central prime, or the two central primes whose mean is p^*. Hence $E_{\mathrm{mid}} = 0$. Assume now $h \ge 3$ and take $M(n) > 0$; the case $M(n) < 0$ follows by a global sign change.

Suppose h is even, so Q is even by Lemma 2.2, $m = Q/2$ and $p^* = \frac{1}{2}(p_m + p_{m+1})$. The window is $\{m - \frac{h}{2} + 1, \ldots, m + \frac{h}{2}\}$. For $j > m$, $p_j - p^* = \frac{1}{2}g_m + \sum_{i=m+1}^{j-1} g_i$; for $j \le m$, $p_j - p^* = -\frac{1}{2}g_m - \sum_{i=j}^{m-1} g_i$. Summing over the window, the $\frac{1}{2}g_m$ terms occur $\frac{h}{2}$ times with each sign and cancel. Counting how many j contain a given g_{m+k} gives the coefficient $\frac{h}{2} - k$, and symmetrically $-(\frac{h}{2} - k)$ for g_{m-k}.

If h is odd then Q is odd, $m = (Q+1)/2$ and $p^* = p_m$, the window being $\{m - \frac{h-1}{2}, \ldots, m + \frac{h-1}{2}\}$. The same count gives the stated coefficients, the two sides again carrying magnitudes $\frac{h-1}{2}, \ldots, 1$. □

Lemma 10.2 (Weight norm)**.**

$$\boxed{\sum_i w_i^2 = \begin{cases} \dfrac{h(h^2-1)}{12}, & h \text{ odd}, \\[2ex] \dfrac{h(h-1)(h-2)}{12}, & h \text{ even}, \end{cases} \qquad \text{so} \qquad \sum_i w_i^2 = \frac{h^3}{12}(1 + O(h^{-1})).} \tag{10.4}$$

Proof. By Theorem 10.1 each side contributes $\sum_{j=1}^{\lfloor (h-1)/2 \rfloor} j^2$. For h odd this is $2 \cdot \frac{(\frac{h-1}{2})(\frac{h+1}{2})h}{6} = \frac{h(h^2-1)}{12}$; for h even it is $2 \cdot \frac{(\frac{h}{2}-1)(\frac{h}{2})(h-1)}{6} = \frac{h(h-1)(h-2)}{12}$. □

Theorem 10.1, equation (10.1), and Lemma 10.2 are exact and unconditional. They are the reason the exponent 3/2 appears below: it is forced by the triangular weights, not fitted.

Remark 10.3. Theorem 10.1 does *not* evade Proposition 8.1: the window position $m(n)$ and its length h are determined by $(M(n), Q(n))$, and the primes are deterministic, so E_{mid} remains a function of $(M(n), Q(n))$. What the theorem changes is what the *size* of E_{mid} is controlled by, namely the local second-order statistics of prime gaps.

Proposition 10.4 (Prime-counting integral form)**.** *Let $h = |M(n)|$, $a = \min(N_-(n), N_+(n))$ and $b = \max(N_-(n), N_+(n)) = a + h$. Retain the convention $p_0 = 0$ and extend $\pi(t)$ by $\pi(t) = 0$ for $0 \le t < 2$. Then, for $h > 0$,*

$$\boxed{E_{\mathrm{mid}}(n) = \mathrm{sgn}(M(n)) \left\{ \int_{p_a}^{p_b} (b - \pi(t))\, dt - h\big(p^*(n) - p_a\big) \right\}.} \tag{10.5}$$

For $h = 0$ both sides are 0 by convention.

Proof. For $j = 0, \dots, b-1$, on $[p_j, p_{j+1})$ one has $\pi(t) = j$ (with the preceding convention when $j = 0$). The sum below is finite, so there is no interchange of infinite sums or limiting operations:

$$\int_{p_a}^{p_b} (b - \pi(t))\, dt = \sum_{j=a}^{b-1} (b - j)(p_{j+1} - p_j) = \sum_{j=a+1}^{b} (p_j - p_a). \tag{10.6}$$

The last expression is the unsigned prime block minus hp_a. Subtracting $h(p^* - p_a)$ gives the unsigned block minus hp^*, and the global factor $\mathrm{sgn}(M)$ gives E_{mid}. □

Thus each value of E_{mid} may also be read as an integrated prime counting discrepancy over the finite interval $[p_a, p_b]$, of exact length $p_b - p_a$. In the numerical range this length is typically on the scale $h \log p_a$; no uniform pointwise asymptotic for such short blocks is asserted. This is genuine information about prime gaps in that specific window, even though the choice of the window itself is already fixed by (M, Q) and hence cannot create a new restriction on M.

11. Two variance models

Theorem 10.1 suggests modelling E_{mid} as a weighted sum of gap fluctuations. Two models are compared; both are heuristics, and both were tested against the data of Section 12.

Definition 11.1 (Independent-gap model)**.** For $h \ge 3$, treat the finitely many g_i carrying nonzero weights near index $m(n)$ as uncorrelated with common local variance $\sigma_g(n)^2$. Then by Lemma 10.2

$$\mathrm{SD}_{\mathrm{ind}}(n) := \sigma_g(n) \sqrt{\textstyle\sum_i w_i^2} \;\approx\; \frac{\sigma_g(n)}{\sqrt{12}}\, h^{3/2}. \tag{11.1}$$

Definition 11.2 (Covariance-corrected model)**.** For $h \ge 3$, let $\rho(d)$ be the local autocorrelation used in the model, normalised by $\rho(0) = 1$, and put $R_{ab} = \rho(|i_a - i_b|)$ on the support of w. Define

$$\kappa(h) := \frac{w^\top R w}{w^\top w}, \qquad \mathrm{SD}_{\mathrm{cov}}(n) := \sigma_g(n) \sqrt{\kappa(h) \textstyle\sum_i w_i^2}. \tag{11.2}$$

For fixed h the correction factor contains no statistic beyond the gap autocorrelation itself. If

$$c_d(h) := \sum_{|i-j|=d} w_i w_j, \qquad c_0(h) = \sum_i w_i^2, \tag{11.3}$$

then exactly

$$\boxed{\kappa(h) = \frac{1}{c_0(h)} \sum_{d=0}^{D(h)} c_d(h) \rho(d),} \tag{11.4}$$

where $D(h)$ is the largest lag between two nonzero weights. The sum is finite. Thus κ is the autocorrelation sequence viewed through a family of triangular antisymmetric kernels. Numerically,

the regenerated exact-extension script gives $\max|\kappa_{\text{direct}}-(C\rho)/c_0| < 2.0\times10^{-15}$ in double precision. Solving the finite linear system in the reverse direction reproduces the first gap correlations

$$-0.0463,\ -0.0225,\ -0.0132,\ -0.0057,\ -0.0069,\ -0.0061,\ -0.0060,\ -0.0064. \tag{11.5}$$

The maximum error reported for $d \le 60$ is 6.8×10^{-4}. The inversion is severely ill-conditioned (condition number about 10^8), so in practice only the first lags are stably recoverable. This reinforces the information statement: $\kappa(h)$ is a compressed linear image of $\rho(d)$, not an additional spectral observable.

Because w is antisymmetric about the centre, a *positive* correlation between gaps on opposite sides of the centre enters $w^\top Rw$ with a negative sign. One therefore expects $\kappa(h) < 1$, i.e. the independent model to *overpredict*. This is what is observed, and it accounts quantitatively for the discrepancy (Section 12, part D).

12. Numerical verification and measured laws

12.1. **Protocol.** μ was computed by a linear sieve for $n \le N = 10^6$; M, Q by prefix sums; a prime table to index 304 071 ($p = 4\,318\,813$); T by prefix sums of the primes. At $N = 10^6$: $M(N) = 212$, $Q(N) = 607\,926$, maximum required prime index 304 069. All scripts and raw outputs accompany this note.

12.2. **A. Exact statements verified.** The following hold with no exceptions for all $1 \le n \le 10^6$: Lemma 2.2; Proposition 3.3 (direct summation of S against the closed form); Proposition 4.1 and both inequalities of Corollary 4.2; $\operatorname{sgn} T = \operatorname{sgn} M$ and $T = 0 \iff M = 0$ (there are 5 361 values of $n \le 10^6$ with $M(n) = 0$); Proposition 9.2. Theorem 10.1 was checked on 4 000 random $n \in [10^3, 10^6]$ with maximum absolute error 0, and Lemma 10.2 for all $h \le 60$ and $h \in \{100, 101, 300, 301, 370\}$.

12.3. **B. Global residual scale,** $100 \le n \le 10^6$**.**

(12.1)

	E	E_{mid}
median $\lvert\cdot\rvert$	1.09×10^3	9.74×10^2
r.m.s.	5.295×10^3	5.066×10^3
max $\lvert\cdot\rvert$	6.2766×10^4	5.7654×10^4
median $\lvert\cdot\rvert/\lvert T\rvert$	1.069×10^{-5}	8.894×10^{-6}

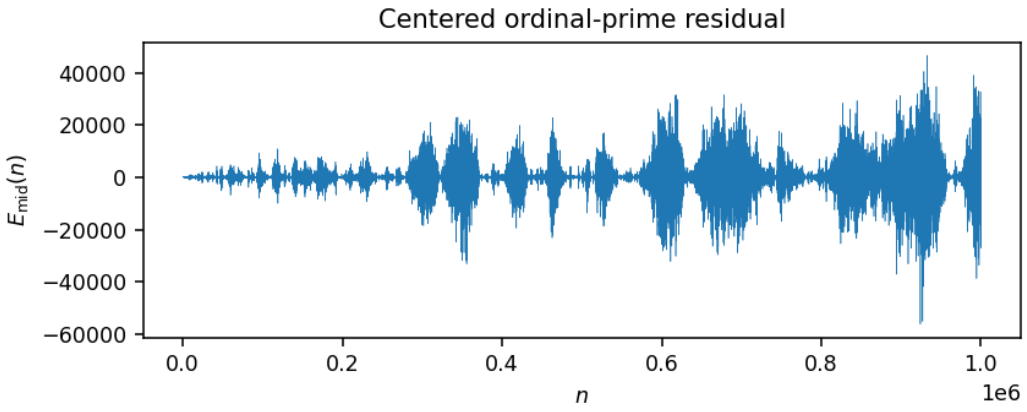


Figure 1. $E_{\text{mid}}(n)$ for $n \le 10^6$.

12.4. **C. The $h^{3/2}$ law and its stationarity.** Put $Z_{3/2}(n) := E_{\mathrm{mid}}(n)/(|M(n)|^{3/2}\log n)$. Conditioning on $h = |M(n)|$ over $n \geq 10^3$:

(12.2)

$h \in$	$[8,12)$	$[12,19)$	$[19,31)$	$[31,51)$	$[51,84)$	$[84,138)$	$[138,225)$	$[225,400)$
median $\lvert Z_{3/2}\rvert$	0.1426	0.1477	0.1589	0.1589	0.1556	0.1546	0.1507	0.1550
count	36 451	58 125	86 041	127 696	152 059	164 880	231 986	65 791

and conditioning on n instead:

(12.3)

$n \in$	RMS $Z_{3/2}$	median $\lvert Z_{3/2}\rvert$	count
$[1.00, 1.58) \times 10^5$	0.2331	0.1549	51 624
$[1.58, 2.51) \times 10^5$	0.2379	0.1574	82 660
$[2.51, 3.98) \times 10^5$	0.2340	0.1541	134 286
$[3.98, 6.31) \times 10^5$	0.2314	0.1530	215 541
$[6.31, 10.0) \times 10^5$	0.2324	0.1552	359 873

Both tables are flat: the normalisation $|M|^{3/2}\log n$ is stable across a factor of 46 in h and a factor of 10 in n. This supports, but does not prove, the hypothesis that $Z_{3/2}$ converges in distribution.

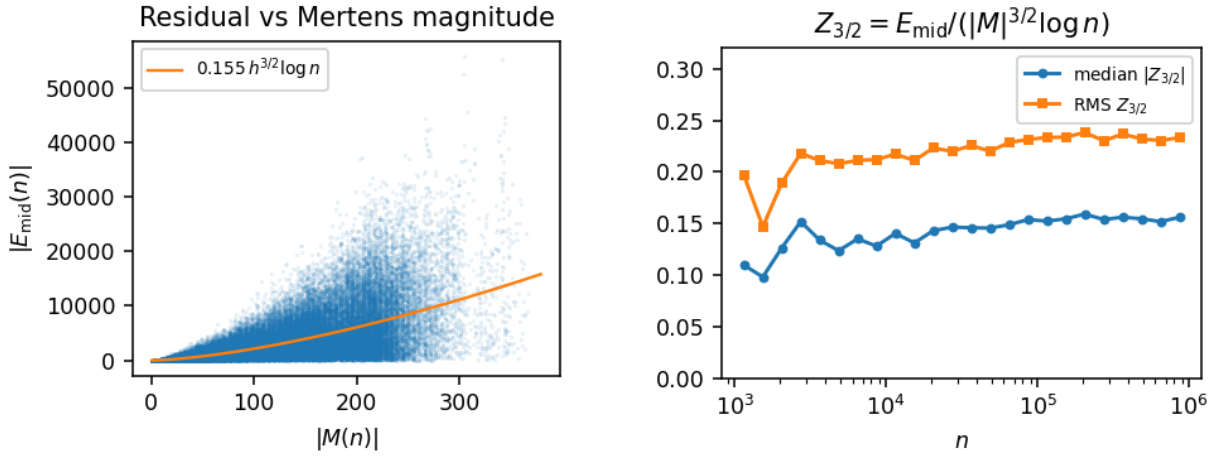


FIGURE 2. Left: $|E_{\mathrm{mid}}|$ against $|M|$ with the envelope $0.155\,h^{3/2}\log n$. Right: stationarity of $Z_{3/2}$ in n.

12.5. **D. Which variance model is right.** Local gap statistics were measured on the prime-index band $[5 \times 10^4, 3.05 \times 10^5)$, which covers the windows for $n \in [10^5, 10^6]$: mean gap 14.5927, $\log p = 14.6984$ at the band centre, $\mathrm{Var}(g) = 145.93$, hence

$$\mathrm{Var}(g)/\log^2 p = 0.6755 \tag{12.4}$$

against the value 1 predicted by a Poisson (Cramér) gap model. The gap autocorrelation is negative at short lag and positive over a long tail:

(12.5)

d	1	2	3	4	5	6
$\rho(d)$	−0.0463	−0.0225	−0.0132	−0.0057	−0.0069	−0.0061

with $\sum_{1\le d\le 420}\rho(d) = +0.4646$. The resulting correction factor decreases steadily,

(12.6)

h	5	10	20	40	80	160	320
$\kappa(h)$	1.0206	0.9532	0.9094	0.8514	0.7869	0.7381	0.6871

Testing both models on the 883 123 values of n with $n \ge 10^4$ and $h \ge 12$:

(12.7)

model	$\mathrm{RMS}(E_{\mathrm{mid}})/\,\mathrm{RMS}(\mathrm{SD})$	RMS Z	mean Z	median $\lvert Z\rvert$
independent gaps	0.8527	0.8963	-1.88×10^{-4}	0.5962
covariance-corrected	1.0031	1.0036	-2.13×10^{-4}	0.6694

The 15% deficit of the independent model is therefore explained, to within 0.3%, by the correlations between prime gaps; and the corrected model reproduces the Gaussian value median $|Z| = 0.6745$ to within 0.8% with no free parameter. Within the measured range, this accounts for the scale discrepancy left by the independent model.

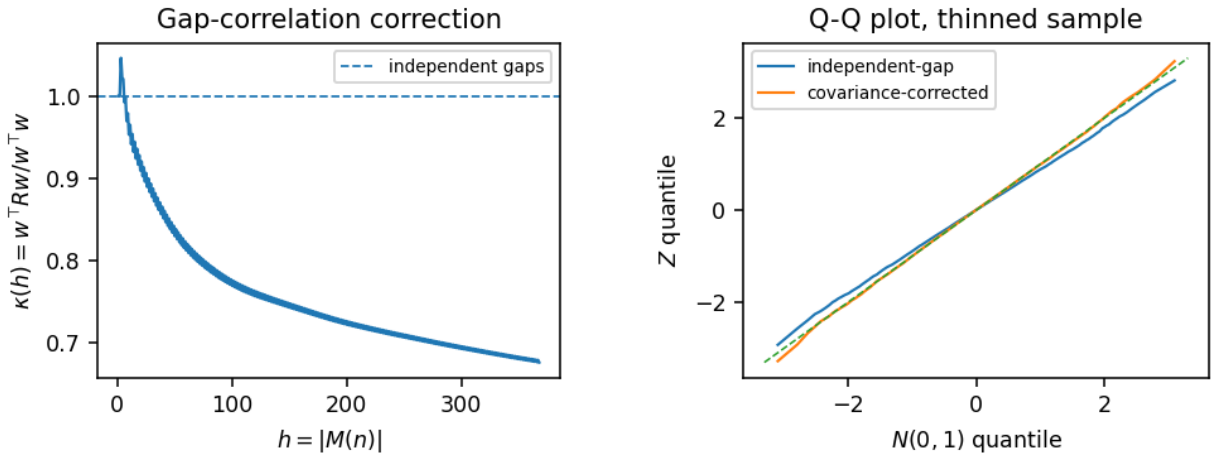


FIGURE 3. Left: the correction factor $\kappa(h)$. Right: quantiles of $Z = E_{\mathrm{mid}}/\mathrm{SD}$ against the normal, on a thinned sample; the independent-gap curve is visibly compressed, the corrected one is not.

12.6. **E. Distribution of the standardised residual.** With $Z := E_{\mathrm{mid}}/\,\mathrm{SD}_{\mathrm{cov}}$, by blocks in n ($h \ge 12$):

(12.8)

$n \in$	count	mean	sd	skew	exc. kurt.	median $\lvert Z\rvert$
$[10^4, 10^5)$	67 120	+0.0043	0.9813	−0.0098	+0.2556	0.6474
$[10^5, 3.16\times10^5)$	183 261	−0.0000	1.0029	−0.0167	+0.1559	0.6705
$[3.16\times10^5, 10^6)$	632 741	−0.0008	1.0061	−0.0188	+0.0957	0.6715

and by blocks in h, the excess kurtosis decreases overall, with a small nonmonotone rise between the fourth and fifth bins,

(12.9)

$h \in$	$[12, 19)$	$[19, 31)$	$[31, 51)$	$[51, 84)$	$[84, 138)$	$[138, 225)$	$[225, 400)$
exc. kurt.	+0.676	+0.338	+0.213	+0.031	+0.058	+0.024	−0.143

which is qualitatively consistent with, but does not establish, the behaviour expected if a central limit theorem governs the sum of Theorem 10.1 as $h \to \infty$. Quantiles of Z over all $n \geq 10^4$, $h \geq 12$:

(12.10)

level	1%	5%	25%	50%	75%	95%	99%
observed	−2.372	−1.652	−0.666	+0.000	+0.673	+1.641	+2.346
$N(0,1)$	−2.326	−1.645	−0.674	0	+0.674	+1.645	+2.326

12.7. **F. The points are not independent.** This is a necessary caveat on the preceding tables. Consecutive n share almost the same window, so the 883 123 values of Z are strongly dependent: the autocorrelation of Z in n is +0.979 at lag 1, +0.692 at lag 10, and first falls below 0.05 at lag 128. The integrated autocorrelation time is ≈ 51, giving an effective sample size of about 1.7×10^4, not 8.8×10^5.

Consequently a Kolmogorov–Smirnov test on the full sample is meaningless. On thinned subsamples:

(12.11)

thinning	count	sd	exc. kurt.	KS	p
1	883 123	1.0036	+0.120	0.00317	3.8×10^{-8}
51	17 317	1.0036	+0.170	0.00451	0.87
128	6 900	1.0057	+0.018	0.00775	0.80

On the two thinned samples the reported KS tests do not reject a standard normal reference law. This is descriptive rather than a proof of normality: thinning reduces dependence but does not establish independence. The full-sample p-value is not interpreted because the observations are strongly dependent.

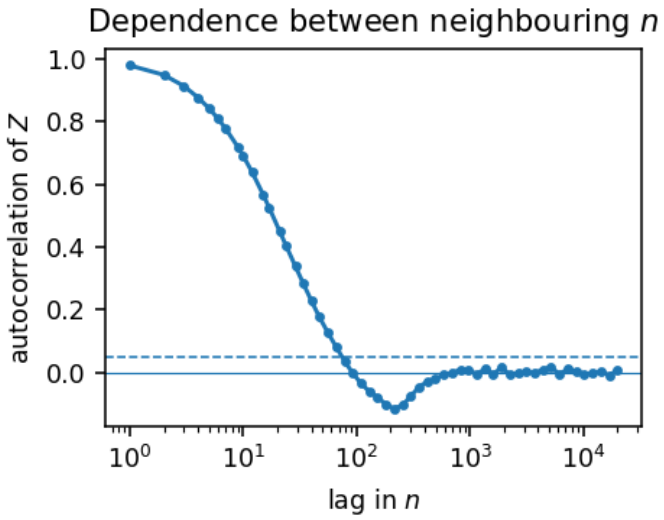


FIGURE 4. Autocorrelation of Z in n; the effective sample is roughly 50 times smaller than the nominal one.

12.8. **G. Relative size of the residual.** Combining $|E_{\mathrm{mid}}| \approx c\,h^{3/2}\log n$ with $|T| \approx h\,p^* \approx \frac{3}{\pi^2} h\,n\log n$ predicts

$$\frac{|E_{\mathrm{mid}}(n)|}{|T(n)|} \approx \frac{c\pi^2}{3}\cdot\frac{\sqrt{|M(n)|}}{n}, \qquad \frac{c\pi^2}{3} = 0.510 \quad (c = 0.155). \tag{12.12}$$

Measured medians of $(|E_{\mathrm{mid}}|/|T|)\,n/\sqrt{|M|}$: 0.4385, 0.4800, 0.4995, 0.4955 on

$$[10^3, 10^4), \quad [10^4, 10^5), \quad [10^5, 3.16\times 10^5), \quad [3.16\times 10^5, 10^6). \tag{12.13}$$

This quantitatively explains why the central term reproduces T so accurately. The measured median satisfies

$$\text{median}\,\frac{|E_{\mathrm{mid}}|}{|T|} = 8.9\times 10^{-6}. \tag{12.14}$$

If $|M(n)|$ is of size roughly $n^{1/2}$, the relative correction is of size $n^{-3/4}$.

12.9. **H. Largest excursions.** The extremes of E_{mid} on $n \le 10^6$ occur in tight clusters, e.g. -5.7654×10^4 at $n = 924\,067$ ($M = -301$) and $+4.6576\times 10^4$ at $n = 932\,737$ ($M = -269$). These are the n at which $|M|$ is locally largest, consistent with the $h^{3/2}$ law and carrying no separate endpoint information.

12.10. **I. The packet envelope.** The visible packets are almost entirely the excursions of $|M(n)|$ passed through the $h^{3/2}$ variance scale. With a moving RMS window of width 2000, the log–log correlation between the measured envelope of E_{mid} and a moving average of $|M(n)|^{3/2}\log n$ is

$$0.9939 \qquad (n \ge 5\times 10^4). \tag{12.15}$$

Replacing the Mertens factor by a trend depending only on n gives correlation 0.4442. Figure 5 displays the resulting amplitude modulation.

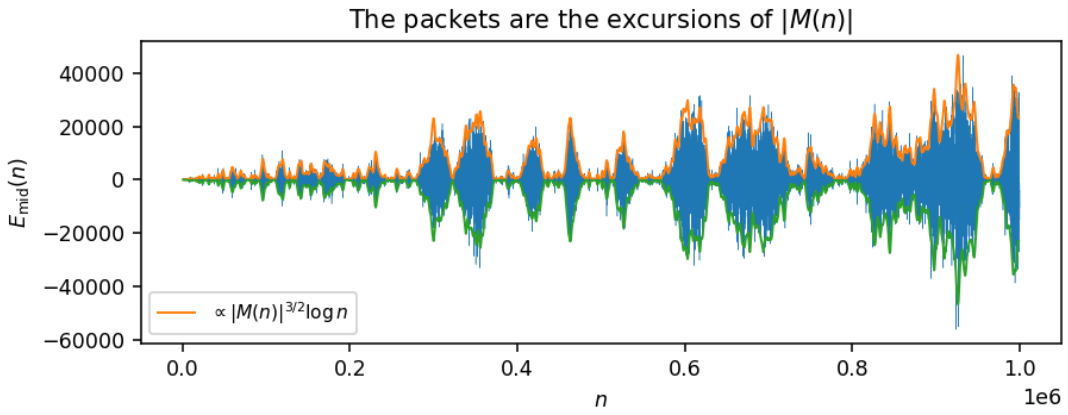


FIGURE 5. The packet structure of E_{mid} follows the excursions of $|M(n)|$. The red curves show a scaled moving average proportional to $|M(n)|^{3/2}\log n$.

12.11. **J. Spectrum on the logarithmic coordinate: a classical check.** Put $u = \log x$ and, for integer samples $x = n$, consider

$$X(u) := e^{-u/2} M(\lfloor e^u \rfloor). \tag{12.16}$$

The classical Perron/explicit-formula treatment of the Mertens function [11], after truncation and contour displacement and under the usual hypotheses on the zeros, has residue contributions of the form

$$\frac{x^\rho}{\rho \zeta'(\rho)}. \tag{12.17}$$

For a simple zero $\rho = \beta + i\gamma$, the corresponding contribution after $x = e^u$ and multiplication by $e^{-u/2}$ is proportional to $e^{(\beta-1/2)u} e^{i\gamma u}$. Under RH this becomes a pure mode $e^{i\gamma u}$. Consequently, within that classical explicit-formula framework, the appearance of power at frequencies γ_k in $X(u)$ is the expected explicit-formula behaviour, not a new law of the transform. Historical numerical uses of zeta zeros to reconstruct prime-counting structure include [6].

The numerical spectrum over $10^3 \le n \le 10^6$ (after the interpolation and windowing used by the supplied script) shows peaks near the first zeta ordinates, while a random-sign surrogate with the same squarefree support does not show the same visual alignment. The original peak-to-nearest-zero summary (mean distances 0.263 for the true sequence and 1.738 for one surrogate) has a selection bias because peaks were selected before matching them to the zeros. It is retained only as a diagnostic and not used as evidence. The proper fixed-frequency question is the power evaluated at preassigned γ_k, or a regression on the preassigned modes.

More fundamentally, the finite logarithmic window itself imposes the resolution

$$\Delta\omega \simeq \frac{2\pi}{u_{\max} - u_{\min}} = \frac{2\pi}{\log(10^6/10^3)} \simeq 0.91. \tag{12.18}$$

Peak offsets of a few tenths are therefore below the resolving power of this finite window and carry no precision information about the known γ_k. Doubling the frequency resolution requires doubling the range in u, hence squaring the range in n; reaching resolution 10^{-3} would require a logarithmic span of order $2\pi \times 10^3$, far outside this computation.

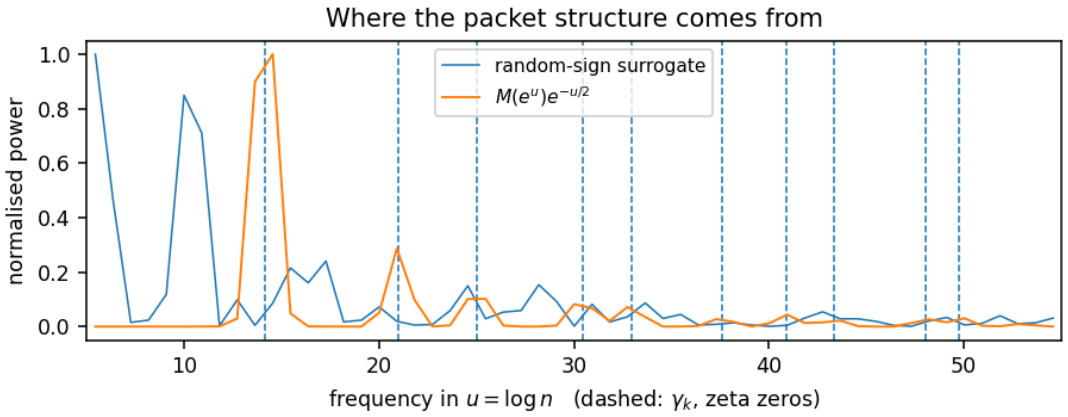


FIGURE 6. Power spectrum of $M(e^u)e^{-u/2}$ on $u = \log n$. Dashed lines are the first zeta ordinates. This is an illustration of the classical explicit-formula modes; the peak-matching statistic is not used inferentially because of selection bias and finite-window resolution.

12.12. **K. Prime-counting variance comparison.** Proposition 10.4 puts the residual on the prime-counting side. At prime scale $X \simeq 4.3\times10^6$, the supplied experiment compares the measured variance of short-interval prime counts with an $H\log(X/H)$ reference scale motivated by Montgomery and Soundararajan for weighted prime counts $\psi(x+H)-\psi(x)$, and with a Poisson/Cramér-type $H\log X$ reference. This is only a scale comparison. It is not a direct application of the Montgomery–Soundararajan statement [3] to the unweighted count $\pi(x+H)-\pi(x)$. The Poisson comparison is in the spirit of Cramér's model [2]. The reported ratios are

(12.19)

H	200	1000	5000	20000
$\mathrm{Var}_{\mathrm{meas}}\,/\,\mathrm{Var}_{\mathrm{MS}}$	1.116	1.202	1.310	1.622
$\mathrm{Var}_{\mathrm{meas}}\,/\,\mathrm{Var}_{\mathrm{Cramer}}$	0.552	0.468	0.372	0.314

The decreasing ratio to the Cramér scale is the expected qualitative signature of an $H\log(X/H)$ variance rather than $H\log X$. The excess over the leading MS term is not interpreted as a new statistic: at $H = 2\times10^4$ the displayed leading logarithmic factor is only about 2.96, versus $\log X \simeq 15.3$, so lower-order terms are not negligible in this preasymptotic range.

12.13. **L. Cross-coupling test.** The only numerical loophole left inside E_{mid} is weaker than new endpoint information. The gap factor in Theorem 10.1 is controlled by prime statistics, while the window length is $h = |M(n)|$. One can ask whether the standardised fluctuation

$$Z(n) = \frac{E_{\mathrm{mid}}(n)}{\mathrm{SD}_{\mathrm{cov}}(n)}, \qquad h = |M(n)| \geq 12, \tag{12.20}$$

is coupled to the sign or to a zeta phase of M. The supplied follow-up tests reported

(12.21)

test	value
$\mathrm{corr}(Z, \mathrm{sgn}\, M)$	$+0.00058$
$\mathrm{corr}(Z, M/\sqrt{n})$	$+0.00110$
$\mathrm{corr}(Z, \cos(\gamma_1 \log n))$	-0.00119
$\mathrm{corr}(Z, \sin(\gamma_1 \log n))$	$+0.00173$
$\overline{Z}\,|_{M>0} - \overline{Z}\,|_{M<0}$	$+0.00027$ $(0.08\sigma$ effective$)$

and an eight-bin phase test gave dispersion of the bin means 0.00070, compared with a typical standard error 0.00499.

These are null results only at the tested resolution. With effective sample $n_{\mathrm{eff}} \simeq 17\,316$, the nominal standard error of a small correlation is $1/\sqrt{n_{\mathrm{eff}}} \simeq 0.0076$, so the experiment excludes only linear couplings of order $|r| \gtrsim 0.015$. Weaker effects, nonlinear dependence, untested functions of the phase, or effects emerging only at much larger n remain outside the sensitivity of this computation. The result therefore does *not* prove decoupling; it says that no coupling was detected in this small specified family.

13. Status of the claims

To keep the register unambiguous:

- *Proved*: Lemmas 2.1, 2.2, 10.2; Propositions 3.3, 4.1, 5.1, 8.1, 9.2, 14.1, 14.2, 16.1, 16.5, 16.7, 17.3, 17.4, and 17.5, 19.1, 19.5, 20.1, and 20.2, 21.1, and 22.1; Corollaries 4.2 and 15.3; Theorems 6.2, 10.1, 14.3, 15.1, 18.1, 18.3, and 19.2, 21.2, 21.3, and 22.2; the exact identities (16.9), (16.15), (17.3), and (17.9); and the mean-square criterion (14.22).
- *Measured*, over $10^4 \leq n \leq 10^6$ with effective sample $\approx 1.7\times10^4$ where relevant: the law $|E_{\mathrm{mid}}| \approx 0.155\,|M|^{3/2}\log n$; the failure of the independent-gap model by a factor 0.853 and its repair by the measured gap covariance; approximate normality of the standardised residual;

the 0.9939 packet-envelope correlation; the exact finite κ–ρ transform (regenerated to double-precision roundoff) and the reported finite-dimensional recovery of the first lags of ρ from κ; the short-interval variance ratios; the null cross-coupling tests of Section 12.13; and the Version 6 quadratic-energy and center-displacement measurements in Experiments 14.5 and 14.6; the signed third-moment measurement in Experiment 16.3, which is explicitly recorded with drift and without an asymptotic exponent claim; and the conditioned active-lifetime comparison in Experiment 17.6. The finite level-collision, excursion, and occupancy counts in Experiment 22.3 are also measured diagnostics; the identities checked there are proved separately and do not depend on those counts.

- *Classical interpretation, not novelty*: the frequencies γ_k in the $u = \log n$ spectrum of $M(e^u)e^{-u/2}$ are the modes supplied by the explicit formula. The displayed peak matching is diagnostic only and has frequency resolution about 0.91.
- *Conjectural*: that $Z_{3/2}$ converges in distribution; that the limit is Gaussian; that a central limit theorem holds for the antisymmetric weighted gap sum of Theorem 10.1; that the observed numerical absence of cross-coupling persists asymptotically; or that $\mathcal{D}_2/(2\mathcal{M}_2) \to 1$. None follows from the present numerics.
- *Closed or classified negative routes*: the quadratic orthogonal split (14.18) is pairing-insensitive, and the proposed uniform contraction $\mathcal{E}_2 \geq \delta\mathcal{D}_2$ is not supported; the reported ratio decreases strongly with scale. The root-of-unity prefix filter reduces exactly to higher-order consecutive correlations by (16.15); universal coercivity can be evenised by Proposition 16.5; the real-axis generating function is not uniformly coercive by Proposition 16.7; and the two-cut survival kernel collapses to endpoint heights by Proposition 17.3. Question 17.2 remains an open mechanism, not a claimed consequence of the large sieve.

14. Ordinal pairing and the quadratic route

This section records a second representation that arises directly from the two sign-specific ordinal counters. Unlike the prime-gap residual, the construction below uses only the order in which the nonzero values of μ occur. It is useful because it gives an exact transport picture of the squarefree-time Mertens walk; it does *not* create an independent bound on that walk.

Let

$$s_1 < s_2 < \cdots \tag{14.1}$$

be the squarefree positive integers, and define the squarefree-time sign sequence and its partial sums by

$$\varepsilon_q := \mu(s_q) \in \{-1, +1\}, \qquad R(q) := \sum_{r \leq q} \varepsilon_r = M(s_q). \tag{14.2}$$

Let α_j be the squarefree ordinal of the j-th occurrence of $+1$, and β_j the squarefree ordinal of the j-th occurrence of -1. Thus

$$\varepsilon_{\alpha_j} = +1, \qquad \varepsilon_{\beta_j} = -1, \qquad d_j := \beta_j - \alpha_j, \qquad c_j := \frac{\alpha_j + \beta_j}{2}, \qquad e_j := c_j - 2j. \tag{14.3}$$

The corresponding separation in the original integer coordinate is

$$\Delta_j := s_{\beta_j} - s_{\alpha_j}. \tag{14.4}$$

The sign of d_j records which member of the ordinal pair occurs first.

Proposition 14.1 (Exact endpoint identities)**.** *For every $j \geq 1$,*

$$\alpha_j + R(\alpha_j) = 2j, \tag{14.5}$$

$$\beta_j - R(\beta_j) = 2j, \tag{14.6}$$

and hence

$$d_j = R(\alpha_j) + R(\beta_j), \tag{14.7}$$

$$2e_j = R(\beta_j) - R(\alpha_j). \tag{14.8}$$

Moreover,

$$|2e_j| \le |d_j|. \tag{14.9}$$

Proof. At $q = \alpha_j$, the number of positive signs is j, so $j = (\alpha_j + R(\alpha_j))/2$; similarly $j = (\beta_j - R(\beta_j))/2$. Adding and subtracting these two endpoint relations gives (14.7) and (14.8). Finally, R changes by exactly ± 1 at each squarefree-time step, hence it is 1-Lipschitz on the integer variable q and

$$|R(\beta_j) - R(\alpha_j)| \le |\beta_j - \alpha_j|. \tag{14.10}$$

Together with (14.8), this is (14.9). □

Proposition 14.2 (Pairing criterion for RH)**.** *The following are equivalent:*

(a) the Riemann hypothesis;

(b) for every $\varepsilon > 0$,

$$d_j = O_\varepsilon\left(j^{1/2+\varepsilon}\right); \tag{14.11}$$

(c) for every $\varepsilon > 0$,

$$\Delta_j = O_\varepsilon\left(j^{1/2+\varepsilon}\right). \tag{14.12}$$

Proof. Under RH, the classical Littlewood criterion gives $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$; since $s_q \asymp q$, equation (14.7) gives (14.11). Conversely, fix q and write $h = |R(q)|$. If $R(q) > 0$, exactly h ordinal pairs are open across the cut after q, namely the pairs whose positive member has occurred and whose negative member has not. For $j = N_-(q) + r$, $1 \le r \le h$, there are at least $h - r$ subsequent positive members before the cut and at least r negative members still required after the cut. Therefore

$$|d_j| \ge h. \tag{14.13}$$

The case $R(q) < 0$ is identical with the signs interchanged. Since every such open index satisfies $j \le q$, (14.11) yields $|R(q)| = O_\varepsilon(q^{1/2+\varepsilon})$. As $R(q) = M(s_q)$ and $s_q \asymp q$, this gives the Littlewood criterion and hence RH.

For the original integer displacement, the standard squarefree counting estimate [1]

$$s_q = \frac{\pi^2}{6} q + O(\sqrt{q}) \tag{14.14}$$

shows that (14.11) implies (14.12). Conversely, monotonicity of s_q gives $|d_j| \le |\Delta_j|$, so (14.12) implies (14.11). □

The point of Proposition 14.2 is representational rather than inferential: testing a $j^{1/2}$ scale numerically is only a compatibility test for RH. It cannot establish the required uniform bound.

14.1. **Quadratic energy on a balanced prefix.** Fix a squarefree-time prefix Q satisfying

$$R(Q) = 0, \qquad J := Q/2. \tag{14.15}$$

Then the first Q squarefree signs contain exactly J positive and J negative values, so $\{\alpha_1, \dots, \alpha_J\}$ and $\{\beta_1, \dots, \beta_J\}$ partition $\{1, \dots, Q\}$. Define

$$\mathcal{M}_2(Q) := \sum_{q \le Q} R(q)^2, \qquad \mathcal{D}_2(J) := \sum_{j \le J} d_j^2, \qquad \mathcal{E}_2(J) := 4 \sum_{j \le J} e_j^2. \tag{14.16}$$

Theorem 14.3 (Exact quadratic identity and comparison)**.** *On every balanced prefix* (14.15)*,*

$$\sum_{j\le J}\big(R(\alpha_j)^2+R(\beta_j)^2\big)=\mathcal{M}_2(Q), \tag{14.17}$$

$$\boxed{\mathcal{D}_2(J)+\mathcal{E}_2(J)=2\mathcal{M}_2(Q)}, \tag{14.18}$$

and

$$\boxed{\mathcal{M}_2(Q)\le\mathcal{D}_2(J)\le 2\mathcal{M}_2(Q)}. \tag{14.19}$$

Proof. Equation (14.17) follows because every $q\in\{1,\dots,Q\}$ occurs exactly once as an α_j or as a β_j. For each ordinal pair, equations (14.7) and (14.8) give the parallelogram identity

$$d_j^2+4e_j^2=2R(\alpha_j)^2+2R(\beta_j)^2. \tag{14.20}$$

Summing (14.20) and using (14.17) proves (14.18). The upper bound in (14.19) follows from $\mathcal{E}_2\ge 0$. The Lipschitz inequality (14.9) gives $\mathcal{E}_2\le\mathcal{D}_2$; substituting this into (14.18) gives the lower bound. □

Remark 14.4 (Why the orthogonal decomposition is not a transport gain)*.* Let $x_j:=R(\alpha_j)$ and $y_j:=R(\beta_j)$. For any permutation π of $\{1,\dots,J\}$,

$$\sum_{j\le J}\Big((x_j+y_{\pi(j)})^2+(y_{\pi(j)}-x_j)^2\Big)=2\sum_{j\le J}x_j^2+2\sum_{j\le J}y_j^2=2\mathcal{M}_2(Q). \tag{14.21}$$

Thus the 45° energy split itself is independent of how the two endpoint sets are paired. The ordinal matching is special only because for $\pi(j)=j$ the two rotated coordinates are exactly d_j and $2e_j$. Consequently (14.18) cannot by itself generate an iterative or multiscale gain.

14.2. **Mean-square criterion and closure of the quadratic route.** The comparison (14.19) shows that, on balanced prefixes, the quadratic transport energy and the squarefree-time mean square of Mertens differ by at most an absolute factor two. Independently of the pairing, the following mean-square criterion is equivalent to RH:

$$\text{RH}\quad\Longleftrightarrow\quad\mathcal{M}_2(Q)=O_\varepsilon\Big(Q^{2+\varepsilon}\Big)\quad\text{for every }\varepsilon>0. \tag{14.22}$$

Indeed RH implies the bound termwise. Conversely, extend R as a step function on the nonnegative reals. If (14.22) holds, Cauchy–Schwarz on dyadic blocks shows that the Mellin integral

$$F(s):=s\int_1^\infty R(t)t^{-s-1}\,dt \tag{14.23}$$

converges absolutely and is holomorphic for $\Re s>1/2$; for $\Re s>1$ it equals $\sum_{q\ge1}\varepsilon_q q^{-s}$ by summation by parts. This first constructs analytic continuation, rather than assuming ordinary series convergence in the larger half-plane. Since (14.14) gives, uniformly on compact subsets of that half-plane,

$$s_q^{-s}=\left(\frac{\pi^2}{6}q\right)^{-s}+O_s\Big(q^{-\Re s-1/2}\Big), \tag{14.24}$$

the difference between $\sum_q\varepsilon_q s_q^{-s}=1/\zeta(s)$ and $(\pi^2/6)^{-s}F(s)$ is absolutely convergent there. Hence $1/\zeta(s)$ has a holomorphic continuation to $\Re s>1/2$, which excludes zeros of ζ in that half-plane; the functional equation then gives RH.

Thus a proof of

$$\mathcal{D}_2(J)=O_\varepsilon\Big(J^{2+\varepsilon}\Big) \tag{14.25}$$

would be useful only if it were obtained by an argument genuinely easier than, or independent of, the corresponding mean-square estimate for R. The exact comparison (14.19) shows that the ordinal pairing does not soften the quadratic problem merely by changing coordinates.

The mean-square behaviour of the Mertens function and its limiting distribution under additional hypotheses on negative moments of $\zeta'(\rho)$ are classical subjects; see, for example, Ng [4]. In particular, one should not identify a finite quadratic fit with a new zeta-statistical law.

Experiment 14.5 (Regenerated controls for the quadratic decomposition)**.** The following controls were regenerated in Python from the Version 9 data arrays by the supplied audit scripts. On the balanced prefix

$$Q = 488868, \qquad J = 244434, \tag{14.26}$$

the exact identities (14.17), (14.7), (14.8), (14.18), and (14.19) were checked without discrepancy. The regenerated energies are

$$\mathcal{D}_2 = 10646184520, \qquad \mathcal{E}_2 = 36823324, \qquad 2\mathcal{M}_2 = 10683007844. \tag{14.27}$$

The resulting fractions are

$$\frac{\mathcal{D}_2}{2\mathcal{M}_2} = 0.99655, \qquad \frac{\mathcal{E}_2}{2\mathcal{M}_2} = 0.00345. \tag{14.28}$$

As a structural control, the total parallelogram energy was unchanged when the endpoint heights were paired ordinally, randomly, or in reverse order, exactly as predicted by Remark 14.4.

On balanced prefixes with $J \geq 8000$ through $J = 244434$, regenerated log–log fits gave

$$\mathcal{D}_2 \sim J^{2.0163}, \qquad \mathcal{E}_2 \sim J^{1.4641}, \qquad \mathcal{M}_2 \sim J^{2.0120}. \tag{14.29}$$

Over the same selected range the ratio $\mathcal{E}_2/\mathcal{D}_2$ decreased from 0.01909 to 0.00346. These finite-range measurements are consistent with $\mathcal{D}_2/(2\mathcal{M}_2) \to 1$, but do not prove that asymptotic statement. In particular they refute, on the measured range and trend, the proposed uniform lower comparison

$$\mathcal{E}_2 \geq \delta \mathcal{D}_2 \qquad (\delta > 0 \text{ fixed}), \tag{14.30}$$

as a plausible multiscale mechanism.

Experiment 14.6 (Scale of the center displacement)**.** Equation (14.8) shows exactly that $2e_j$ is the increment of the same squarefree-time Mertens walk across the ordinal-pair window, whose length is $|d_j|$. The regenerated conditional RMS values were

$\lvert d\rvert$ bin	15–23	56–87	136–211	328–510
$\mathrm{RMS}(2e)/\lvert d\rvert^{1/2}$	1.006	1.023	0.979	0.980
$\mathrm{RMS}(2e)/\lvert d\rvert$	0.231	0.121	0.075	0.049

A separate log-binned median fit gave exponents 0.2400 for $|e_j|$ and 0.4689 for $|d_j|$. These finite-range fits are consistent with the heuristic relation

$$|e_j| \text{ of typical size } |d_j|^{1/2}, \tag{14.31}$$

not with e_j providing a second matching energy at the same scale. No independence is asserted: both quantities are increments of the same walk.

Remark 14.7 (Negative result: no quadratic recursion)*.* Two proposed mechanisms are therefore closed. First, (14.18) is a pairing-insensitive parallelogram identity, so its orthogonal form contains no recursive transport information. Second, the candidate lower bound (14.30) is contradicted by the regenerated energy split, while e_j has the same number of terms as d_j and is merely the half-increment (14.8) of the original walk. The quadratic ordinal representation remains mathematically exact, but at this level it is a re-expression of the classical Mertens mean-square problem rather than a new source of leverage.

15. Local ordinal correlations beyond quadratic energy

The quadratic comparison in Section 14 closes the route through norms of the displacement sequence, but it does not exhaust the ordinal pairing. Local statistics depend on the order of the squarefree signs rather than only on the multiset of heights $\{R(q)\}$. This section isolates the first unconditional result of that type.

For fixed $h \geq 1$ define the squarefree-ordinal two-point correlation

$$C_h^{\mathrm{sf}}(Q) := \frac{1}{\log Q} \sum_{q \leq Q} \frac{\varepsilon_q \varepsilon_{q+h}}{q}. \tag{15.1}$$

The averaging in (15.1) is logarithmic. The corresponding natural average is not claimed here. Results of Matomäki–Radziwiłłon cancellation and sign changes of multiplicative functions in short intervals concern the natural integer clock [8]; they do not directly imply the squarefree-ordinal statement below, whose proof instead returns to fixed affine forms and uses Tao's logarithmically averaged two-point theorem.

Theorem 15.1 (Logarithmic two-point Chowla in squarefree ordinal time)**.** *For every fixed integer $h \geq 1$,*

$$\boxed{\sum_{q \leq Q} \frac{\mu(s_q)\mu(s_{q+h})}{q} = o(\log Q)} \qquad (Q \to \infty). \tag{15.2}$$

Proof. Write

$$c := \frac{\pi^2}{6}. \tag{15.3}$$

The squarefree inverse estimate (14.14) gives

$$\frac{1}{q} = \frac{c}{s_q} + O(q^{-3/2}), \tag{15.4}$$

so replacing $1/q$ by c/s_q changes the sum in (15.2) by $O(1)$.

For each q put

$$r_q := s_{q+h} - s_q. \tag{15.5}$$

If $n = s_q$ and $r = r_q$, then n and $n+r$ are squarefree and there are exactly $h-1$ squarefree integers strictly between them. Since the factor $\mu(n)\mu(n+r)$ already vanishes unless both endpoints are squarefree, define

$$W_{h,r}(n) := \mathbf{1}\left\{\sum_{a=1}^{r-1} \mu(n+a)^2 = h-1\right\}. \tag{15.6}$$

After the change of clock, the contribution with fixed physical separation r is therefore

$$\sum_{n \leq X} \frac{\mu(n)\mu(n+r)}{n} W_{h,r}(n), \tag{15.7}$$

up to endpoint terms which are $O(1)$ after logarithmic weighting.

Fix $D \geq 2$ and introduce the periodic squarefree approximation

$$\nu_D(m) := \mathbf{1}_{\{p^2 \nmid m \text{ for every prime } p \leq D\}}. \tag{15.8}$$

It differs from $\mu(m)^2$ only when $p^2 \mid m$ for some prime $p > D$. Consequently

$$\sum_{m \leq X} \frac{|\nu_D(m) - \mu(m)^2|}{m} \ll \frac{\log X}{D} + O(1). \tag{15.9}$$

For fixed h and r, replacing every interior $\mu(n+a)^2$ in (15.6) by $\nu_D(n+a)$ therefore incurs

$$\sum_{n\le X}\frac{|W_{h,r}(n)-W_{h,r}^{(D)}(n)|}{n}\ll_r \frac{\log X}{D}+O_r(1), \tag{15.10}$$

where

$$W_{h,r}^{(D)}(n):=\mathbf{1}\left\{\sum_{a=1}^{r-1}\nu_D(n+a)=h-1\right\}. \tag{15.11}$$

Set

$$M_D:=\prod_{p\le D}p^2. \tag{15.12}$$

The function $W_{h,r}^{(D)}$ is periodic modulo M_D. Hence there is a fixed set of residue classes $\mathcal{A}_{h,r,D}\subseteq \mathbf{Z}/M_D\mathbf{Z}$ such that

$$W_{h,r}^{(D)}(n)=\sum_{a\in\mathcal{A}_{h,r,D}}\mathbf{1}_{n\equiv a \pmod{M_D}}. \tag{15.13}$$

For one such residue class, write $n=M_Dm+a$. The two Möbius factors become

$$\mu(M_Dm+a)\mu(M_Dm+a+r). \tag{15.14}$$

The two affine forms have determinant

$$M_D(a+r)-M_Da=M_Dr\neq 0. \tag{15.15}$$

Tao's logarithmically averaged two-point Elliott theorem applies to fixed nondegenerate affine forms and, in particular, to the Möbius function [9]. Thus, with D,r,a fixed,

$$\sum_{m\le X}\frac{\mu(M_Dm+a)\mu(M_Dm+a+r)}{m}=o(\log X). \tag{15.16}$$

Moreover

$$\frac{1}{M_Dm+a}=\frac{1}{M_Dm}+O_{D,a}(m^{-2}), \tag{15.17}$$

so the same conclusion holds with the original weight $1/n$. Because D and r are fixed, only finitely many residue classes occur; no uniformity in a growing modulus is required. Combining this with (15.10), letting $X\to\infty$ first and then $D\to\infty$, gives

$$\sum_{n\le X}\frac{\mu(n)\mu(n+r)}{n}W_{h,r}(n)=o(\log X) \tag{15.18}$$

for every fixed h and r.

It remains to remove the truncation in r. The first moment of r_q is controlled exactly by telescoping:

$$\sum_{q\le Q}(s_{q+h}-s_q)=\sum_{i=1}^{h}s_{Q+i}-\sum_{i=1}^{h}s_i, \tag{15.19}$$

$$\frac{1}{Q}\sum_{q\le Q}r_q\longrightarrow ch. \tag{15.20}$$

Therefore, for

$$A_R(Q):=\#\{q\le Q: r_q>R\}, \tag{15.21}$$

Markov's inequality gives

$$A_R(Q) \le \left(\frac{ch}{R} + o(1)\right) Q. \tag{15.22}$$

Partial summation yields

$$\limsup_{Q\to\infty} \frac{1}{\log Q} \sum_{\substack{q\le Q \\ r_q > R}} \frac{1}{q} \le \frac{ch}{R}. \tag{15.23}$$

For fixed R, the finitely many contributions $h \le r \le R$ vanish after normalisation by $\log Q$ by (15.18); then (15.23) tends to zero as $R \to \infty$. This proves (15.2). □

Remark 15.2 (Qualitative character). By the prime number theorem,

$$M_D = \prod_{p\le D} p^2 = \exp\big((2 + o(1))D\big). \tag{15.24}$$

This causes no logical difficulty because D is held fixed while Tao's theorem is applied, but it makes the reduction qualitative and quantitatively very poor. No uniformity in growing moduli is used, and no effective decay rate for (15.2) is claimed.

Corollary 15.3 (Fixed-window second moment)**.** *For every fixed integer $h \ge 1$,*

$$\boxed{\frac{1}{\log Q} \sum_{q\le Q} \frac{(R(q+h) - R(q))^2}{q} \longrightarrow h}. \tag{15.25}$$

Proof. Since

$$R(q+h) - R(q) = \sum_{i=1}^{h} \varepsilon_{q+i}, \tag{15.26}$$

we have

$$(R(q+h) - R(q))^2 = h + 2 \sum_{1\le i<j\le h} \varepsilon_{q+i}\varepsilon_{q+j}. \tag{15.27}$$

For fixed $i < j$, the logarithmically weighted average of the second term vanishes by Theorem 15.1 after a finite shift of the summation index. There are only $\binom{h}{2}$ such pairs. □

Remark 15.4 (What two points do not give). Theorem 15.1 does not imply a geometric run-length law, nor does it imply $\mathrm{Var}(d_{j+1} - d_j) = 4$. A run of length k is a pattern of $k+1$ signs and requires correlations of orders up to $k+1$. Higher-order logarithmic Chowla is known in several important cases, including all odd orders in the natural integer clock [10], but this does not turn the full squarefree-ordinal run law into a consequence of the two-point theorem.

Experiment 15.5 (Measured local ordinal statistics)**.** The following measurements were regenerated for the squarefree-time sequence and are recorded as measured rather than proved. For gap lengths $k = 1, \dots, 6$, the frequencies were

$$0.50054,\ 0.24911,\ 0.12500,\ 0.06260,\ 0.03149,\ 0.01576, \tag{15.28}$$

against the Bernoulli values 2^{-k}. The regenerated variance of $d_{j+1} - d_j$ was 4.01203, against the fair-coin value 4. At fixed ordinal lags $1, 2, 3, 5, 10, 50$, the measured correlations $\langle \varepsilon_q \varepsilon_{q+h} \rangle$ were respectively

$$0.001087,\ -0.001405,\ -0.001354,\ 0.001485,\ 0.000237,\ -0.001938. \tag{15.29}$$

These finite natural averages are compatible with local decorrelation, but they are not consequences of Theorem 15.1, whose averaging is logarithmic.

16. Generating function, parity cancellation, and higher-order filters

The displacement sequence also has an exact generating function. Its main value is structural: it makes explicit a parity cancellation and shows that a root-of-unity filter repackages higher-order sign correlations rather than bypassing them.

Define

$$D(z) := \sum_{j\geq 1} d_j z^j, \qquad |z| < 1. \tag{16.1}$$

With

$$N_+(q) = \frac{q + R(q)}{2}, \qquad N_-(q) = \frac{q - R(q)}{2}, \tag{16.2}$$

level counting gives

$$\boxed{D(z) = \frac{z}{1-z} \sum_{q\geq 0} \left(z^{N_-(q)} - z^{N_+(q)} \right)}. \tag{16.3}$$

Equivalently, for $z = e^{-t}$ with $t > 0$,

$$\boxed{D(e^{-t}) = \frac{2e^{-t}}{1-e^{-t}} \sum_{q\geq 0} e^{-tq/2} \sinh\left(\frac{tR(q)}{2} \right)}. \tag{16.4}$$

Proposition 16.1 (Parity cancellation)**.** *Under the involution $\varepsilon_q \mapsto -\varepsilon_q$ one has*

$$R \mapsto -R, \qquad \alpha_j \leftrightarrow \beta_j, \qquad d_j \mapsto -d_j, \qquad D(z) \mapsto -D(z). \tag{16.5}$$

Consequently every formal expansion of the linear functional D in powers of R contains only odd powers; in particular

$$D(e^{-t}) = \frac{2e^{-t}}{1-e^{-t}} \sum_{m\geq 0} \frac{(t/2)^{2m+1}}{(2m+1)!} \sum_{q\geq 0} R(q)^{2m+1} e^{-tq/2}. \tag{16.6}$$

Proof. The first three transformations in (16.5) follow from interchanging the positive and negative sign counters. The last follows by linearity of (16.1). Equation (16.6) is the Taylor expansion of the hyperbolic sine in (16.4). □

Remark 16.2 (Status of the cancellation). Proposition 16.1 is an exact algebraic cancellation, but by itself it gives no nontrivial bound on R or M. Biased sign sequences may have $R(q) \asymp q$ while satisfying the same involution and the same absence of even powers. Thus the cancellation is a structural property of the antisymmetric matching, not an RH estimate.

For the first nontrivial odd chaos, define

$$H_3(q) := R(q)^3 - (3q-2)R(q). \tag{16.7}$$

Since $\varepsilon_r^2 = 1$,

$$6 \sum_{1\leq a<b<c\leq q} \varepsilon_a \varepsilon_b \varepsilon_c = H_3(q), \tag{16.8}$$

and

$$H_3(q) - H_3(q-1) = 3\varepsilon_q (R(q-1)^2 - (q-1)). \tag{16.9}$$

This identifies the cubic component with an integrated three-point object, but the shifts in (16.8) range over the full triangle $a < b < c \leq q$. Fixed-shift three-point Chowla does not by itself control this integrated quantity.

Experiment 16.3 (Signed third moment)**.** The Version 9 Python regeneration gives a stable-looking finite-range exponent about 2.481 for $\sum_{q\le Q}|R(q)|^3$, close to the Brownian reference 5/2. By contrast, a log–log fit to $|\sum_{q\le Q}R(q)^3|$ is highly protocol-sensitive because the signed cumulative sum has large oscillations and near-zeros; the supplied regeneration returns an exponent about 2.57 on its fixed logarithmic endpoint grid, while earlier fits with different endpoint choices gave substantially different values. No signed exponent is therefore promoted to a measured law. The more robust ratio

$$\frac{|\sum_{q\le Q}R(q)^3|}{\sum_{q\le Q}|R(q)|^3} \tag{16.10}$$

increases from approximately 0.164 near $Q = 4.85\times10^4$ to 0.278 near $Q = 6.08\times10^5$. This records finite-range signed cancellation but does not support a specific asymptotic exponent.

Remark 16.4 (Standard explicit-formula interpretation)*.* The finite-range suppression in Experiment 16.3 is not claimed as a new asymptotic phenomenon. Under the usual explicit-formula model for $M(x)/\sqrt{x}$, together with linear independence of the zero ordinates and the auxiliary convergence assumptions required to pass to a limiting distribution, the limiting law is symmetric; odd limiting moments therefore vanish when those moments exist. This is the standard heuristic background for a cancellation such as $\sum R(q)^3 = o(Q^{5/2+\varepsilon})$, not a theorem proved by the present transform; compare the distributional framework in [4]. The drift in the measured ratio is the reason no asymptotic claim is made here.

16.1. **The mod–4 root-of-unity filter.** Because

$$R(q) = q - 2N_-(q), \tag{16.11}$$

for odd q one has

$$(-1)^{(R(q)-1)/2} = (-1)^{(q-1)/2}(-1)^{N_-(q)}. \tag{16.12}$$

Define

$$P(q) := (-1)^{N_-(q)}. \tag{16.13}$$

Since $N_-(q)$ counts the negative terms among $\varepsilon_1,\dots,\varepsilon_q$,

$$\boxed{P(q) = \prod_{r=1}^{q}\varepsilon_r}. \tag{16.14}$$

Therefore, for every $h \ge 1$,

$$\boxed{P(q)P(q+h) = \prod_{r=q+1}^{q+h}\varepsilon_r}. \tag{16.15}$$

Thus the mod–4 filter selected by the root-of-unity specialization does not avoid higher-order Chowla: its lag-h autocorrelation is exactly an h-point consecutive correlation of the squarefree-time sign sequence. In particular, the root-of-unity filter repackages the higher-order correlation hierarchy rather than bypassing it.

Proposition 16.5 (Universal coercivity can be evenised)**.** *Let σ denote the sign flip $\varepsilon\mapsto-\varepsilon$, so that $d(\sigma\varepsilon) = -d(\varepsilon)$. Suppose a real-valued functional F on finite sign sequences satisfies*

$$F(\varepsilon) \ge c\,\Phi(\|d(\varepsilon)\|_\infty) \tag{16.16}$$

for every admissible sign sequence, where $c > 0$ and $\Phi \ge 0$. Then the symmetrised functional

$$F_{\mathrm{ev}}(\varepsilon) := \frac{1}{2}(F(\varepsilon) + F(-\varepsilon)) \tag{16.17}$$

is σ-even and satisfies the same lower bound

$$F_{\mathrm{ev}}(\varepsilon) \geq c\,\Phi(\|d(\varepsilon)\|_\infty)\,. \tag{16.18}$$

Thus odd parity cancellation cannot, by itself, be the essential source of a universal coercive estimate for $\|d\|_\infty$.

Proof. The norm $\|d\|_\infty$ is invariant under σ. Apply (16.16) to both ε and $-\varepsilon$ and average the two inequalities. □

Remark 16.6 (What the involution obstruction does and does not say). If an evenised functional admits an analytic expansion in the height variables $R(q)$, only even total degrees can occur. When its second variation is nonzero, the first nonconstant term is a quadratic form $\sum_{q,r} K(q,r)R(q)R(r)$, putting the first coercive approximation in a weighted mean-square class. Proposition 16.5 does not assert that every even coercive functional must possess a nonzero quadratic term: higher even orders or nonanalytic functionals are not excluded. Its exact conclusion is narrower and decisive for the parity mechanism: the odd-only cancellation of Proposition 16.1 cannot alone furnish a universal lower bound for the size of d. A route that uses the arithmetic Möbius sequence specifically could escape this universal symmetrisation only by introducing arithmetic input not contained in the abstract sign matching.

Proposition 16.7 (Exact real-axis cancellation counterexample)**.** *For every integer $H \geq 1$ there is a finite sign sequence with*

$$\max_q |R(q)| = \|d\|_\infty = H \tag{16.19}$$

and whose lifetime generating polynomial is

$$D_H(z) = H\,\frac{z(1-z^H)^2}{1-z}. \tag{16.20}$$

Consequently $D_H(z) \to 0$ as $z \uparrow 1$ while $\|d\|_\infty = H$ remains fixed. In particular there is no positive constant c for which

$$|D(z)| \geq c\,\|d\|_\infty \tag{16.21}$$

holds uniformly for all finite sign sequences and all z sufficiently close to 1.

Proof. Take the sign word

$$\varepsilon^{(H)} = (+1)^H(-1)^{2H}(+1)^H. \tag{16.22}$$

Its first H ordinal pairs have $d_j = H$ and its next H pairs have $d_{H+j} = -H$ for $1 \leq j \leq H$. Thus

$$D_H(z) = H\sum_{j=1}^{H} z^j - H\sum_{j=H+1}^{2H} z^j \tag{16.23}$$

$$= H\,\frac{z(1-z^H)^2}{1-z}, \tag{16.24}$$

and (16.19) follows directly from the same word. □

17. Excursion blocks, active pairs, and prime residue channels

The identities above provide structure but not yet a new estimate for M. A more coercive use of d_j would have to exploit the simultaneous activity of many ordinal pairs during a large excursion.

For a cut at squarefree time q, define

$$a(q) := \min\{N_+(q), N_-(q)\}, \qquad h(q) := |R(q)|. \tag{17.1}$$

Exactly the indices

$$a(q)+1,\dots,a(q)+h(q) \tag{17.2}$$

correspond to ordinal pairs having one endpoint on each side of the cut. Moreover their orientations all agree with $\operatorname{sgn} R(q)$. Consequently, for any finitely supported weight sequence (w_j),

$$\boxed{\sum_j w_j d_j = \sum_q \operatorname{sgn} R(q) \sum_{j=a(q)+1}^{a(q)+|R(q)|} w_j}. \tag{17.3}$$

Equation (17.3) is a double-counting identity: each oriented pair contributes w_j once for every squarefree-time cut crossed by the interval between α_j and β_j.

For primes $p>2$, the only reduced residue classes modulo 4 are 1 and 3. Set, for $j\ge 2$,

$$\eta_j := \chi_4(p_j) = \begin{cases} +1, & p_j \equiv 1 \pmod 4,\\ -1, & p_j \equiv 3 \pmod 4. \end{cases} \tag{17.4}$$

The prime number theorem in arithmetic progressions gives

$$\frac{1}{J}\sum_{j\le J}\eta_j \longrightarrow 0. \tag{17.5}$$

Applying (17.3) with $w_j=\eta_j$ yields

$$\sum_j \chi_4(p_j) d_j = \sum_q \operatorname{sgn} R(q)\, B_4(a(q),|R(q)|), \tag{17.6}$$

where

$$B_4(a,h) := \sum_{j=a+1}^{a+h} \chi_4(p_j). \tag{17.7}$$

Thus $B_4(a,h)$ is the difference between the numbers of primes congruent to 1 and 3 modulo 4 in the selected block of h consecutive prime indices.

Remark 17.1 (Why a single residue channel is not coercive). The asymptotic equality of the two prime residue classes is useful for centering, but it does not give a lower bound for $|B_4(a,h)|$. An even block may have exactly equal counts and hence $B_4(a,h)=0$ no matter how large h is. Therefore the mod–4 channel alone cannot bound $|R(q)|$ from above. The identity (17.6) is retained as a possible cross-arithmetic observable, not as a proved Mertens estimate.

More generally, for Dirichlet characters χ modulo m define

$$B_\chi(a,h) := \sum_{j=a+1}^{a+h} \chi(p_j). \tag{17.8}$$

If n_r denotes the number of primes in the block lying in a reduced residue class $r \pmod m$, character orthogonality gives

$$\sum_{\chi \bmod m} |B_\chi(a,h)|^2 = \varphi(m) \sum_{r\in(\mathbf{Z}/m\mathbf{Z})^\times} n_r^2. \tag{17.9}$$

Removing the principal character gives

$$\sum_{\substack{\chi \bmod m\\ \chi\neq\chi_0}} |B_\chi(a,h)|^2 = \varphi(m)\sum_r n_r^2 - h^2, \tag{17.10}$$

provided the block contains no prime divisor of m; the finitely many exceptional primes can be separated. The right side can again be small when the block is nearly equidistributed. Hence no fixed modulus supplies the coercivity required to deduce a bound for $|R|$.

Question 17.2 (Multimodulus coercivity)**.** Can a family of nonprincipal characters over growing moduli satisfy a uniform lower estimate on every block of h consecutive primes of the form

$$\sum_{m\le L}\sum_{\substack{\chi \bmod m\\ \chi\neq\chi_0}} |B_\chi(a,h)|^2 \ge \Phi(h,L), \tag{17.11}$$

with $\Phi(h,L)$ strong enough, together with independently available upper bounds for the corresponding weighted lifetime sums, to constrain $h=|R(q)|$?

Question 17.2 is deliberately stated as an open mechanism, not as a consequence of the large sieve. Standard large-sieve inequalities naturally give upper bounds; the lower, block-uniform coercivity needed here is the nontrivial direction.

17.1. **Survival kernel of the ordinal pairs.** A natural attempt to use the two inverse clocks is to count pairs that survive across two cuts. For $u<v$ define the signed survival kernel

$$K(u,v) := \#\{j : \alpha_j \le u < v < \beta_j\} - \#\{j : \beta_j \le u < v < \alpha_j\}. \tag{17.12}$$

The monotone inverse matching makes this kernel collapse exactly to the two endpoint heights.

Proposition 17.3 (Two-cut survival collapse)**.** *For every $u<v$,*

$$K(u,v) = \big(N_+(u)-N_-(v)\big)_+ - \big(N_-(u)-N_+(v)\big)_+ \tag{17.13}$$

$$= \frac{\operatorname{sgn}(R(u)+R(v))}{2}\Big(|R(u)+R(v)|-(v-u)\Big)_+. \tag{17.14}$$

The unsigned number of pairs whose interval contains all of $[u,v]$ is

$$C(u,v) := \#\{j : [u,v]\subseteq[\min(\alpha_j,\beta_j),\max(\alpha_j,\beta_j))\} = \frac{1}{2}\Big(|R(u)+R(v)|-(v-u)\Big)_+. \tag{17.15}$$

Proof. The first count in (17.12) requires $j\le N_+(u)$ and $j>N_-(v)$, hence equals $(N_+(u)-N_-(v))_+$. The second equals $(N_-(u)-N_+(v))_+$. Substitution of $N_\pm=(q\pm R(q))/2$ gives (17.14); since at most one of the two positive parts can be nonzero, their sum gives (17.15). □

Proposition 17.4 (First active-lifetime moment also collapses)**.** *For a cut q, let*

$$\mathcal{W}(q) := \{j : \min(\alpha_j,\beta_j)\le q < \max(\alpha_j,\beta_j)\}. \tag{17.16}$$

Extend the unsigned survival count symmetrically by

$$C^*(u,q) := C(\min\{u,q\},\max\{u,q\}) = \frac{1}{2}\Big(|R(u)+R(q)|-|u-q|\Big)_+. \tag{17.17}$$

Then

$$\boxed{\sum_{j\in\mathcal{W}(q)}|d_j| = \sum_{u\ge0}C^*(u,q) = \frac{1}{2}\sum_{u\ge0}\Big(|R(u)+R(q)|-|u-q|\Big)_+}. \tag{17.18}$$

For a finite prefix, the sums are truncated at the final available cut, with the usual boundary convention for pairs not yet closed.

Proof. For each $j\in\mathcal{W}(q)$, the integer $|d_j|=|\beta_j-\alpha_j|$ is the number of squarefree-time cuts crossed by the half-open interval between its two endpoints. Hence

$$\sum_{j\in\mathcal{W}(q)}|d_j| = \sum_{u\ge0}\#\{j : I_j \text{ crosses both } u \text{ and } q\}. \tag{17.19}$$

The inner count is exactly $C^*(u,q)$, and Proposition 17.3 gives (17.17). □

Proposition 17.5 (All ordered cut-count kernels reduce to the two extremes)**.** *Let $q_1 < \cdots < q_k$ be any finite ordered family of cuts. The number of matched intervals crossing every one of them is exactly*

$$\#\{j : I_j \text{ crosses every } q_i\} = C(q_1, q_k), \tag{17.20}$$

and the oriented count is

$$\#\{j : \alpha_j \le q_1 < q_k < \beta_j\} - \#\{j : \beta_j \le q_1 < q_k < \alpha_j\} = K(q_1, q_k). \tag{17.21}$$

Thus every statistic that only counts which ordinal-pair intervals cross a prescribed ordered set of cuts is already a two-point statistic of the extreme cuts.

Proof. Each I_j is an interval. It contains all $q_1, \dots, q_k$ if and only if it contains the two extreme cuts q_1 and q_k. Apply Proposition 17.3. □

Experiment 17.6 (Shape of the active lifetime multiset)**.** The full multiset

$$\mathcal{L}(q) := \{\, |\beta_j - \alpha_j| : \ j \in \mathcal{W}(q) \,\} \tag{17.22}$$

is not determined by the single height $h = |R(q)|$. Reported measurements, conditioned on height bins and compared with an iid fair-sign walk under the same conditioning, were as follows. Each entry is Möbius / iid.

h bin	$\langle \text{mean}\, \mathcal{L} \rangle / h$	between-cut dispersion	$\langle \text{sd}(\mathcal{L}) / \text{mean}(\mathcal{L}) \rangle$
8–16	2.066/2.068	0.364/0.365	0.102/0.102
32–64	2.010/2.025	0.170/0.172	0.055/0.054
128–400	1.997/2.000	0.087/0.082	0.028/0.026

The regenerated active-lifetime tail frequencies were

threshold	$1.5h$	$2.5h$	$3.5h$
Möbius	0.99754	0.00841	0.00037
iid	0.99954	0.00217	0.00007

(17.23)

Thus the part of the inverse matching that escapes the endpoint collapse is, on the tested range, reproduced to within a few percent by the corresponding conditioned random walk. The relative within-cut dispersion decreases from about 0.102 to 0.028 as the conditioned height increases across the three bins, consistent with a concentration scale close to $h^{-1/2}$ on this finite range. This is a measured statement, not an asymptotic theorem.

Remark 17.7 (Structural closure of cut geometry)*.* Propositions 17.3, 17.4, and 17.5 close a whole class of possible uses of the inverse matching. The cardinality of the active set, the first moment of its lifetimes, and every kernel that merely asks how many intervals cross a finite ordered collection of cuts all reduce to one- or two-point height data. The full shape of the multiset (17.22) does retain history, but Experiment 17.6 indicates that this extra information is generic conditioned-walk structure rather than a detected Möbius-specific signal. In the present analysis the only inverse-pair observable that both escapes these collapses and carries proved arithmetic content is the local increment $d_{j+1} - d_j$, through the squarefree-ordinal correlation theorem of Section 15.

18. Universal decoding, square-root recovery, and endpoint verification

The transform has two distinct identification properties. The first is universal and combinatorial: a complete SS trajectory recovers the underlying ternary sign sequence. The second uses the arithmetic identity for squarefree counting and shows that a short genuine prefix, of length $\sqrt{N}$, together with one endpoint value, recovers $(M(N), Q(N))$.

Set $SS(0) := 0$ and write

$$\Delta SS(k) := SS(k) - SS(k-1) = S(k). \tag{18.1}$$

Define the ternary decoder

$$\mathfrak{d}(x) := \begin{cases} 0, & x = 1, \\ +1, & x > 1, \\ -1, & x < 0. \end{cases} \tag{18.2}$$

For the Möbius transform one has $\mathfrak{d}(S(k)) = \mu(k)$ exactly.

Theorem 18.1 (Universal ordinal-prime coding)**.** *Let $(a_k)_{k\geq 1}$ be any sequence in $\{-1, 0, +1\}$. Define separate occurrence counters $A_\pm$ and attach $+p_j$ to the jth $+1$, $-p_j$ to the jth -1, and 1 to every zero; let SS_a be the cumulative sum. Then the map $a \mapsto SS_a$ is injective, with local inverse*

$$a_k = \mathfrak{d}(SS_a(k) - SS_a(k-1)) \,. \tag{18.3}$$

Moreover a trajectory $X_0 = 0, X_1, X_2, \ldots$ lies in the image if and only if, with $\Delta_k = X_k - X_{k-1}$, every step is one of

$$1, \qquad p_{1+\#\{r<k:\Delta_r>1\}}, \qquad -p_{1+\#\{r<k:\Delta_r<0\}}. \tag{18.4}$$

Proof. The forward implication is the definition of the ordinal-prime assignment. Conversely, the sign of every nonunit increment and the distinguished unit increment recover a_k by (18.3). The two running counts then force the absolute value of the next nonzero increment to be the next prime in the appropriate sign counter, which is exactly (18.4). □

Remark 18.2 (Identification is not authentication)*.* Theorem 18.1 is deliberately universal. Hence the ordinal-prime recursion alone cannot certify that a decoded ternary trajectory is the Möbius sequence: every ternary sequence has a valid transform. This also means that the construction by itself is not a cryptographic cipher; differencing SS exposes the ternary symbols through (18.3). Any verification claim for Möbius must therefore include an independent arithmetic check of the decoded prefix.

An isolated endpoint does not have this identification property on the formal counter domain. Indeed, write $a = N_+$ and $b = N_-$. Then

$$SS(n) - n = \big(\mathcal{P}(a) - a\big) - \big(\mathcal{P}(b) - b\big). \tag{18.5}$$

For every $n \geq 6$, the two formally admissible counter pairs

$$(M, Q) = (-1, 3), \qquad (M, Q) = (0, 6) \tag{18.6}$$

both give $SS(n) = n - 6$. Thus a single endpoint $SS(n)$ does not determine the two counters on the unrestricted formal lattice. The recovery theorem below excludes these false pairs by recovering the genuine $Q(N)$ from a short arithmetic prefix of the same SS trajectory.

The required arithmetic check can nevertheless be restricted to square-root depth. Recall the elementary identity

$$Q(N) = \sum_{d \leq \sqrt{N}} \mu(d) \left\lfloor \frac{N}{d^2} \right\rfloor. \tag{18.7}$$

It follows from $\mu^2(m) = \sum_{d^2 | m} \mu(d)$ after summing over $m \leq N$.

Theorem 18.3 (Square-root recovery from the genuine SS trajectory)**.** *Let $N \geq 1$ and suppose*

$$SS(0), SS(1), \ldots, SS(\lfloor \sqrt{N} \rfloor), \qquad SS(N) \tag{18.8}$$

are values of the actual Möbius ordinal-prime transform. Then these data determine $Q(N)$ and $M(N)$ uniquely. Explicitly,

$$\boxed{Q(N) = \sum_{d \leq \sqrt{N}} \mathfrak{d}(SS(d) - SS(d-1)) \left\lfloor \frac{N}{d^2} \right\rfloor.} \tag{18.9}$$

If q denotes this value, then $M(N)$ is the unique integer m satisfying

$$|m| \le q, \qquad m \equiv q \pmod 2, \tag{18.10}$$

and

$$SS(N) - N + q = \mathcal{P}\Big(\frac{q+m}{2}\Big) - \mathcal{P}\Big(\frac{q-m}{2}\Big). \tag{18.11}$$

Proof. For $d \le \sqrt{N}$, (18.1) and (18.2) recover $\mu(d)$ from the genuine trajectory. Substitution into (18.7) gives (18.9). Fix $q = Q(N)$ and put

$$F_q(m) := \mathcal{P}\Big(\frac{q+m}{2}\Big) - \mathcal{P}\Big(\frac{q-m}{2}\Big) \tag{18.12}$$

for the parity-compatible integers in (18.10). If $a = (q+m)/2$ and $b = (q-m)/2$, then whenever $m+2$ is admissible,

$$F_q(m+2) - F_q(m) = p_{a+1} + p_b > 0. \tag{18.13}$$

Thus F_q is strictly increasing on its admissible lattice. The closed form of Proposition 3.3 is exactly (18.11), so its solution is unique. □

Remark 18.4 (Verification protocol and its actual cost). Theorem 18.3 reduces the Möbius-side verification depth from N to $\sqrt{N}$, but it does not remove independent arithmetic work. A verifier presented with a claimed prefix must check that the decoded values in (18.2) equal the genuine $\mu(d)$ for $d \le \sqrt{N}$; the universal coding theorem shows why this check cannot be omitted. The prime-side quantities in (18.11) must also be evaluated or independently certified. The result is therefore a sublinear-depth recovery/checksum identity on the Möbius side, not a stand-alone proof certificate requiring no auxiliary computation.

19. Self-iteration of the ordinal-prime transform

The absolute transform can be iterated on squarefree inputs. Let

$$\mathcal{S} := \{n \ge 1 : \mu(n) \ne 0\}, \qquad \varphi : \mathcal{S} \to \mathbb{P}, \qquad \varphi(n) := |S(n)|, \tag{19.1}$$

where $\mathbb{P}$ is the set of primes. Since every prime is squarefree, φ maps $\mathcal{S}$ into its own domain after one step.

The ordinal variables α_j, β_j of Section 14 are positions in the squarefree clock. Their corresponding positions in the original integer coordinate are

$$A_j := s_{\alpha_j}, \qquad B_j := s_{\beta_j}. \tag{19.2}$$

Thus A_j is the integer at which the jth $+1$ value of μ occurs and B_j is the integer at which the jth -1 value occurs. This distinction is essential in the dynamics below.

Proposition 19.1 (Exact fibres)**.** *For every $j \ge 1$,*

$$\boxed{\varphi^{-1}(p_j) = \{A_j, B_j\} = \{s_{\alpha_j}, s_{\beta_j}\}.} \tag{19.3}$$

Consequently every prime has indegree 2 in the functional digraph of φ, every squarefree composite has indegree 0, and every squarefree integer has outdegree 1.

Proof. By definition the jth positive and jth negative Möbius occurrences are mapped to $+p_j$ and $-p_j$, respectively, and no other occurrence receives absolute value p_j. The integer positions of those two occurrences are $A_j = s_{\alpha_j}$ and $B_j = s_{\beta_j}$. □

The first negative squarefree composite is

$$30 = 2 \cdot 3 \cdot 5, \qquad \mu(30) = -1. \tag{19.4}$$

Every squarefree composite below 30 has exactly two prime factors and hence Möbius value $+1$. This elementary observation completely classifies the periodic dynamics.

Theorem 19.2 (Complete periodic-orbit classification)**.** *The periodic points of φ are exactly*

$$\boxed{2, 3, 5, 7, 11, 13, 17, 19, 23, 29.} \tag{19.5}$$

All ten are fixed points. There are no cycles of length greater than one. More precisely, if a squarefree n does not map in one step to one of the primes in (19.5)*, then*

$$\varphi(n) \geq 31, \qquad \varphi^{\circ(k+1)}(n) > \varphi^{\circ k}(n) \quad (k \geq 1), \tag{19.6}$$

and the orbit tends to infinity.

Proof. Before 30, the negative Möbius occurrences are precisely the primes. Hence for $1 \leq j \leq 10$ the integer position of the jth negative occurrence is

$$B_j = p_j, \tag{19.7}$$

so $\varphi(p_j) = p_j$ for $j \leq 10$. Now let $p \geq 31$ be prime. Every prime not exceeding p contributes one negative occurrence, and 30 contributes an additional one. Therefore

$$N_-(p) \geq \pi(p) + 1. \tag{19.8}$$

Since $\mu(p) = -1$,

$$\varphi(p) = p_{N_-(p)} \geq p_{\pi(p)+1} > p. \tag{19.9}$$

Every image of φ is prime. Thus once an orbit reaches a prime at least 31, (19.9) makes all subsequent terms strictly increasing, so the orbit cannot cycle and must escape. By Proposition 19.1, a squarefree starting point either maps to one of the ten fixed primes or to a prime at least 31, proving the stated classification. □

The finite basins can also be written explicitly. The first ten positive occurrence positions in the original integer coordinate are

$$A_1, \ldots, A_{10} = 1, 6, 10, 14, 15, 21, 22, 26, 33, 34, \tag{19.10}$$

whereas $B_j = p_j$ for $j \leq 10$. No A_j is prime: $A_1 = 1$, and a prime has Möbius value -1, never $+1$. Hence each fixed point p_j, $j \leq 10$, has full basin

$$\boxed{\{p_j, A_j\}.} \tag{19.11}$$

No further preimage can attach to A_j, because the image of φ is always prime.

The restriction of the dynamics to prime inputs has an exact one-dimensional index map. Define

$$f(j) := N_-(p_j), \qquad j \geq 1, \tag{19.12}$$

and let

$$C_-(x) := \#\{m \leq x : m \text{ composite and } \mu(m) = -1\}. \tag{19.13}$$

Proposition 19.3 (Prime-index dynamics)**.** *For every $j \geq 1$,*

$$\varphi(p_j) = p_{f(j)}, \tag{19.14}$$

$$f(j) = j + C_-(p_j). \tag{19.15}$$

The map $f : \mathbb{N} \to \mathbb{N}$ is strictly increasing,

$$f(j) = j \quad (1 \leq j \leq 10), \qquad f(j) > j \quad (j \geq 11). \tag{19.16}$$

Proof. Since $\mu(p_j) = -1$, the absolute transform at p_j is the prime with index $N_-(p_j)$, which gives (19.14). The negative occurrences up to p_j consist of the j primes up to p_j together with the negative squarefree composites, giving (19.15). Passing from p_j to p_{j+1} always adds the new negative occurrence p_{j+1}, hence $f(j+1) \geq f(j)+1$ and f is strictly increasing. There is no negative composite below 30, while $30 \leq p_j$ for every $j \geq 11$; this gives (19.16). □

Proposition 19.3 gives an exact component structure. The positive preimage A_j is always a nonprime leaf. The second preimage B_j is a prime precisely when j lies in the image $f(\mathbb{N})$; in that case, if $j = f(k)$, then $B_j = p_k$. Since f is injective, a prime has at most one prime predecessor. Thus the prime-restricted graph consists of the ten fixed points and a disjoint family of strictly increasing one-way rays. A ray starts at each index

$$j \notin f(\mathbb{N}), \qquad j \geq 11. \tag{19.17}$$

In the full squarefree graph, one +1 leaf A_j is attached to every prime on such a ray; at its initial prime there is also the negative-composite leaf B_j.

The set of ray sources has density one among prime indices.

Proposition 19.4 (Asymptotic density of prime-ray sources)**.** *As $j \to \infty$,*

$$f(j) \sim \frac{3}{\pi^2} j \log j. \tag{19.18}$$

Consequently

$$\#\big(f(\mathbb{N}) \cap [1, J]\big) \sim \frac{\pi^2}{3} \frac{J}{\log J}, \tag{19.19}$$

and hence

$$\frac{1}{J} \#\{1 \leq j \leq J : j \notin f(\mathbb{N})\} \longrightarrow 1. \tag{19.20}$$

Proof. By (19.12),

$$f(j) = N_-(p_j) = \frac{Q(p_j) - M(p_j)}{2}. \tag{19.21}$$

The squarefree counting asymptotic $Q(x) \sim 6x/\pi^2$ and the prime-number- theorem consequence $M(x) = o(x)$ give $f(j) \sim 3p_j/\pi^2$. Since $p_j \sim j \log j$, this proves (19.18). Because f is strictly increasing, the left side of (19.19) is the largest k with $f(k) \leq J$. Inverting $f(k) \sim (3/\pi^2) k \log k$ gives (19.19); subtracting from J gives (19.20). □

The asymptotic speed of every escaping orbit is also unconditional.

Proposition 19.5 (Growth of escaping orbits)**.** *Let $x_0 \in \mathcal{S}$ have an escaping orbit and put $x_{k+1} = \varphi(x_k)$. Then, as $k \to \infty$,*

$$x_{k+1} \sim \frac{3}{\pi^2} x_k \log x_k, \tag{19.22}$$

$$\log x_k \sim k \log k, \tag{19.23}$$

$$x_k = k^{\,k+o(k)}. \tag{19.24}$$

Proof. After at most one step all x_k are primes and tend to infinity by Theorem 19.2. For a prime x,

$$N_-(x) = \frac{Q(x) - M(x)}{2} \sim \frac{3}{\pi^2} x, \tag{19.25}$$

using $Q(x) \sim 6x/\pi^2$ and the prime-number-theorem consequence $M(x) = o(x)$. The prime number theorem in the form $p_m \sim m \log m$ then gives (19.22). Writing $y_k = \log x_k$ yields

$$y_{k+1} = y_k + \log y_k + \log\left(\frac{3}{\pi^2}\right) + o(1). \tag{19.26}$$

Elementary comparison first gives $y_k \asymp k \log k$; substituting this back into (19.26) and applying Stolz–Cesàro gives $y_k/(k \log k) \to 1$. Exponentiation gives (19.24). □

Remark 19.6 (Forward versus inverse geometry). The graph is not an infinite full binary tree. Its inverse fibres over primes have size two by Proposition 19.1, but a squarefree composite has no inverse fibre. Prime-restricted dynamics is described by the strictly increasing map f: ten fixed points are followed by a forest of disjoint escaping rays, and Proposition 19.4 shows that a proportion tending to one of prime indices begin such rays. The full squarefree graph adds nonprime leaves to that prime skeleton as described above. This decomposition is specific to the self-iteration of $|S|$.

20. Collisions and elementary modular rigidity of SS

The endpoint sequence is not injective, but its collisions obey simple exact constraints. The first two increments are $S(1) = 2$ and $S(2) = -2$. Thereafter every increment is odd: zeros contribute 1, and both sign counters have already consumed the unique even prime 2.

Proposition 20.1 (Parity rigidity)**.** *For every $n \geq 2$,*

$$\boxed{SS(n) \equiv n \pmod 2.} \tag{20.1}$$

Consequently, if $m, n \geq 2$ and $SS(m) = SS(n)$, then $m \equiv n \pmod 2$.

Proof. Since $SS(2) = 0$ and every $S(k)$ is odd for $k \geq 3$, the sum $SS(n) - SS(2)$ has the parity of $n - 2$. □

The smallest possible nontrivial collision length is characterised by the ordinal matching itself.

Proposition 20.2 (Length-two collisions)**.** *Let*

$$A_j := s_{\alpha_j}, \qquad B_j := s_{\beta_j}, \qquad \Delta_j := B_j - A_j = s_{\beta_j} - s_{\alpha_j}. \tag{20.2}$$

Then for every $n \geq 0$,

$$\boxed{SS(n+2) = SS(n) \iff \text{there exists } j \text{ with } \{A_j, B_j\} = \{n+1, n+2\} \iff |\Delta_j| = 1.} \tag{20.3}$$

Proof. The equality is equivalent to $S(n+1) + S(n+2) = 0$. The unit value cannot participate in such a cancellation. A positive and a negative transform value have equal absolute value exactly when they are the two integer occurrences carrying the same ordinal prime p_j, namely A_j and B_j. They occupy consecutive integer positions exactly when $|B_j - A_j| = |\Delta_j| = 1$. □

The sequence is not Sidon. From

$$SS(1), \ldots, SS(6) = 2, 0, -3, -2, -7, -4 \tag{20.4}$$

one already has

$$2 + (-4) = 0 + (-2). \tag{20.5}$$

Thus the stronger $B_2[1]$ property fails. Whether the value set of SS has uniformly bounded additive multiplicity $B_2[g]$ for some finite g is not settled here.

Modulo 2, Proposition 20.1 completely determines the trajectory. Modulo 4 the first genuinely nontrivial residue interaction appears, because every odd prime lies in the classes 1 or 3 modulo 4; that problem is distinct from the mod–4 character channel of Section 17 and no equidistribution theorem for $SS \pmod 4$ is claimed here.

21. Level crossings, returns, and negative-excursion energy

The squarefree clock s_q, its signs ε_q, and the walk $R(q) = M(s_q)$ were defined in (14.1)–(14.2). In this section set $s_0 = 0$, $R(0) = 0$, and $SS(s_0) = 0$. The transform has an especially rigid form when the walk reaches or returns to a prescribed integer level.

Proposition 21.1 (Exact crossing values)**.** *Suppose that the squarefree step q reaches the level h, so $R(q) = h$. If the step is upward, then*

$$S(s_q) = p_{(q+h)/2}. \tag{21.1}$$

If it is downward, then

$$S(s_q) = -p_{(q-h)/2}. \tag{21.2}$$

In particular, every crossing into zero has the direction-independent magnitude

$$|S(s_q)| = p_{q/2}, \tag{21.3}$$

and at every physical position n with $M(n) = 0$ one has

$$SS(n) = n - Q(n). \tag{21.4}$$

Proof. At squarefree time q the two occurrence counts are $(q+h)/2$ and $(q-h)/2$. An upward step carries the newest positive ordinal prime and a downward step carries the negative of the newest negative ordinal prime. Equations (21.1) and (21.2) follow. Setting $h = 0$ proves (21.3); equation (21.4) follows from Proposition 3.3 because the two prime-prefix sums cancel. □

Theorem 21.2 (Exact return formula)**.** *Let $0 \le a < b$ satisfy $R(a) = R(b) = h$, put $b - a = 2r$, and let*

$$Z(a, b) := s_b - s_a - (b - a) \tag{21.5}$$

be the number of nonsquarefree physical positions in $(s_a, s_b]$. Write

$$A = \frac{a+h}{2}, \qquad B = \frac{a-h}{2}. \tag{21.6}$$

Then

$$\boxed{SS(s_b) - SS(s_a) = Z(a, b) + \sum_{t=1}^{r} (p_{A+t} - p_{B+t}).} \tag{21.7}$$

Consequently:

(i) if $h > 0$, every nontrivial return has $SS(s_b) > SS(s_a)$;
(ii) if $h = 0$, then $SS(s_b) - SS(s_a) = Z(a, b)$;
(iii) if $h = -k < 0$, then

$$SS(s_b) - SS(s_a) = Z(a, b) - \sum_{t=1}^{r} (p_{A+k+t} - p_{A+t}). \tag{21.8}$$

Thus a same-level collision below zero is possible exactly when the zero count equals the displayed positive prime-separation sum.

Proof. Between squarefree times a and b there are equally many positive and negative signs, namely r of each. They consume respectively the ordinal prime blocks $p_{A+1}, \dots, p_{A+r}$ and $p_{B+1}, \dots, p_{B+r}$, while the $Z(a, b)$ zero positions each contribute 1. This proves (21.7). If $h > 0$, then $A = B + h$ and every prime difference is positive. If $h = 0$, the two blocks cancel term by term. If $h = -k$, then $B = A + k$, giving (21.8). □

The negative case supplies a one-sided quantity absent from the positive and zero levels. A complete negative squarefree excursion is an interval $[q_0, q_1]$ with $R(q_0) = R(q_1) = 0$ and $R(q) < 0$ for $q_0 < q < q_1$. Its depth is

$$H = -\min_{q_0<q<q_1} R(q). \tag{21.9}$$

For $1 \le k < H$, let u_k and v_k be the first and last visits to $-k$ in the excursion, set $v_k - u_k = 2r_k$, and put

$$\mathcal{A}_k := Z(u_k, v_k) - \big(SS(s_{v_k}) - SS(s_{u_k})\big), \qquad \mathcal{A} := \sum_{k=1}^{H-1} \mathcal{A}_k. \tag{21.10}$$

Theorem 21.3 (Coercive energy of a negative excursion)**.** *With the preceding notation, if $a_k = (u_k - k)/2$, then*

$$\boxed{\mathcal{A}_k = \sum_{t=1}^{r_k} (p_{a_k+k+t} - p_{a_k+t}) > 0.} \tag{21.11}$$

Moreover

$$r_k \ge H - k \tag{21.12}$$

and therefore

$$\boxed{\mathcal{A} \ge 2 \sum_{k=1}^{H-1} k(H-k) = \frac{H^3 - H}{3}.} \tag{21.13}$$

More sharply, if

$$G(k) := \inf_{m \ge 2} (p_{m+k} - p_m), \tag{21.14}$$

then

$$\mathcal{A} \ge \sum_{k=1}^{H-1} (H-k) G(k). \tag{21.15}$$

Proof. Formula (21.11) is (21.8) applied to the first and last visits to $-k$. To descend from the first visit to $-k$ to depth $-H$ and return to the last visit to $-k$ requires at least $H - k$ downward and $H - k$ upward steps, so $v_k - u_k \ge 2(H - k)$, proving (21.12). Since $\mu(1) = 1$, the lower prime index $a_k + t$ in (21.11) is at least 2. All consecutive prime gaps from p_2 onward are at least 2, and hence $p_{a_k+k+t} - p_{a_k+t} \ge 2k$. Summing first over t and then over k gives (21.13). Replacing $2k$ by the defining infimum $G(k)$ gives (21.15). □

Remark 21.4 (What the coercive inequality does not yet prove)*.* The lower bound (21.13) is unconditional and uses the actual prime magnitudes essentially. It does not alone upper-bound H: such a conclusion would require an independently obtained upper bound for $\mathcal{A}$ that is stronger than one obtained by substituting a prior bound for M. The theorem therefore identifies a positive channel at $h < 0$ without claiming a new Mertens estimate.

22. Removing the primes: exact ordinal and positional controls

Let $w_1, w_2, \ldots$ be arbitrary positive weights and $W(m) = \sum_{j \le m} w_j$. Replacing the ordinal primes by these weights gives the universal endpoint identity

$$SS_w(n) = W\Big(\frac{Q+M}{2}\Big) - W\Big(\frac{Q-M}{2}\Big) + n - Q. \tag{22.1}$$

There are two distinct meanings of "remove the primes."

Proposition 22.1 (Positional and ordinal controls)**.** *If $w_j = 1$, so that only the sign positions and the convention $\mu = 0 \mapsto 1$ remain, then*

$$\boxed{SS_{\mathrm{pos}}(n) = n - Q(n) + M(n).} \tag{22.2}$$

If $w_j = j$, which preserves the complete ordinal architecture but removes primality, then

$$\boxed{SS_{\mathrm{ord}}(n) = n - Q(n) + \frac{Q(n)+1}{2} M(n).} \tag{22.3}$$

Proof. For $w_j = 1$, equation (22.1) gives the difference $N_+ - N_- = M$. For $w_j = j$, use $W(m) = m(m+1)/2$ and

$$\frac{N_+(N_+ + 1) - N_-(N_- + 1)}{2} = \frac{(N_+ - N_-)(N_+ + N_- + 1)}{2} = \frac{M(Q+1)}{2}.$$

□

Define the accumulated prime displacement

$$D(m) := \sum_{j=1}^{m} (p_j - j), \qquad D(0) := 0. \tag{22.4}$$

Theorem 22.2 (Exact prime contribution and amplification)**.** *For every n,*

$$\boxed{SS(n) - SS_{\mathrm{ord}}(n) = D\Big(\frac{Q+M}{2}\Big) - D\Big(\frac{Q-M}{2}\Big).} \tag{22.5}$$

If $a = (Q - |M|)/2$, this is equivalently

$$SS(n) - SS_{\mathrm{ord}}(n) = \mathrm{sgn}(M) \sum_{j=a+1}^{a+|M|} (p_j - j). \tag{22.6}$$

In particular it has the sign of M and vanishes exactly when $M = 0$. Along any sequence $n \to \infty$ on which $M(n) \neq 0$,

$$\boxed{\frac{SS(n) - n + Q(n)}{SS_{\mathrm{ord}}(n) - n + Q(n)} \sim \log\Big(\frac{Q(n)}{2}\Big).} \tag{22.7}$$

Consequently, for the squared amplitudes

$$\boxed{\frac{(SS(n) - n + Q(n))^2}{(SS_{\mathrm{ord}}(n) - n + Q(n))^2} \sim \log^2\Big(\frac{Q(n)}{2}\Big).} \tag{22.8}$$

Proof. Subtract (22.3) from the prime-prefix formula of Proposition 3.3; this proves (22.5) and (22.6). Every summand $p_j - j$ is positive.

By the prime number theorem and the classical squarefree asymptotic, $p_j \sim j \log j$, $Q(n) \sim 6n/\pi^2$, and $M(n) = o(n) = o(Q(n))$. Every index in the block between $(Q-|M|)/2$ and $(Q+|M|)/2$ is therefore $(1 + o(1))Q/2$, uniformly across that block. Hence

$$\sum_{j=a+1}^{a+|M|} p_j \sim |M| \frac{Q}{2} \log \frac{Q}{2}, \qquad \sum_{j=a+1}^{a+|M|} j = |M| \frac{Q+1}{2}.$$

Their signed versions are respectively $SS - n + Q$ and $SS_{\mathrm{ord}} - n + Q$, proving (22.7); squaring proves (22.8). □

For completeness, an occupancy entropy cannot be decomposed into prime and nonprime terms by the preceding algebra. For an integer-valued path $X(1),\dots,X(N)$, let $L_X(v)=\#\{n\le N: X(n)=v\}$. The exact formulae are

$$H_1(X;N)=\log_2 N-\frac{1}{N}\sum_v L_X(v)\log_2 L_X(v), \tag{22.9}$$

$$\mathcal{E}_2^{\mathrm{occ}}(X;N)=\frac{1}{N^2}\sum_v L_X(v)^2=\frac{N+2C_X(N)}{N^2}, \tag{22.10}$$

where $C_X(N)$ counts unordered equal-value pairs. Thus an analytic entropy comparison requires a theorem controlling the complete collision set; the prime number theorem alone does not provide one. Assigning an effective temperature $T_N=T_0N$ merely multiplies $k_{\mathrm{B}}\log(2)H_1(X;N)$ by the common scale T_0N and creates no new arithmetic information.

Experiment 22.3 (Finite audit of the exact controls)**.** At $N=10^6$, the script `code/v11_level_energy_entropy.py` verifies all identities above with zero integer error. The endpoint data are

$$Q=607926,\quad M=212,\quad T_{\mathrm{ord}}=64440262,\quad T=915270668, \tag{22.11}$$

so the exact amplitude ratio is 14.2033976833 and the squared ratio is 201.736505749. The finite occupancy data are

path	distinct values	maximum multiplicity	H_1 (bits)	collision pairs
M	680	5433	8.878645	1263476491
SS_{pos}	392288	22	18.254030	1475777
SS_{ord}	982577	11	19.894001	21778
SS	997816	4	19.927180	2212

The small Shannon increment from SS_{ord} to SS is a ceiling effect: the ordinal control is already close to the maximum $\log_2 N$. In contrast, the primes reduce its residual collision-pair count by 19566, from 21778 to 2212. These are finite diagnostics and are not used to prove Theorem 22.2.

The same audit finds, among positive squarefree times up to 10^6, same-level collisions only at $h=0$ (348 pairs) and $h=-1$ (60 pairs). It also verifies (21.11) exactly on all 1637 complete negative excursions, of which 804 have depth at least 2. The deepest excursion has $H=368$, physical endpoints 804155 and 960379, and

$$\mathcal{A}=22400089790,\qquad \frac{H^3-H}{3}=16611888. \tag{22.12}$$

23. What the construction does and does not deliver

It delivers an exact, elementary dictionary between SS and (M,Q), a two-sided transfer of the Mertens criterion (Theorem 6.2), a residual whose size obeys a clean measured scaling law, and an ordinal-pairing representation whose pointwise $j^{1/2+\varepsilon}$ criterion is equivalent to RH (Proposition 14.2). The quadratic energy of that pairing is, however, equivalent up to absolute constants to the squarefree-time Mertens mean square (Theorem 14.3), so it does not by itself supply an independent route. Beyond that quadratic closure, Theorem 15.1 gives an unconditional logarithmic two-point decorrelation theorem in squarefree ordinal time, with the fixed-window second moment in Corollary 15.3. These are local ordering statements rather than norms of M. Version 9 adds two independent structural uses of the transform itself: the recovery theorem of Section 18, and the completely classified functional dynamics of $|S|$ in Section 19. Neither is a new bound on M, but both use information specific to the ordinal-prime construction rather than a norm of the Mertens path. Version 11 adds the exact level-return law of Theorem 21.2. Its negative-level branch yields the positive excursion functional $\mathcal{A}$ and the cubic lower bound (21.13). This is a genuine coercive statement on negative excursions, but it becomes a bound on their depth only if an independent

upper estimate for $\mathcal{A}$ is supplied. The exact nonprime controls of Proposition 22.1 also show that the prime weights amplify the ordinal M-signal by $\log(Q/2)$ in amplitude and $\log^2(Q/2)$ in squared energy.

It does not deliver an independent arithmetic constraint on $M(n)$, by Proposition 8.1. For a transformed representation to become useful it would have to produce at least one of:

(i) a cancellation unavailable in the original variables;
(ii) a positivity or monotonicity principle strong enough to yield new bounds;
(iii) a spectral representation turning a Mertens bound into an operator inequality;
(iv) an inequality between independently controlled components;
(v) a mechanism by which the prime side constrains the Möbius side beyond what (M, Q) already encodes.

None of these has yet produced a new bound on M. Proposition 16.5 shows that odd-parity cancellation alone cannot be the essential source of a universal coercive inequality, Proposition 16.7 rules out a simple real-axis Tauberian lower bound for D, and Proposition 17.3 shows that the two-cut survival count collapses to two endpoint heights, Proposition 17.4 shows that even the first moment of the active lifetime multiset is determined by two-point heights, and Proposition 17.5 shows that adding arbitrarily many ordered cuts does not escape this reduction. These are structural obstructions, not merely failed numerical tests. The measured law of Section 12 is of type (iv) in form only: it bounds E_{mid} in terms of $|M|$, which is the wrong direction to be useful. The cross-coupling experiment tests whether the prime-gap fluctuation might nevertheless provide an independently controlled handle on the Mertens side; no such handle is detected, but the numerical sensitivity is only at the 10^{-2} correlation scale and therefore does not close every conceivable nonlinear or asymptotic coupling.

24. A variant that is not an endpoint statistic (speculative)

The obstruction identified in Proposition 8.1 is caused by the use of two *separate* ordinal counters, which destroys the ordering of the ± 1 occurrences. The minimal modification that removes it is a *shared* counter:

$$\widetilde{S}(k) := \mu(k)\, p_{Q(k)}, \qquad \widetilde{SS}(n) := \sum_{k \le n} \mu(k)\, p_{Q(k)}. \tag{24.1}$$

Here the prime index depends on the position k through the shared squarefree counter rather than on a sign-specific ordinal. As a functional of a ternary history, it is not an endpoint statistic: for example the histories $(+1,-1,+1)$ and $(+1,+1,-1)$ have the same terminal counts $(M, Q) = (1, 3)$ but shared-counter sums $2 - 3 + 5 = 4$ and $2 + 3 - 5 = 0$, respectively. For the actual Möbius sequence, finite summation by parts with $w(k) = p_{Q(k)}$ gives exactly

$$\widetilde{SS}(n) = M(n)w(n) - \sum_{k<n} M(k)\big(w(k+1) - w(k)\big), \tag{24.2}$$

The increments $w(k+1) - w(k)$ vanish when $\mu(k+1) = 0$ and otherwise equal the next prime gap, so the right-hand side samples the preceding values $M(k)$ along the whole path. Moreover $Q(k) \sim 6k/\pi^2$ and the prime number theorem imply $w(k) \sim (6/\pi^2)k \log k$. This is offered as a non-endpoint direction, not as a proved source of new bounds; asymptotically its bulk is close to a classical weighted Mertens sum with weight $k \log k$.

25. Novelty status

The three zeta ingredients are classical. The spectral occurrence of the γ_k in $M(e^u)e^{-u/2}$ is also classical explicit-formula content, and $\kappa(h)$ is exactly a linear functional of the prime-gap autocorrelation sequence, not a new GUE-type statistic. GUE concerns zero spacings; the prime gaps entering Theorem 10.1 belong to a different arithmetic statistical problem.

A search did not locate the sequence S $(2,-2,-3,1,-5,3,-7,1,1,5,-11,1,\dots)$ as a named sequence or standard transform. Absence of a search result is weak evidence and the search was not exhaustive; this is recorded as an observation about the formulation, not a claim about its consequences, which by Proposition 8.1 follow from (M,Q). Likewise, the explicit ordinal-pair displacement d_j was not located as a standard named object in the searches performed for this project, but Proposition 14.2 and Theorem 14.3 show that its principal pointwise and quadratic bounds are reparametrisations of classical Mertens problems. Versions 8–9 further record negative structural results: universal coercive functionals can be evenised under the sign-flip involution (Proposition 16.5), and the natural two-cut survival kernel of the inverse matching collapses exactly to $R(u), R(v)$ and $v-u$ (Proposition 17.3). Version 8 also records that the first moment of the active lifetime multiset collapses to a sum of the same two-point kernel (Proposition 17.4), and that every finite ordered cut-count kernel reduces to its two extreme cuts by interval convexity (Proposition 17.5). The full lifetime multiset remains history-dependent, but the conditioned comparison in Experiment 17.6 is closely reproduced by an iid fair-sign walk on the measured range. The local increments $d_{j+1}-d_j$ therefore remain the principal inverse-pair statistic here with proved arithmetic content beyond the cut geometry. Version 9 adds the universal coding theorem, the square-root recovery identity, and the exact dynamical classification of $\varphi(n)=|S(n)|$. These statements are proved here from the defining transform and standard facts about μ, squarefree counting, and primes; no claim of priority is made without a dedicated literature search. Version 11 adds the exact fixed-level return formula, its one-sided negative excursion energy, and the exact decomposition obtained by replacing p_j by 1 or by j. The cubic coercive inequality is elementary after the return identity. It is recorded as a structural property of this transform, not as a new estimate for M, because the necessary independent upper control of the excursion energy is presently absent. The entropy table is explicitly a finite diagnostic and not a novelty claim. The shared-counter variant remains a direction rather than a result. The additional comparison in Section 27 distinguishes the rank-prime encoding from the classical analytic tools used to study it. In particular, monotone matching and its first absolute transport cost are not claimed as new. The nonpolynomial two-time duration observable retains order information, but an arithmetic estimate for its long weighted tail is not supplied by the fixed-lag correlation theorem.

26. Summary

$$\mu \longrightarrow (N_-, N_0, N_+) \longrightarrow S \longrightarrow SS, \qquad N_\pm = \frac{Q \pm M}{2}, \quad N_0 = n - Q, \quad Q \equiv M \ (2), \tag{26.1}$$

$$\boxed{SS(n) = \mathcal{P}\Big(\frac{Q+M}{2}\Big) - \mathcal{P}\Big(\frac{Q-M}{2}\Big) + n - Q} \tag{26.2}$$

$$\boxed{T(n) = SS(n) - n + Q(n) = \operatorname{sgn}(M) \sum_{j \in W(n)} p_j, \quad |W(n)| = |M(n)|, \ \text{centre } \tfrac{Q(n)+1}{2}} \tag{26.3}$$

$$\boxed{T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big), \quad n \geq 2, \quad \theta(n) \to 0} \tag{26.4}$$

$$\boxed{\text{RH} \iff T(n) = O_\varepsilon\big(n^{3/2+\varepsilon}\big) \ \ \forall \varepsilon > 0} \tag{26.5}$$

$$\boxed{\begin{aligned} T(n) &= M(n)\,p^{*}(n) + E_{\mathrm{mid}}(n),\\ E_{\mathrm{mid}}(n) &= \operatorname{sgn}(M)\sum_{i} w_i g_i,\\ \sum_{i} w_i^{2} &= \frac{h^{3}}{12}(1+O(h^{-1})). \end{aligned}} \tag{26.6}$$

$$\boxed{d_j = \beta_j - \alpha_j = R(\alpha_j) + R(\beta_j), \qquad \mathrm{RH} \iff d_j = O_\varepsilon(j^{1/2+\varepsilon}) \ \ \forall \varepsilon > 0} \tag{26.7}$$

$$\boxed{R(Q) = 0,\ J = Q/2: \qquad \mathcal{M}_2(Q) \le \mathcal{D}_2(J) \le 2\mathcal{M}_2(Q), \qquad \mathcal{D}_2 + \mathcal{E}_2 = 2\mathcal{M}_2} \tag{26.8}$$

$$\boxed{\mathrm{SD}(E_{\mathrm{mid}}) \ = \ \sigma_g(n)\sqrt{w^{\top} R\, w}\,, \qquad |E_{\mathrm{mid}}(n)| \approx 0.155\,|M(n)|^{3/2}\log n} \tag{26.9}$$

$$\boxed{\forall h \ge 1 \text{ fixed}: \qquad \sum_{q \le Q} \frac{\mu(s_q)\mu(s_{q+h})}{q} = o(\log Q)} \tag{26.10}$$

$$\boxed{\forall h \ge 1 \text{ fixed}: \qquad \frac{1}{\log Q}\sum_{q \le Q} \frac{(R(q+h)-R(q))^{2}}{q} \to h} \tag{26.11}$$

$$\boxed{D(e^{-t}) = \frac{2e^{-t}}{1-e^{-t}} \sum_{q \ge 0} e^{-tq/2} \sinh\!\left(\frac{tR(q)}{2}\right)} \tag{26.12}$$

$$\boxed{K(u,v) = \frac{\operatorname{sgn}(R(u)+R(v))}{2}\big(|R(u)+R(v)| - (v-u)\big)_{+}} \tag{26.13}$$

$$\boxed{SS(0), \dots, SS(\lfloor\sqrt{N}\rfloor), SS(N) \Longrightarrow Q(N) \Longrightarrow M(N) \quad \text{uniquely}} \tag{26.14}$$

$$\boxed{\mathrm{Per}(\varphi) = \{2,3,5,7,11,13,17,19,23,29\}, \qquad \varphi(n) = |S(n)|} \tag{26.15}$$

$$\boxed{x_{k+1} = \varphi(x_k),\ x_k \to \infty: \qquad x_{k+1} \sim \frac{3}{\pi^{2}} x_k \log x_k, \qquad x_k = k^{k+o(k)}} \tag{26.16}$$

$$\boxed{\varphi(p_j) = p_{f(j)}, \qquad f(j) = N_{-}(p_j) = j + C_{-}(p_j), \qquad \#(f(\mathbb{N}) \cap [1,J]) \sim \frac{\pi^{2}}{3}\frac{J}{\log J}.} \tag{26.17}$$

$$\boxed{\begin{aligned} SS(n) &\equiv n \pmod 2 \qquad (n \ge 2),\\ SS(n+2) = SS(n) &\iff |\Delta_j| = 1 \quad \text{for the corresponding integer pair.} \end{aligned}} \tag{26.18}$$

$$\boxed{R(a) = R(b) = h,\ b-a = 2r: \quad SS(s_b) - SS(s_a) = Z(a,b) + \sum_{t=1}^{r}(p_{A+t} - p_{B+t})} \tag{26.19}$$

$$\boxed{\text{complete negative excursion of depth } H: \qquad \mathcal{A} \ge \frac{H^{3}-H}{3}} \tag{26.20}$$

$$\boxed{SS_{\mathrm{pos}} = n - Q + M, \qquad SS_{\mathrm{ord}} = n - Q + \frac{Q+1}{2}M} \tag{26.21}$$

$$\boxed{\frac{SS - n + Q}{SS_{\mathrm{ord}} - n + Q} \sim \log\frac{Q}{2}, \qquad \frac{(SS - n + Q)^2}{(SS_{\mathrm{ord}} - n + Q)^2} \sim \log^2\frac{Q}{2}} \tag{26.22}$$

The gap representation and its prime-counting integral form are exact; the 0.155 scaling and distributional statements are measured over $10^4 \le n \le 10^6$ and are not proved. The packets are the excursions of $|M|$ amplified by the $h^{3/2}$ weight norm. Their zeta frequencies on $u = \log n$ are classical explicit-formula modes, and the tested standardised gap factor shows no detectable linear coupling to the sign or first zeta phase of M at current sensitivity.

The pointwise transform quantities $SS(n)$, $T(n)$, $E(n)$ and $E_{\mathrm{mid}}(n)$ are deterministic functions of $(M(n), Q(n))$. This proves that they contain no additional endpoint data. The complete SS trajectory is nevertheless losslessly identifying as a code for its underlying ternary sequence, and the genuine square-root prefix gives the separate recovery result of Theorem 18.3. It does not, by itself, rule out an analytic inequality obtained independently from the prime-gap representation; such an inequality would have to supply leverage not already equivalent to a bound on M. The finite coupling tests above find no such mechanism in the tested family. The shared-counter variant of Section 24 is genuinely non-endpoint as a functional of sign histories and is therefore outside this endpoint obstruction.

27. Research programme through alpha: exact pairing and analytic obligations

This independently readable chapter states the research programme, proves its intermediate identities, and compares its remaining obligations with established approaches. The comparison concerns the cited works, not every proposed RH proof. An equivalence theorem, a conditional theorem, and a speculative model are not all failed mathematics.

The active route is exact cancellation between sign-specific ordinal pairs, retaining unfinished pairs at every integer cutoff. No estimate of RH strength is assumed. The prime number theorem (PNT) and elementary squarefree counting are used explicitly as unconditional background. Statements are separated into proved results, classical transfers, unresolved targets, and excluded universal mechanisms.

27.1. **Step 0: independent notation and target.** Let μ be the usual Möbius function, p_j the jth prime, and $p_0 = 0$. For real $x \geq 1$ define the step functions

$$N_{\pm}(x) = \#\{n \leq x : \mu(n) = \pm 1\}, \quad Q(x) = N_+(x) + N_-(x), \quad M(x) = N_+(x) - N_-(x). \tag{27.1}$$

The transform and its cumulative sum are

$$S(n) = \begin{cases} p_{N_+(n)}, & \mu(n) = 1, \\ -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \end{cases} \qquad SS(x) = \sum_{n \leq x} S(n). \tag{27.2}$$

Set $w_j = p_j - j$ and $D_w(m) = \sum_{j=1}^{m} w_j$, with $D_w(0) = 0$. For $\mu(n) \neq 0$, let $j(n)$ be the occurrence number within its own sign. The index $j(n)$ is not used at nonsquarefree integers. Define

$$\alpha(s) = \sum_{n \geq 1} S(n) n^{-s}, \qquad \delta(s) = \sum_{\substack{n \geq 1 \\ \mu(n) \neq 0}} \mu(n) w_{j(n)} n^{-s}. \tag{27.3}$$

Let $s_1 < s_2 < \cdots$ enumerate the squarefree integers. Write

$$\begin{aligned} &\varepsilon_q = \mu(s_q), && R(q) = \sum_{\ell \leq q} \varepsilon_\ell = M(s_q), \qquad R(0) = 0, \\ &\alpha_j = \text{ordinal position of the } j\text{th } +1, && \beta_j = \text{ordinal position of the } j\text{th } -1, \\ &a_j = s_{\alpha_j}, && b_j = s_{\beta_j}, \\ &d_j = \beta_j - \alpha_j, && \Delta_j = b_j - a_j. \end{aligned} \tag{27.4}$$

The function $\alpha(s)$ differs from the ordinal α_j. The physical positions a_j, b_j avoid a clash with the analytic blocks $A(s), B(s)$. Never identify d_j, Δ_j, and w_j. The notation D_w also differs from a lifetime generating function $D(z)$. Finally set

$$c_0 = \frac{3}{\pi^2}, \qquad \Theta = \sup_{\rho \text{ nontrivial}} \Re\rho, \qquad \mathcal{H} = \{s : \Re s > 3/2\}. \tag{27.5}$$

The target is exclusion of nontrivial zeros with $\Re\rho > 1/2$. The functional equation supplies the other half of the strip.

27.2. **Step 1: decomposition and genuine pole statements.**

Proposition 27.1 (Exact decomposition and domains)**.** *The series in* (27.3) *converge absolutely for $\Re s > 2$, and in that domain*

$$\boxed{\alpha(s) = A(s) + B(s) + C(s) + \delta(s),} \tag{27.6}$$

where

$$A(s) = \zeta(s) - \frac{\zeta(s)}{\zeta(2s)}, \tag{27.7}$$

$$B(s) = c_0 \frac{s}{(s-1)\zeta(s-1)}, \tag{27.8}$$

$$C(s) = \frac{s}{2}\int_1^\infty M(x)(Q(x)+1-2c_0x)x^{-s-1}\,dx. \tag{27.9}$$

The blocks A, B are meromorphic on $\mathbb{C}$. The integral for C is holomorphic on $\mathcal{H}$ unconditionally; more precisely, its domain includes $\Re s > \Theta + 1/2$.

Proof. The bound $p_j \ll j\log(2j)$ majorises both coefficient sequences by $O(n\log(2n))$. The signed ordinal coefficients have the partial sum

$$\sum_{\substack{n\le x\\ \mu(n)\neq 0}} \mu(n)j(n) = \frac{N_+(N_++1) - N_-(N_-+1)}{2} = \frac{M(x)(Q(x)+1)}{2}. \tag{27.10}$$

Partial summation gives the integral of this expression against $sx^{-s-1}\,dx$. Splitting $Q+1 = 2c_0x + (Q+1-2c_0x)$ and using

$$\int_1^\infty M(x)x^{-z-1}\,dx = \frac{1}{z\zeta(z)} \qquad (\Re z > 1) \tag{27.11}$$

gives $B + C$. The zero row contributes A. The unconditional estimates $Q(x) = 2c_0x + O(\sqrt{x})$ and $|M(x)| \le x$ give local uniform absolute convergence of the C integral for $\Re s > 3/2$. The classical bound $M(x) = O_\epsilon(x^{\Theta+\epsilon})$ gives the refined domain [11]. □

Proposition 27.2 (Poles of the ordinal main block)**.** *A nontrivial zero ρ of ζ gives a pole of B at $1+\rho$ with the same order. For a simple zero,*

$$\operatorname*{Res}_{s=1+\rho} B(s) = c_0 \frac{1+\rho}{\rho\zeta'(\rho)}. \tag{27.12}$$

There is also a simple pole at $s = 1$ of residue $-2c_0$, and the shifted trivial-zero poles $s = 1-2k$, $k \ge 1$. The point $s = 2$ is a zero, not a pole.

Proof. Use the local factorisation of $\zeta(s-1)$ at its zeros. The prefactor $c_0 s/(s-1)$ is finite and nonzero at every shifted zero. At $s = 1$ use $\zeta(0) = -1/2$; at $s = 2$ use the pole of ζ at 1. □

No corresponding pole count for the full α has been proved. Critical-line zero theorems transfer to B, not automatically to α: continuation and noncancellation of the other blocks must first be checked. Holomorphy of δ alone does not remove a pole of B from α. Nor does extending δ across $3/2$ extend C across that line.

27.3. **Step 2: exact pairing without linearisation.**

Proposition 27.3 (Pairs, oriented integrals, and unit cuts)**.** *For $\Re s > 2$,*

$$\delta(s) = \sum_{j\ge 1} w_j(a_j^{-s} - b_j^{-s}). \tag{27.13}$$

For each pair and every complex s,

$$a_j^{-s} - b_j^{-s} = s\int_{a_j}^{b_j} x^{-s-1}\,dx. \tag{27.14}$$

With

$$u_q = \frac{q+R(q)}{2}, \quad v_q = \frac{q-R(q)}{2}, \quad U(q) = D_w(u_q) - D_w(v_q), \tag{27.15}$$

the initial half-plane has the representation

$$\boxed{\delta(s) = \sum_{q\geq 1} U(q)(s_q^{-s} - s_{q+1}^{-s}).} \tag{27.16}$$

Proof. Absolute convergence allows occurrence matching. Differentiating x^{-s} proves the oriented integral, including $s = 0$. Telescoping gives

$$a_j^{-s} - b_j^{-s} = \operatorname{sgn}(d_j) \sum_{q=\min(\alpha_j,\beta_j)}^{\max(\alpha_j,\beta_j)-1} (s_q^{-s} - s_{q+1}^{-s}). \tag{27.17}$$

At cut q the open labels are exactly $\min(u_q, v_q) < j \leq \max(u_q, v_q)$, all with orientation $\operatorname{sgn} R(q)$. Their signed weight is $U(q)$. The bound $|U(Q(x))| \ll x^2 \log(2x)$ gives an integrable absolute majorant $|s|x^{1-\Re s}\log(2x)$ when $\Re s > 2$, justifying the rearrangement. □

In particular,

$$U(q) = \operatorname{sgn} R(q) \sum_{j=\min(u_q,v_q)+1}^{\max(u_q,v_q)} w_j. \tag{27.18}$$

The kernel in (27.16) is local, but its coefficient still contains every long open pair. At one cut all these contributions have the same sign. Useful cancellation must occur across different cuts.

PNT already gives $\alpha_j, \beta_j \sim 2j$ and $a_j, b_j \sim j/c_0$. Thus $d_j = o(j)$ and $\Delta_j = o(j)$ unconditionally: eventually excluding a macroscopic relative separation does not require RH. This relative smallness supplies no square-root cancellation, and a Taylor argument would still need uniform summable errors. The exact formulas avoid that requirement entirely.

27.4. **Step 3: the nonpolynomial duration observable.** For this step take a balanced finite word of length $Q = 2J$, so $R(Q) = 0$. All its J pairs are complete. Define half-open cut intervals and duration tails:

$$I_j = [\min(\alpha_j, \beta_j), \max(\alpha_j, \beta_j)), \qquad V_J(r) = \sum_{j=1}^{J} (|d_j| - r)_+. \tag{27.19}$$

Proposition 27.4 (Two-cut identity)**.** *For $0 \leq r < Q$,*

$$\boxed{V_J(r) = \frac{1}{2} \sum_{q=0}^{Q-r-1} (|R(q) + R(q+r)| - r)_+.} \tag{27.20}$$

For $r \geq Q$ use an empty sum. In particular,

$$V_J(0) = \sum_{q=0}^{Q-1} |R(q)| \geq J. \tag{27.21}$$

For $0 < z < 1$,

$$\Phi_J(z) := \sum_{j=1}^{J} z^{|d_j|} = J - (1-z)V_J(0) + (1-z)^2 \sum_{r=1}^{Q-1} z^{r-1} V_J(r). \tag{27.22}$$

Proof. Using $N_\pm(q) = (q \pm R(q))/2$ in the ordinal clock, pairs opening with plus and crossing both cuts have labels $N_-(q+r) < j \leq N_+(q)$. Those opening with minus have labels $N_+(q+r) < j \leq N_-(q)$. Their disjoint counts give

$$C_r(q) = (N_+(q) - N_-(q+r))_+ + (N_-(q) - N_+(q+r))_+ = \tfrac{1}{2}(|R(q) + R(q+r)| - r)_+. \tag{27.23}$$

An interval of length ℓ contains both cuts for $(\ell-r)_+$ choices of q. Double counting proves (27.20). The case $r=0$ and $|d_j|\ge 1$ give (27.21). Finally, sum the finite identity

$$z^\ell = 1-(1-z)\ell+(1-z)^2\sum_{r=1}^{\ell-1} z^{r-1}(\ell-r) \tag{27.24}$$

over the pairs. □

Proposition 27.5 (Order survives a one-time height histogram)**.** *The words* $\mathrm{I}=++-+--+-$ *and* $\mathrm{II}=++--++--$ *have the same height histogram but different duration tails and transforms.*

Proof. Their height lists are $(1,2,1,2,1,0,1,0)$ and $(1,2,1,0,1,2,1,0)$: both contain two zeros, four ones, and two twos. Their durations are $(2,3,2,1)$ and $(2,2,2,2)$, respectively. Therefore

$$V_{\mathrm{I}}(2)=1,\quad V_{\mathrm{II}}(2)=0,\qquad \Phi_{\mathrm{I}}(z)-\Phi_{\mathrm{II}}(z)=z(1-z)^2>0. \tag{27.25}$$

Their quadratic duration sums also differ, 18 versus 16. Not every quadratic functional is insensitive to order. □

Order sensitivity is not yet arithmetic specificity. If F_+, F_- are the distribution functions of the empirical probability measures $J^{-1}\sum_j \delta_{\alpha_j}$ and $J^{-1}\sum_j \delta_{\beta_j}$, then

$$\frac{V_J(0)}{J}=\frac{1}{J}\sum_j|\alpha_j-\beta_j|=\int_{\mathbb{R}}|F_+(t)-F_-(t)|\,dt. \tag{27.26}$$

This is the classical one-dimensional monotone Wasserstein transport identity [19]. Neither monotone matching nor its first absolute cost is claimed as new.

27.4.1. *What fixed-lag arithmetic input actually controls.* The squarefree-clock two-point theorem obtained from Tao's logarithmic two-point theorem [9] gives, for every fixed $r\ge 1$,

$$\frac{1}{\log X}\sum_{q\le X}\frac{(R(q+r)-R(q))^2}{q}\longrightarrow r. \tag{27.27}$$

The clock transfer keeps a physical separation and a squarefree-sieve modulus fixed when applying the two-affine-form theorem. The error from omitted prime squares tends to zero as the sieve cutoff subsequently tends to infinity. Large physical separations have vanishing logarithmic tail because $\sum_{q\le X}(s_{q+r}-s_q)=O_r(X)$ by telescoping. Expanding the increment square then gives r diagonal terms and finitely many correlations. These successive limits do not give uniformity in growing r.

The precise relation to the duration observable is

$$\left|C_r(q)-\left(|R(q)|-\frac{r}{2}\right)_+\right|\le\frac{1}{2}|R(q+r)-R(q)|. \tag{27.28}$$

It follows from the 1-Lipschitz property of absolute value and positive part. Cauchy–Schwarz gives

$$\limsup_{X\to\infty}\frac{1}{\log X}\sum_{q\le X}\frac{|C_r(q)-(|R(q)|-r/2)_+|}{q}\le\frac{\sqrt{r}}{2}. \tag{27.29}$$

The unestimated main term depends on the absolute height. A theorem on fixed increments is therefore not automatically a bound for $V_J(r)$, even for fixed r. Moreover V_J is unsigned and lacks the weights w_j; both transfers to δ require additional argument.

Remark 27.6 (Height, area, and prime energy are different). Let $H=\max_{q\le Q}|R(q)|$. The H pairs active at a cut of height $\pm H$ each have duration at least H: for the tth active label, at least $H-t$ steps separate its opening from the cut and at least t steps separate the cut from its closing. Hence

$$V_J(r)\ge H(H-r)_+. \tag{27.30}$$

In particular $V_J(0)$ is an area, not a height. The bound $V_J(0) \ll_\epsilon J^{1/2+\epsilon}$ for every $\epsilon > 0$ is impossible because $V_J(0) \geq J$. A square-root-plus-ϵ height bound instead gives an area bound $O_\epsilon(J^{3/2+\epsilon})$. The cubic negative-excursion prime-separation energy is a different functional.

27.5. **Step 4: holomorphy for a fixed duration cutoff.**

Theorem 27.7 (Unconditional bounded-duration holomorphy)**.** *For every fixed $L \geq 1$ the series*

$$\delta_{\leq L}(s) = \sum_{|d_j| \leq L} w_j (a_j^{-s} - b_j^{-s}) \tag{27.31}$$

converges normally by pairs in $\Re s > 1$ and is holomorphic there. For $\sigma = \Re s > 1$,

$$\sum_{|d_j| \leq L} w_j |a_j^{-s} - b_j^{-s}| \ll L|s| \left(\frac{\log 2}{\sigma - 1} + \frac{1}{(\sigma-1)^2} \right). \tag{27.32}$$

The proof uses no Mertens estimate, density theorem, or linearisation.

Proof. At physical cut x, put $q = Q(x)$. An active ordinal interval of length at most L opened within the last L positions. There are at most L such intervals, and each has label $j \leq q \leq x$, so $w_j \ll x \log(2x)$. The exact integral and Tonelli give

$$\begin{aligned} \sum_{|d_j| \leq L} w_j |a_j^{-s} - b_j^{-s}| &\leq |s| \int_1^\infty x^{-\sigma-1} \sum_{\substack{|d_j| \leq L \\ \min(a_j,b_j) \leq x < \max(a_j,b_j)}} w_j \, dx \\ &\ll L|s| \int_1^\infty x^{-\sigma} \log(2x) \, dx. \end{aligned} \tag{27.33}$$

The last integral equals the parenthesis in (27.32). On any compact set use its minimum real part and maximum $|s|$ to obtain a summable majorant for the suprema of the paired terms. □

This theorem also holds for arbitrary sign-occurrence lists in an increasing integer clock and weights $|w_j| \ll j \log(2j)$. Its generality is useful, but it is not an arithmetic cancellation theorem.

For $\Re s > 2$, dominated convergence gives $\delta_{\leq L} \to \delta$ locally uniformly as $L \to \infty$. The displayed bound grows with L; it does not justify this limit on $\mathcal{H}$.

27.6. **Step 5: long pairs and the sign involution.** Large rank and large duration are different: an alternating word has $|d_j| = 1$ at every rank, however large w_j becomes.

Proposition 27.8 (Universal antisymmetry does not delete the long tail)**.** *Sign reversal and ordinal matching alone do not justify discarding all pairs of duration greater than a fixed L.*

Proof. For $H > L$, take the auxiliary balanced word $(+1)^H(-1)^H$ on any increasing positive clock. Every duration equals H, but for real $\sigma > 0$,

$$\delta_{\leq L}(\sigma) = 0, \qquad \sum_{j=1}^{H} w_j (s_j^{-\sigma} - s_{H+j}^{-\sigma}) > 0. \tag{27.34}$$

There are no short pairs. This disproves the claimed universal deletion, not a possible cancellation specific to the genuine Möbius word. □

The global sign flip gives

$$R \mapsto -R, \qquad d_j \mapsto -d_j, \qquad \delta(s) \mapsto -\delta(s). \tag{27.35}$$

It has no action on s and gives no equation under $s \mapsto 3 - s$. The zero row of α is unchanged, so the whole transform does not simply change sign either.

If a universal functional satisfies $F(\varepsilon) \geq c\,\Psi(\|d(\varepsilon)\|_\infty)$ for both signs, then its evenisation $[F(\varepsilon) + F(-\varepsilon)]/2$ has the same lower bound. Thus odd cancellation alone cannot explain such universal coercivity. This does not exclude all even, nonpolynomial, or arithmetic-specific functionals.

27.7. **Step 6: the precise original-order target.** For integer X let

$$\delta_X(s) = \sum_{j\ge 1} w_j(\mathbf{1}_{a_j\le X}a_j^{-s} - \mathbf{1}_{b_j\le X}b_j^{-s}). \tag{27.36}$$

This is a finite sum in the original integer ordering.

27.7.1. *The unfinished-pair term.* Put $m_X = \min(N_+(X), N_-(X))$ and $h_X = |N_+(X) - N_-(X)|$. Exactly,

$$\delta_X(s) = \sum_{j\le m_X} w_j(a_j^{-s} - b_j^{-s}) + E_X(s), \tag{27.37}$$

where

$$E_X(s) = \begin{cases} \displaystyle\sum_{j=m_X+1}^{m_X+h_X} w_j a_j^{-s}, & N_+(X) > N_-(X), \\ \displaystyle -\sum_{j=m_X+1}^{m_X+h_X} w_j b_j^{-s}, & N_-(X) > N_+(X), \\ 0, & N_+(X) = N_-(X). \end{cases} \tag{27.38}$$

At a real argument all terms of E_X have the same sign. Convergence along balanced prefixes or after completing every pair does not justify convergence at arbitrary integer cutoffs.

27.7.2. *Unresolved Cauchy estimate.* The central target is, for every compact $K \subset \mathcal{H}$,

$$\boxed{\lim_{X\to\infty} \sup_{\substack{Y\in\mathbb{N}\\ Y\ge X}} \sup_{s\in K} |\delta_Y(s) - \delta_X(s)| = 0.} \tag{27.39}$$

A concrete sufficient estimate is $|\delta_Y(s) - \delta_X(s)| \le \eta_K(X)$ for all such Y, s, where an independently controlled $\eta_K(X)$ tends to zero. Complete short pairs, complete long pairs, and pairs cut by either endpoint must be handled with compatible uniformity.

This is locally uniform convergence of original partial sums, not absolute convergence of individual integer terms, whose threshold remains 2. A different grouping that gives analytic continuation is not automatically a proof of (27.39).

27.8. **Step 7: the rigorous transfer to RH.**

Proposition 27.9 (The exact coefficient sum and its scale)**.** *Let* $W(x) = \sum_{n\le x,\ \mu(n)\ne 0} \mu(n) w_{j(n)}$. *Then*

$$W(x) = D_w(N_+(x)) - D_w(N_-(x)) = SS(x) - \lfloor x\rfloor + Q(x) - \frac{M(x)(Q(x)+1)}{2}. \tag{27.40}$$

There is $e(x) \to 0$ *such that, for sufficiently large* x,

$$W(x) = c_0 x \log x\, M(x)(1 + e(x)), \tag{27.41}$$

with both sides zero when $M(x) = 0$.

Proof. The difference of cumulative weights gives the first identity, and (27.10) gives the second. PNT implies $Q(x) \sim 2c_0 x$ and $M(x) = o(x)$, hence both counters are $c_0 x(1 + o(1))$. Uniformly between them, $w_j = p_j - j \sim j\log j \sim c_0 x \log x$. The block has signed length $M(x)$. This uses PNT, not an RH-sized bound. □

Theorem 27.10 (Original-order criterion and RH)**.** *The Cauchy target* (27.39) *is equivalent to RH. It is also equivalent to convergence of the ordinary series* $\delta(\sigma)$ *for every real* $\sigma > 3/2$.

Proof. Real convergence and Kronecker's lemma give $W(x) = o(x^\sigma)$. Consequently (27.41) gives

$$M(x) = o\left(\frac{x^{\sigma-1}}{\log x}\right). \tag{27.42}$$

Taking $\sigma = 3/2 + \epsilon/2$ for each $\epsilon > 0$ yields the Littlewood criterion for RH [7, 11]. Its needed direction follows because these estimates make $z\int_1^\infty M(x)x^{-z-1}\,dx$ holomorphic for $\Re z > 1/2$. It agrees with $1/\zeta(z)$ for $\Re z > 1$, so no zero is possible to the right of $1/2$; symmetry completes RH.

Conversely, under RH, $M(x) = O_\eta(x^{1/2+\eta})$ gives $W(x) = O_\eta(x^{3/2+\eta}\log x)$. For compact $K \subset \mathcal{H}$ choose $0 < \eta < \min_{s\in K}(\Re s - 3/2)$. Partial summation gives exactly

$$\delta_Y(s) - \delta_X(s) = W(Y)Y^{-s} - W(X)X^{-s} + s\int_X^Y W(x)x^{-s-1}\,dx. \tag{27.43}$$

All terms tend to zero uniformly in $Y \geq X$ and $s \in K$. □

This is a classical transfer in new coefficients, not a proof of its unresolved premise. In the forward direction the Mertens estimate is a conclusion. Assuming it while proving the premise would be circular.

The standard abscissa formula and the classical identification of the Mertens growth exponent with Θ also yield

$$\sigma_c(\alpha) = \sigma_c(\delta) = 1 + \Theta. \tag{27.44}$$

Indeed W has growth exponent $1 + \Theta$; the signed prime part of SS has the same exponent, whereas $\lfloor x \rfloor - Q(x) = O(x)$. The abscissae refer to the original ordering, not to arbitrary paired rearrangements. An abscissa is not automatically a line of poles.

The ordinary series $\alpha(3/2)$ diverges. Otherwise Kronecker's lemma would give $SS(x) = o(x^{3/2})$ and the same signed prime-block asymptotic would force $M(x) = o(\sqrt{x}/\log x)$, contrary to $M(x) = \Omega_\pm(\sqrt{x})$ [4, 5]. The stronger constants of the Mertens-conjecture disproof are unnecessary. Divergence of this series does not decide whether a continuation could have a finite value at the real point $3/2$.

27.9. **Step 8: the separate continuation-and-poles route.** The meromorphic route has a different sufficient objective:

$$\boxed{\begin{aligned}&\text{RH} \iff B \in \mathcal{O}(\mathcal{H})\\ &\quad\iff \alpha - \delta \text{ admits holomorphic continuation to } \mathcal{H}.\end{aligned}} \tag{27.45}$$

The first equivalence follows from Proposition 27.2 and zeta symmetry; the second uses $A, C \in \mathcal{O}(\mathcal{H})$ and (27.6), initially on $\Re s > 2$. Separate holomorphy of both α and δ suffices; holomorphy of δ alone does not.

Subtracting δ leaves the zero row and the ordinal part. Without a new independent estimate, this returns to a classical shifted reciprocal-zeta problem. There is no assumed global bijection between the poles of the full α and the zeros of ζ.

27.9.1. *8A: contours.* A contour route must establish continuation, control away from poles, and a count excluding poles. Uniform polynomial growth on a strip containing poles is impossible without removing their neighbourhoods or their principal parts. Furthermore,

$$\frac{1}{2\pi i}\oint_\Gamma \frac{\alpha'}{\alpha}\,ds = N_{\text{zeros}}(\Gamma) - N_{\text{poles}}(\Gamma). \tag{27.46}$$

A zero integral is not a zero pole count. A lower bound on a separating curve does not supply the missing interior count. For RH it suffices to work in the open half-plane $\mathcal{H}$; a strip crossing its boundary is not intrinsically necessary.

27.9.2. *8B: positivity and the correctly shifted Li map.* One first needs a completed function with proved divisor, controlled growth, and an appropriate convergent product. For candidate poles $\varpi = 1 + \rho$, the transferred expression is

$$\lambda_n^{\alpha} = \lim_{T\to\infty} \sum_{\substack{\varpi=1+\rho \\ |\Im\varpi|\le T}} \left[1 - \left(\frac{\varpi - 2}{\varpi - 1}\right)^n\right], \tag{27.47}$$

with multiplicities and the displayed summation convention. The map is not an arbitrary $1-\varpi_0/\varpi$:

$$\left|\frac{\varpi - 2}{\varpi - 1}\right| = 1 \quad\Longleftrightarrow\quad \Re\varpi = \frac{3}{2}. \tag{27.48}$$

Li's theorem gives a criterion, not automatic positivity [12]. An ordinal implementation must derive the relevant form and prove every required inequality. Generic Gram positivity is insufficient. The sign involution supplies neither the divisor nor the completion.

27.9.3. *8C: spectral realisation.* A spectral programme would require:

(i) a specified Hilbert space and densely defined self-adjoint operator;
(ii) a justified determinant, regularised when necessary;
(iii) an exact relation between its divisor and the relevant analytic divisor;
(iv) an affine spectral map taking real spectrum to $\Re s = 3/2$.

Reality of a spectrum matters only after the divisor relation is proved. An arbitrary real-spectrum matrix, or an operator defined using already known zero ordinates, does not provide that relation independently.

27.10. **Step 9: corrected rules and excluded shortcuts.**

R1. Do not assume an RH-equivalent estimate for M, SS, W, d_j or a corresponding abscissa while trying to prove it. An independently proved equivalent assertion is not circular; neither is an unconditional bound.
R2. Distinguish original-order convergence, absolute convergence by pairs, and analytic continuation. Retain unfinished-pair boundary terms.
R3. Use the exact integral (27.14). The presence of a displacement in an identity is not circular; assuming an RH-sized bound for it would be.
R4. Do not transfer poles or a functional equation from a single summand to α. Coefficient size alone does not force a symmetry line.
R5. A power-saving prime-sum remainder requires a specified main term. A finite inverse-logarithm expansion does not supply it. This obstructs that approximation scheme, not every possible analytic continuation.
R6. A prime-index block of length $|M(x)|$ is not automatically a physical prime interval of length $\sqrt{x}$. No argument here identifies a single missing ϵ in a prime-interval bound as the sole obstacle.
R7. Fixed-lag logarithmic cancellation does not estimate the absolute height in (27.28), growing lags, or the full signed weighted tail without further proof.
R8. An identity valid for every sign word is not by itself arithmetic specificity. Auxiliary words test a universal mechanism; they are not asserted to be the Möbius sequence.
R9. Quadratic comparison theorems prove their stated comparisons only. Proposition 27.5 refutes the claim that every quadratic functional loses order.
R10. The off-boundary assertion

$$\int_1^{\infty} M(x)^2 x^{-2\sigma-1}\,dx < \infty \quad \text{for every } \sigma > 1/2 \tag{27.49}$$

is equivalent to RH without simplicity of zeros. RH implies finiteness by choosing a smaller exponent. Conversely, Cauchy–Schwarz makes the Mellin integral of M analytic to the right

of 1/2. The stronger weak Mertens condition $\int_1^X M(x)^2 x^{-2}\,dx = O(\log X)$ has additional simplicity and negative-moment consequences [4]. The active route here does not assume a quadratic estimate.

R11. Balanced finite words justify combinatorial identities, not replacement of all integer cutoffs by balanced cutoffs in an infinite analytic argument.

27.11. **Step 10: comparison with established research programmes.** Each comparison separates what the cited work proves from what would still be needed for RH. The missing implication identified here is a logical assessment, not a claim that the programme is impossible or that its authors' valid theorem is erroneous.

27.11.1. *10A: Mertens criteria and conditional moments.* The Mellin identity (27.11) and the Littlewood criterion connect cancellation of μ to zero exclusion [7, 11]. Replacing a Mertens bound by one for SS or W does not independently prove it. The additional occurrence-pair representation is the distinction here, not a weaker convergence threshold.

Ng's distributional and moment theorems explicitly use RH and additional negative-moment hypotheses [4]. Using their conclusions without those hypotheses to prove RH would reverse the logical direction. Here finite pairing identities and bounded-duration holomorphy are unconditional, but the infinite-tail estimate remains missing.

27.11.2. *10B: Nyman–Beurling and Báez-Duarte.* Let $\{y\} = y - \lfloor y \rfloor$. In $L^2(0,\infty)$ set

$$\rho_a(x) = \left\{ \frac{1}{ax} \right\}, \qquad \mathcal{B}_{\mathbb{N}} = \operatorname{span}\{\rho_a : a \in \mathbb{N}\}. \tag{27.50}$$

Báez-Duarte's strengthening makes RH equivalent to $\mathbf{1}_{(0,1)} \in \overline{\mathcal{B}_{\mathbb{N}}}$ [14]. The missing RH step is the approximation, not nonnegativity of a squared norm.

The present route has two actual occurrence clocks and fixed prime weights rather than freely selected approximating coefficients. Its target is nevertheless an infinite convergence statement. An arithmetic definition without mentioning zeta zeros is not unique to this construction, nor does it remove that obligation.

27.11.3. *10C: Li–Weil positivity.* Bombieri and Lagarias relate Li's criterion to general inequalities for complex multisets and to the Guinand–Weil explicit formula [13]. A positivity reformulation need not be specific to the original zeta notation. The unsupplied step is positivity for every required index or admissible test.

Ordinal pairing gives explicit finite kernels, but their automatic Gram nonnegativity is not the needed arithmetic positivity. No equality between an ordinal form and the required Li–Weil form is established here. The shift $\rho \mapsto 1 + \rho$ in (27.47) is an exact coordinate change, not a new positivity mechanism.

27.11.4. *10D: Hilbert–Pólya, Berry–Keating, and Connes.* Berry and Keating study an xp-based spectral interpretation and explicitly identify substantial speculative gaps [15]. Matching a spectral density or a semiclassical count would not identify every eigenvalue with a zeta zero. The exact divisor relation is essential.

Connes supplies a different framework: critical zeros appear as an absorption spectrum, and hypothetical off-critical zeros as resonances. His work reduces RH to a trace-formula assertion [16]. The existence of that spectral interpretation does not already exclude the resonances.

Here the starting data are discrete occurrence lists and an exact finite summation identity, without postulating an operator. This changes the starting object but supplies no self-adjoint realisation. Adopting a spectral route would still require all four obligations of Step 8C.

27.11.5. *10E: de Bruijn–Newman deformation.* In the de Bruijn–Newman family all zeros are real precisely beyond a threshold Λ. RH is equivalent to $\Lambda \leq 0$. Rodgers and Tao prove $\Lambda \geq 0$ [17]. That proves Newman's conjecture, not the complementary inequality needed for RH; it is not a defect in their theorem.

The present duration cutoff L is not a heat-flow parameter. Holomorphy for every fixed L does not give monotonic zero motion or a threshold theorem. Any such additional structure would need a proof.

27.11.6. *10F: local correlations and short intervals.* Tao's logarithmic two-point theorem and the short-interval results of Matomäki and Radziwiłłestablish genuine arithmetic cancellation in their respective regimes [9, 8]. Neither states the weighted original-order Cauchy estimate (27.39). Equation (27.28) identifies a specific unestimated height term, so describing the observable as local does not license an exchange of limits.

The distinctive interface is an exact relation between matched-pair tails and two-time heights. The inherited theorem controls its increment error, not automatically its absolute height, sign, rank weight, or full tail.

27.11.7. *10G: rigorous computation and statistical models.* Platt and Trudgian rigorously verify RH through height $3 \cdot 10^{12}$ [18]. Their theorem certifies that finite interval; it does not quantify over all heights.

Likewise, the present finite-word checks validate finite identities, and the numerical controls test mechanisms over a bounded range. Neither bounds the supremum over all $Y \geq X$ in (27.39). A statistical resemblance, or its absence, does not supply a deterministic infinite-tail estimate.

27.12. **Step 11: precise distinctiveness and limits of novelty.** The combination studied here has six demonstrated features:

(i) Ordinary primes are attached to separately numbered positive and negative occurrences of μ, with an explicit zero row.
(ii) Position, ordinal growth, and prime displacement separate exactly: the ordinal partial sum is $M(Q+1)/2$, and the excess is $D_w(N_+) - D_w(N_-)$.
(iii) Integer-clock, squarefree-clock, inverse-pair, and active-cut descriptions have exact transition identities.
(iv) Nonpolynomial duration observables distinguish paths having the same one-time height histogram.
(v) Every fixed-duration paired subseries is holomorphic for $\Re s > 1$, with an explicit cutoff-dependent majorant.
(vi) The original-order boundary term and the two-time height remainder remain visible instead of being suppressed by a rearrangement.

These features describe this construction; they do not establish priority for each ingredient. Monotone transport, layer counting, Mellin transforms, partial summation, convergence-abscissa formulae, and positivity criteria have classical antecedents. The sign flip is generic, and the bounded-duration theorem applies to much more than the Möbius word.

The complete transform is a lossless encoding, not additional independent arithmetic data: $\mu(n) = 0$ is detected by $S(n) = 1$, and otherwise $\mu(n)$ is the sign of $S(n)$. At one endpoint, $SS(n)$ depends only on $(M(n), Q(n))$. Duration data distinguish different histories, not two copies of an identical full input trajectory.

The defensible difference from the cited programmes is the particular rank-prime encoding and its exact pair/cut interfaces. No exhaustive literature theorem establishes a difference from every other construction, and no theorem here proves the remaining RH-equivalent estimate easier than the classical one. Its potential value would be a new, independently controlled arithmetic mechanism at those interfaces; that mechanism is still unresolved.

27.13. **Step 12: complete execution order and present status.**

P1. Fix clocks, weights, and finite-cut conventions: completed.
P2. Establish the initial series and $A + B + C + \delta$: completed.
P3. Separate poles of B from unproved claims about α: completed.
P4. Prove the exact pair integral and unit-cut reorganisation: completed.
P5. Prove the duration identity and order-sensitive example: completed.
P6. Determine the scope of the fixed-lag theorem: completed; an absolute-height remainder is left unestimated.
P7. Prove fixed-duration holomorphy: completed.
P8. Exclude universal tail deletion by sign antisymmetry: completed; Möbius-specific cancellation is not excluded.
P9. Establish weighted signed cancellation uniformly over durations, including unfinished pairs: unresolved.
P10. Deduce (27.39): unresolved because P9 is missing.
P11. Apply Theorem 27.10 only after proving its premise: the implication is proved; the premise is not.
P12. Alternatively prove (27.45). Contours, positivity, and spectral constructions are admissible only with their additional obligations verified; none is a completed substitute.

The next mathematical work is P9, not declaring a pole line or promoting a finite computation into an infinite bound. This chapter is an audited research programme with explicit intermediate results, not a proof of RH.

27.14. **Step 13: the energy of SS and its exact critical threshold.** For real x, the correct step-function subtraction is

$$T(x) = SS(x) - \lfloor x \rfloor + Q(x) = \mathcal{P}(N_+(x)) - \mathcal{P}(N_-(x)), \qquad \mathcal{P}(m) = \sum_{j \le m} p_j. \tag{27.51}$$

Define the nonnegative, possibly infinite quantity

$$\boxed{\mathcal{I}(\sigma) = \int_1^\infty \left(SS(x) - \lfloor x \rfloor + Q(x)\right)^2 x^{-2\sigma-1}\, dx.} \tag{27.52}$$

Using x instead of $\lfloor x \rfloor$ changes T by $-\{x\}$. Since $\{x\}$ is bounded, this change does not affect finiteness of (27.52) for $\sigma > 0$.

Proposition 27.11 (Energy comparison, including the logarithmic weight)**.** *For sufficiently large $x_0 > 1$ and any real σ,*

$$\mathcal{I}(\sigma) < \infty \quad \Longleftrightarrow \quad \int_{x_0}^\infty M(x)^2 (\log x)^2 x^{-2(\sigma-1)-1}\, dx < \infty. \tag{27.53}$$

More precisely, the two nonnegative integrands have asymptotic ratio $c_0^2 = 9/\pi^4$, wherever $M(x) \neq 0$, and both vanish when $M(x) = 0$.

Proof. The exact prime block in (27.51) has $|M(x)|$ terms. PNT gives $M(x) = o(x)$, $Q(x) \sim 2c_0 x$, and $p_j \sim j \log j$. Every index in the block is therefore $(1 + o(1))c_0 x$, uniformly over that block. Consequently

$$T(x) = c_0 x \log x\, M(x)(1 + o(1)). \tag{27.54}$$

This is a two-sided comparison in absolute value for all sufficiently large x, with the zero case treated exactly. Squaring and integrating proves the assertion. The finite initial interval causes no issue. □

Theorem 27.12 (A zero forces energy divergence even at its endpoint)**.** *If $\zeta(\rho) = 0$ for a nontrivial zero $\rho = \beta + i\gamma$, then*

$$\boxed{\mathcal{I}(\sigma) = \infty \qquad (\sigma \le 1 + \beta).} \tag{27.55}$$

In particular, unconditionally $\mathcal{I}(3/2) = \infty$.

Proof. Suppose instead that $\mathcal{I}(1 + \beta) < \infty$. By Proposition 27.11,

$$J_\beta := \int_{x_0}^{\infty} M(x)^2 (\log x)^2 x^{-2\beta-1}\, dx < \infty. \tag{27.56}$$

For $\varepsilon \ge 0$, Cauchy–Schwarz gives the uniform estimate

$$\begin{aligned} \left| \int_{x_0}^{\infty} M(x) x^{-\beta-\varepsilon-i\gamma-1}\, dx \right| &\le J_\beta^{1/2} \left(\int_{x_0}^{\infty} \frac{x^{-2\varepsilon-1}}{(\log x)^2}\, dx \right)^{1/2} \\ &\le \frac{J_\beta^{1/2}}{\sqrt{\log x_0}}. \end{aligned} \tag{27.57}$$

The same bound with absolute values inside the first integral proves absolute convergence at the boundary. The full integral $F(z) = \int_1^\infty M(x) x^{-z-1}\, dx$ is holomorphic for $\Re z > \beta$ and remains bounded as $z = \rho + \varepsilon$ tends to ρ from the right. On $\Re z > 1$, it equals $1/(z\zeta(z))$. The identity theorem, applied to $z\zeta(z)F(z) = 1$ away from $z = 1$, propagates that identity throughout $\Re z > \beta$. It either already excludes a zero in this open half-plane, or contradicts boundedness when $z \to \rho$. Thus the endpoint integral diverges. For $\sigma \le 1 + \beta$, its integrand is at least that of the endpoint because $x \ge 1$. Existence of a critical-line zero gives the final claim. □

Corollary 27.13 (RH expressed as the energy of SS)**.**

$$\boxed{\text{RH} \iff \mathcal{I}(\sigma) < \infty \text{ for every } \sigma > 3/2.} \tag{27.58}$$

More generally,

$$\inf\{\sigma \in \mathbb{R} : \mathcal{I}(\sigma) < \infty\} = 1 + \Theta. \tag{27.59}$$

Proof. Under RH, $M(x) \ll_\epsilon x^{1/2+\epsilon}$. For a given $\sigma > 3/2$, choose $\epsilon < \sigma - 3/2$ and use (27.53). Conversely, a zero with $\beta > 1/2$ contradicts the claimed finiteness at any $3/2 < \sigma \le 1 + \beta$, by Theorem 27.12. The functional equation then gives RH. The same argument with $M(x) \ll_\epsilon x^{\Theta+\epsilon}$ proves (27.59). If the supremum Θ is attained by a zero, the integral diverges at $1+\Theta$; no endpoint assertion is inferred from a supremum that is not known to be attained. □

This proof transfers through M using an explicit two-sided comparison; it does not assume a pole correspondence for the Mellin transform of T. Indeed the exact initial-half-plane identity is

$$s \int_1^\infty T(x) x^{-s-1}\, dx = \alpha(s) - A(s) = B(s) + C(s) + \delta(s). \tag{27.60}$$

The term δ cannot be silently included in a holomorphic remainder. The logarithmic factor is also useful: it makes the boundary Cauchy–Schwarz integral $\int dx/(x \log^2 x)$ finite in (27.57). Off-boundary finiteness for every $\sigma > 3/2$ does not impose simplicity of zeta zeros; it is different from a bounded critical mean-square condition.

27.15. **Step 14: explicit poles and an attempted isolation of delta's singularities.** This step separates the meromorphic blocks from singularity claims that still depend on the remainder.

27.15.1. *14A: what the two explicit zeta blocks actually imply.* The pole statement for B is Proposition 27.2. For A, the pole at $s = 1$ has residue $1 - 1/\zeta(2)$. At $s_0 = \rho/2$, where ρ is a nontrivial zero of multiplicity m, the potential pole order is

$$\max\{m - \operatorname{ord}_{s=\rho/2} \zeta(s), 0\}. \tag{27.61}$$

For a simple zero with $\zeta(\rho/2) \neq 0$ its residue is

$$\operatorname*{Res}_{s=\rho/2} A(s) = -\frac{\zeta(\rho/2)}{2\zeta'(\rho)}. \tag{27.62}$$

At negative odd integers A has simple poles arising from $\zeta(2s)$; at negative even integers the numerator's simple zero cancels the denominator's simple zero. Both $s = 0$ and $s = 1/2$ are regular. These are poles of the displayed block, not assertions about their survival in α outside the established common domain. In particular, claiming that all poles lie on $3/2$ would already require specifying nontrivial poles and proving what happens to the other blocks.

27.15.2. *14B: a differentiated ordinal block.* The partial sum of the coefficients of δ is exactly

$$W(x) = D_w(N_+(x)) - D_w(N_-(x)) = T(x) - \frac{M(x)(Q(x)+1)}{2}. \tag{27.63}$$

For $\Re s > 2$, introduce

$$F_\zeta(z) = \frac{1}{z\zeta(z)}, \qquad P_\delta(s) = -c_0 s F'_\zeta(s-1) = \frac{B(s)}{s} - B'(s). \tag{27.64}$$

Differentiation under the absolutely convergent Mellin integral gives the exact decomposition

$$\delta(s) = P_\delta(s) + E_\delta(s), \tag{27.65}$$

$$E_\delta(s) = s\int_1^\infty (W(x) - c_0 x \log x\, M(x)) x^{-s-1}\, dx. \tag{27.66}$$

The factor $B(s)/s$ has its natural removable interpretation at $s = 0$.

Proposition 27.14 (Explicit higher-order pole candidates)**.** *At $s_0 = 1 + \rho$, if ρ is a zero of multiplicity m, the explicit meromorphic function P_δ has a pole of order $m+1$. For a simple zero, with $r_\rho = c_0(1+\rho)/(\rho\zeta'(\rho))$,*

$$P_\delta(s) = \frac{r_\rho}{(s-s_0)^2} + \frac{r_\rho/s_0}{s-s_0} + O(1). \tag{27.67}$$

Proof. Insert $B(s) = r_\rho/(s-s_0) + b_0 + O(s-s_0)$ into $B(s)/s - B'(s)$. In higher multiplicity, differentiation raises the leading pole order by one with a nonzero leading coefficient. □

This locates poles of P_δ, and supplies a precise candidate principal part for δ. It does not establish any pole of δ. For example, holomorphy of E_δ in a neighbourhood of s_0 would make (27.67) the principal part of δ there. No such neighbourhood is presently proved. The comparison $W(x) = c_0 x \log x\, M(x)(1+o(1))$ alone gives no power saving in the Mellin remainder. The integral (27.66) is justified for $\Re s > 1 + \Theta$ by the classical growth bound, but this does not cross the candidate singularities.

For orientation, the first five positive critical ordinates give the following numerical coefficients. These are evaluations of the explicit blocks, not a computation of a continuation of δ.

γ in $\rho = 1/2 + i\gamma$	$\Re r_\rho$	$\Im r_\rho$
14.1347251417	0.3751535752	−0.0871438181
21.0220396388	0.2639742227	0.0461618613
25.0108575801	0.2065891363	−0.0811247572
30.4248761259	0.2042801902	0.1128180994
32.9350615877	0.1811319824	−0.1250906057

At each listed $s_0 = 3/2 + i\gamma$, r_ρ is the residue of B and the double-pole coefficient of P_δ. The accompanying code evaluates the Laurent limits independently.

27.15.3. *14C: why a relative asymptotic does not prove meromorphy.* Here is an explicit control example, unrelated to any assumed formula for the true W. For $x \geq e$ take $W_{\mathrm{mod}}(x) = x^{s_0}(\log x + \log\log x)$. The second term is $o(x^{s_0}\log x)$ in modulus. Nevertheless, putting $z = s - s_0$ and $\Re z > 0$ gives

$$\begin{aligned}\int_e^\infty W_{\mathrm{mod}}(x)x^{-s-1}\,dx &= \int_1^\infty e^{-zu}(u+\log u)\,du \\ &= \frac{e^{-z}(1+z)}{z^2} + \frac{E_1(z)}{z},\end{aligned} \tag{27.68}$$

where the exponential integral satisfies $E_1(z) = -\gamma_E - \log z + O(z)$ near zero in a slit domain [21]. Thus a double-pole main term can coexist with a logarithmic branch singularity in a relatively smaller remainder. An actual meromorphic continuation for δ must exclude such possibilities by additional information, rather than by interpreting an asymptotic logarithmic weight as an exact pole order.

27.16. **Step 15: audit of five routes suggested by the Connes survey.** The source for this comparison is Connes's 2026 survey, specifically Sections 2.1.3, 2.2.3–2.2.4, 4.1, and 6.4–6.6 [20]. Its conditional programmes and proved equivalences are not failed theorems. The following distinctions determine which parts transfer to the present object.

27.16.1. *15A: stability of poles is local and conditional.* If F is meromorphic near s_0 and H is holomorphic there, the negative powers of the Laurent series of $F+H$ equal those of F. This elementary fact proves stability of every pole and its principal part under a holomorphic additive perturbation. The Borchsenius–Jessen result discussed in the survey illustrates very different behaviour for zeros after adding a nonzero constant. It does not prove that every conceivable perturbation destroys a zero alignment.

For $\alpha = B + (A + C + \delta)$, the stability argument is available precisely where $A + C + \delta$ is known to be holomorphic. It supplies no holomorphy of that perturbation and no unconditional bijection from the zeros of ζ to the poles of α. The exact pole bijection currently belongs to B.

27.16.2. *15B: Voronin universality does not forbid a polynomial lower bound.* Voronin's theorem gives approximation to fixed nonvanishing targets on suitable compact subsets of $1/2 < \Re z < 1$ by vertical translates of ζ. The approximation tolerance and the translation height are different quantifiers. Qualitative universality does not assert an error smaller than every inverse power of that height.

There is a concrete countercheck to the proposed obstruction. Euler summation gives

$$\zeta(z) = \frac{z}{z-1} - z\int_1^\infty \{x\}x^{-z-1}\,dx \qquad (\Re z > 0,\ z \neq 1). \tag{27.69}$$

On $3/2 + \eta \leq \Re s \leq 2 - \eta$, for $0 < \eta < 1/4$ and $|\Im s| \geq 1$, it implies $|\zeta(s-1)| \leq C_\eta(1 + |\Im s|)$. Hence at every regular point of B in this closed substrip,

$$\boxed{|B(s)| \geq \frac{c_\eta}{1+|\Im s|}.} \tag{27.70}$$

Thus the factor containing $1/\zeta(s-1)$ actually has a polynomial lower bound compatible with universality. This proves nothing comparable for the sum α, where cancellation between summands is possible. It does disprove the assertion that universality alone closes the entire contour route. The other contour obligations of this chapter remain.

27.16.3. *15C: a finite prime window is not yet Weil positivity.* Use the symmetric multiplicative convention of the survey's Section 4.1: $g^*(x) = \overline{g(x^{-1})}$, Haar measure dx/x, and Mellin–Fourier transform $\widehat{g}(t) = \int_0^\infty g(x)x^{-it}\,dx/x$. The Weil criterion has the form

$$\sum_v W_v(g * g^*) \le 0 \quad \text{for all admissible } g, \qquad \widehat{g}(i/2) = \widehat{g}(-i/2) = 0, \tag{27.71}$$

with the survey's sign convention. Its nonarchimedean terms are

$$W_p(f) = (\log p) \sum_{m\ge 1} p^{-m/2}(f(p^m) + f(p^{-m})). \tag{27.72}$$

If g is supported in $[\lambda^{-1}, \lambda]$, then $g * g^*$ is supported in $[\lambda^{-2}, \lambda^2]$; only prime powers $p^m \le \lambda^2$ occur. The archimedean term is still essential, and RH requires all windows and all admissible tests.

By contrast, $E_{\mathrm{mid}} = \operatorname{sgn}(M)\sum_i \tau_i g_i$ is linear in the consecutive prime gaps g_i. Its square is a rank-one quadratic expression in that gap vector. It is not the Weil form merely because only finitely many primes occur. Here τ_i denotes the triangular gap weight of the residual, distinct from $w_j = p_j - j$. Even asking whether a finite vector belongs to the domain of QW_λ requires an embedding into the function space. A quadratic form is not itself a subspace to which a vector can be orthogonal.

A well-posed finite test would specify a linear map

$$\mathcal{J}_\lambda : \mathbb{R}^k \longrightarrow \{g \in C_c^\infty((\lambda^{-1}, \lambda)) : \widehat{g}(\pm i/2) = 0\} \tag{27.73}$$

and calculate the pulled-back matrix $H_{ij} = QW_\lambda(\mathcal{J}_\lambda e_i, \mathcal{J}_\lambda e_j)$. Then $\tau^{\mathsf{T}} H \tau$ is defined and checkable. No canonical embedding, equality with E_{mid}, or implication covering all admissible tests is established by the finite-window observation. That is the precise missing identification.

27.16.4. *15D: an exact determinant with entries zero or signed primes.* The proposed finite algebraic construction is possible. First recall why the classical Redheffer determinant works. Let $Z_{ij} = \mathbf{1}_{i|j}$ for $1 \le i, j \le n$. It is upper triangular with determinant 1, and $(Z^{-1})_{ij} = \mu(j/i)$ when $i \mid j$, zero otherwise. Replacing its first column by the all-ones vector gives $R_n = Z + (\mathbf{1} - e_1)e_1^{\mathsf{T}}$. The determinant lemma yields

$$\det R_n = 1 + e_1^{\mathsf{T}} Z^{-1}(\mathbf{1} - e_1) = \sum_{j\le n} \mu(j) = M(n). \tag{27.74}$$

This is the classical divisor-incidence construction recalled in [20].

For the present transform, a different elementary encoding suffices. Put

$$U = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, \qquad U^{-1} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}, \qquad \det U = 1. \tag{27.75}$$

Given column blocks $u_i, v_i \in \mathbb{Z}^2$, form

$$\mathcal{R}(u, v) = \begin{pmatrix} U & 0 & \cdots & 0 & u_1 \\ 0 & U & \cdots & 0 & u_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & U & u_m \\ v_1^{\mathsf{T}} & v_2^{\mathsf{T}} & \cdots & v_m^{\mathsf{T}} & 0 \end{pmatrix}. \tag{27.76}$$

Theorem 27.15 (Prime-entry determinant representation)**.** *There are explicit integer matrices $\mathcal{R}_T(n)$ of size $2Q(n)+1$ and $\mathcal{R}_{SS}(n)$ of size $2n+1$, all of whose entries belong to $\{0\}\cup\{\pm p_j : j \ge 1\}$, such that*

$$\boxed{\det \mathcal{R}_T(n) = T(n), \qquad \det \mathcal{R}_{SS}(n) = SS(n).} \tag{27.77}$$

Proof. The Schur complement of the diagonal block $D = \operatorname{diag}(U, \dots, U)$ gives

$$\det \mathcal{R}(u, v) = -\sum_{i=1}^{m} v_i^{\mathsf{T}} U^{-1} u_i. \tag{27.78}$$

For any zero or signed prime t_i, choose $u_i = (t_i, 0)^{\mathsf{T}}$ and $v_i = (-2, -3)^{\mathsf{T}}$. Then $v_i^{\mathsf{T}} U^{-1} u_i = (-10 + 9)t_i = -t_i$. For $\mathcal{R}_T(n)$ use the $Q(n)$ signed-prime summands $S(s_q)$.

For $\mathcal{R}_{SS}(n)$ use one block per integer $i \le n$. If $\mu(i) \neq 0$, use the preceding choice with $t_i = S(i)$. If $\mu(i) = 0$, choose instead $u_i = (5, 7)^{\mathsf{T}}$ and $v_i = (-2, -7)^{\mathsf{T}}$. In this case $U^{-1}u_i = (4, -1)^{\mathsf{T}}$ and $-v_i^{\mathsf{T}} U^{-1} u_i = 1 = S(i)$. Every entry is zero or a signed prime, and summing the contributions in (27.78) proves both identities. □

For example, at $n = 4$ the transformed sequence is $(2, -2, -3, 1)$, giving $\det \mathcal{R}_T(4) = -3$ and $\det \mathcal{R}_{SS}(4) = -2$.

This settles the existence question, but the block construction encodes any sum of signed primes, with a separate gadget for a unit summand. It is not a newly proved divisor-incidence constraint. The already established transfer gives

$$\text{RH} \iff |\det \mathcal{R}_T(n)| \ll_\epsilon n^{3/2+\epsilon} \quad \text{for every } \epsilon > 0. \tag{27.79}$$

The determinant growth estimate remains unproved. Neither the representation nor the general Hadamard determinant inequality supplies this bound. The construction is finite and testable; its RH-strength estimate still requires the original arithmetic cancellation.

27.16.5. *15E: what regularising the pair measure actually gives.* The Guinand–Weil formula applies to an admissible test function. A signed atomic measure is not such a function until regularised. The finite regularisation can be worked out exactly, with no continuation of δ assumed. Set $c_n = \mu(n) w_{j(n)}$ at squarefree n and $c_n = 0$ otherwise, and

$$\nu_X = \sum_{n \le X} c_n \delta_{\log n}, \qquad h_{X,\varepsilon} = \varphi_\varepsilon * \nu_X, \qquad f_{X,\varepsilon,s}(x) = x^{-s} h_{X,\varepsilon}(\log x), \tag{27.80}$$

where $\varphi_\varepsilon \in C_c^\infty(\mathbb{R})$. This ν_X retains unfinished pairs; it is the logarithmic pushforward of the original-order coefficient measure. Write

$$\Phi_\varepsilon(v) = \int_{\mathbb{R}} \varphi_\varepsilon(u) e^{vu}\, du, \qquad \delta_X(s) = \sum_{n \le X} c_n n^{-s}. \tag{27.81}$$

For the unnormalised Mellin convention $\widetilde{f}(z) = \int_0^\infty f(x) x^{z-1}\, dx$, direct substitution gives

$$\widetilde{f}_{X,\varepsilon,s}(z) = \Phi_\varepsilon(z - s)\delta_X(s - z). \tag{27.82}$$

Using the survey's Section 2.1.3 convention, the explicit formula becomes

$$\sum_\rho \Phi_\varepsilon(\rho - s)\delta_X(s - \rho) = \Phi_\varepsilon(-s)\delta_X(s) + \Phi_\varepsilon(1 - s)\delta_X(s - 1) - \sum_v W_v(f_{X,\varepsilon,s}). \tag{27.83}$$

Zeros are counted with multiplicity. At fixed X, ε, s the smooth compact support gives rapid vertical decay of the Mellin transform, so the zero sum is legitimate. Here the unnormalised local terms are

$$W_p(f) = (\log p) \sum_{m \ge 1} \left(f(p^m) + p^{-m} f(p^{-m})\right); \tag{27.84}$$

the $p^{-m/2}$ in (27.72) belongs to the different, symmetric normalisation.

Equation (27.83) is a proved finite regularised identity. It is not a closed continuation formula for $\delta(s)$: it also contains $\delta_X(s - 1)$, all the shifted values $\delta_X(s - \rho)$, and local terms depending on the regularisation and cutoff. Passing to $X \to \infty$ and then removing the regularisation requires uniform

summability or another justified limiting procedure. That arithmetic estimate is not supplied by admissibility of each finite test. This route remains meaningful, with the limiting step explicit, rather than bypassing the unresolved control of δ.

27.16.6. *15F: the survey's current programme and the remaining work here.* The survey's Sections 6.4–6.6 distinguish several established constructions from further obligations: a simplicity and evenness property for the lowest eigenfunction of the truncated Weil form, and sufficient control relating that eigenfunction to another family of approximants. Convergence for one family cannot simply be transferred to another because they appear related. This comparison is recorded without pursuing an operator route in the present programme.

The same logical issue occurs here: every fixed-duration $\delta_{\leq L}$ can be holomorphic in a large half-plane while the limit as $L \to \infty$ has singularities. The new determinant identity, the finite Guinand–Weil identity, and the energy equivalence are established intermediate statements. None supplies the missing uniform arithmetic estimate.

What remains unresolved is the arithmetic estimation of the signed, weighted two-time terms, including unfinished pairs. The two-cut identity is exact; by itself it does not control δ. The explicit poles of B and P_δ have been located. The singularities of the full δ, and their effect on α, remain unproved without control of E_δ or another valid continuation mechanism.

References

[1] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer-Verlag, New York, 1976.

[2] H. Cramér, On the order of magnitude of the difference between consecutive prime numbers, *Acta Arith.* **2** (1936), 23–46. DOI: 10.4064/aa-2-1-23-46.

[3] H. L. Montgomery and K. Soundararajan, Primes in short intervals, *Comm. Math. Phys.* **252** (2004), 589–617. DOI: 10.1007/s00220-004-1222-4; arXiv:math/0409258.

[4] N. Ng, The distribution of the summatory function of the Möbius function, *Proc. London Math. Soc.* (3) **89** (2004), 361–389. DOI: 10.1112/S0024611504014741; arXiv:math/0310381.

[5] A. M. Odlyzko and H. J. J. te Riele, Disproof of the Mertens conjecture, *J. Reine Angew. Math.* **357** (1985), 138–160. DOI: 10.1515/crll.1985.357.138.

[6] H. Riesel and G. Göhl, Some calculations related to Riemann's prime number formula, *Math. Comp.* **24** (1970), 969–983. DOI: 10.1090/S0025-5718-1970-0277489-3.

[7] J. E. Littlewood, Quelques conséquences de l'hypothèse que la fonction $\zeta(s)$ de Riemann n'a pas de zéros dans le demi-plan $\Re s > 1/2$, *C. R. Acad. Sci. Paris* **154** (1912), 263–266.

[8] K. Matomäki and M. Radziwiłł, Multiplicative functions in short intervals, *Ann. of Math.* (2) **183** (2016), no. 3, 1015–1056. DOI: 10.4007/annals.2016.183.3.6; arXiv:1501.04585.

[9] T. Tao, The logarithmically averaged Chowla and Elliott conjectures for two-point correlations, *Forum Math. Pi* **4** (2016), e8, 36 pp. DOI: 10.1017/fmp.2016.6; arXiv:1509.05422.

[10] T. Tao and J. Teräväinen, Odd order cases of the logarithmically averaged Chowla conjecture, *J. Théor. Nombres Bordeaux* **30** (2018), no. 3, 997–1015. DOI: 10.5802/jtnb.1062; arXiv:1710.02112.

[11] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.

[12] X.-J. Li, The positivity of a sequence of numbers and the Riemann hypothesis, *J. Number Theory* **65** (1997), no. 2, 325–333. DOI: 10.1006/jnth.1997.2137.

[13] E. Bombieri and J. C. Lagarias, Complements to Li's criterion for the Riemann hypothesis, *J. Number Theory* **77** (1999), no. 2, 274–287. DOI: 10.1006/jnth.1999.2392.

[14] L. Báez-Duarte, A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis, preprint, 2002. arXiv:math/0202141.

[15] M. V. Berry and J. P. Keating, $H = xp$ and the Riemann zeros, in *Supersymmetry and Trace Formulae*, NATO ASI Series, vol. 370, 1999, pp. 355–367. DOI: 10.1007/978-1-4615-4875-1_19.

[16] A. Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, preprint, 1998. arXiv:math/9811068.

[17] B. Rodgers and T. Tao, The de Bruijn–Newman constant is non-negative, preprint, 2018, revised version 5, 2021. arXiv:1801.05914v5.

[18] D. Platt and T. Trudgian, The Riemann hypothesis is true up to $3 \cdot 10^{12}$, *Bull. Lond. Math. Soc.* **53** (2021), 792–797. DOI: 10.1112/blms.12460; arXiv:2004.09765.
[19] S. S. Vallender, Calculation of the Wasserstein distance between probability distributions on the line, *Theory Probab. Appl.* **18** (1974), no. 4, 784–786. DOI: 10.1137/1118101.
[20] A. Connes, The Riemann Hypothesis: Past, Present and a Letter Through Time, preprint, version 1, February 2026. arXiv:2602.04022v1.
[21] NIST Digital Library of Mathematical Functions, Sections 6.6 and 25.10, consulted 7 September 2026. dlmf.nist.gov/6.6; dlmf.nist.gov/25.10.